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ANTI-AIRCRAFT MISSILE COMPLEX “TOR-M1”



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Study guide

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The manual contains a technical description of the complex, the missile and its components. It describes the features of the operation of the means of detecting targets and guiding the missile, the flight trajectory, the target guidance algorithm, the principles of operation and functioning of the main units of the missile, the sequence of operation of the missile as part of the complex and its main elements.

Intended for full-time and evening students studying the subject "Fundamentals of Rocket Design and Construction" in the faculties of "Rocket and Space Technology" and "Information and Control Systems".

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Reviewer of the Department of Rocket Science, PhD in Engineering, Professor *A.L. Isakov*

*Approved
editorial and publishing
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INTRODUCTION

The main requirements that must be taken into account when creating anti-aircraft missile systems (SAM) are determined primarily by the characteristics of the targets. The adoption of high-precision weapons, which have their own specific characteristics, has expanded the range of these requirements.

These characteristics are:

- small effective target dispersion surface, especially in the forward hemisphere, for the centimeter 0.1 m² wavelength range (1.5–5 cm);

- a wide range of altitudes (30–60 m – the lower level of flight altitudes) and angles (45 – 60° and more) the approach of the air defense missile system to the protected objects;

- a wide range of flight speeds (200 – 700 m/s) and available lateral overloads (8 – 10);

- high level of protection of weapons, especially such types as guided bombs, shells.

Taking these characteristics into account, the following general requirements for the air defense missile system can be identified:

- high degree of readiness (3–4 sec);

- short time for accelerating the rocket to maximum speeds (3–5 s) and maintaining these speeds grow until the moment of hitting the air target;

- high maneuverability (possible lateral overloads not less than 10);

- the corresponding combat equipment of the missile, capable of destroying heavily protected goals;

- low cost of the air defense system (at the level of the high-precision weapons used).

These requirements can be achieved through:

- high automation of the complex;

- the use of effective radar means of target detection and transmission commands on board the rocket;

- the use of powerful algorithms and computing tools for information processing, tomized missile tracking;

- design solutions that allow finding a compromise between the price and effectiveness of the air defense missile system. The

Tor-M1 system satisfies these requirements.

This manual provides general information about the Tor-M1 complex, its operation and the missile it is equipped with. The device, features of operation are examined in detail.

missiles and their elements during preparation for launch, during launch, in flight and upon meeting the target.

The manual is intended for students studying in specialties whose curricula contain disciplines related to the study of the structure and functioning of rockets. To facilitate the assimilation of the information presented in the manual, questions are given at the end of the text for self-control.

Based on these questions, an electronic version of the students' knowledge test was compiled.

The author expresses gratitude to S.A. Chirikov and K.A. Afanasyev for technical assistance in creating this manual.

List of accepted abbreviations

AP – autopilot,
ASA – automatic start-up equipment,
BIGG – hot gas source unit, BRU – on-board radio
control equipment,
BM – combat vehicle,
BC – warhead,
BB – explosive substance,
GG – gas generator,
ZGURD – single-action protective and sealing device,
ZIP – spare parts, tools, accessories, K1, K2 – main control commands
for the autopilot,
KV – command to arm the radio fuse, KZAZ – command to
prohibit asynchronous launch of the transponder,
KP – command to switch the autopilot operating mode, KRAZ – command to
allow asynchronous launch of the transponder,
KU – ejection device, KUV – command to control
the arming of the radio fuse, KOS – command for the relative speed of
approach of the missile to the target, KPO – command to switch the limitation of the rudder
angle, KTO – set of rigging equipment, KU – ejection device,

NLC – low-flying target,
PIM – safety actuator mechanism,
PP – passive interference,
PU – starting device,
RV – radio fuse,
RDTT – solid propellant rocket engine, SVR – missile sighting
station,
UHF – ultra-high frequency,
SPK – command transmission station,
SRP – computing and decision device,
TZM – transport and loading vehicle, TPK – transport
and launch container,
UPR – command "Control", UPT – amplifier
– current converter,
HIT – chemical current source,
FAR – phased antenna array, EMP – electric machine
converter.

1. GENERAL INFORMATION

1.1. Complex 9K331 and its operation

The autonomous self-propelled anti-aircraft missile system (SAM) 9K331 ("Tor-M1") is designed to effectively cover troops and military facilities from modern and future air attack weapons, primarily high-precision weapons, as well as aircraft, helicopters, cruise missiles, guided aerial bombs and remotely piloted aircraft. It is a modernization of the 9K330 ("Tor") system, the testing of which began in 1983-84. The tentative date for accepting the 9K331 system into service is 1988.

The mobile all-weather short-range complex "Tor-M1" provides air defense of troops in mobile combat and on the march, as well as in areas of troop concentration, and protects the most important military points (command posts, communications centers, radio-technical equipment, bridges, airfields).

The Tor-M1 air defense missile system differs from its predecessors in that all of its combat information, communications and weapons are mounted on a single tracked chassis (Fig. 1) and represent a compact, functionally complete and technically advanced tactical unit – a combat vehicle capable of autonomously or as part of an air defense system performing a combat mission during a modern battle without additional refueling and loading, and at the same time, round-the-clock (with refueling) all-weather combat duty and support of troops in battle or on the march.



Fig. 1. Autonomous military anti-aircraft missile system "Tor-M1"

The complex includes:

- combat vehicle 9A331 (Fig. 2) and anti-aircraft missile module 9M334 (9M331 missiles in transport and launch container 9Y281, Fig. 3);
- transport and loading vehicle 9T244;
- transport vehicle 9T245;
- maintenance vehicles 9V887M and 9V888-1M;
- set of rigging equipment 9Y116;
- group spare parts kit vehicle 9F399-1M1;
- autonomous electronic training device for combat vehicle operators 9F678.

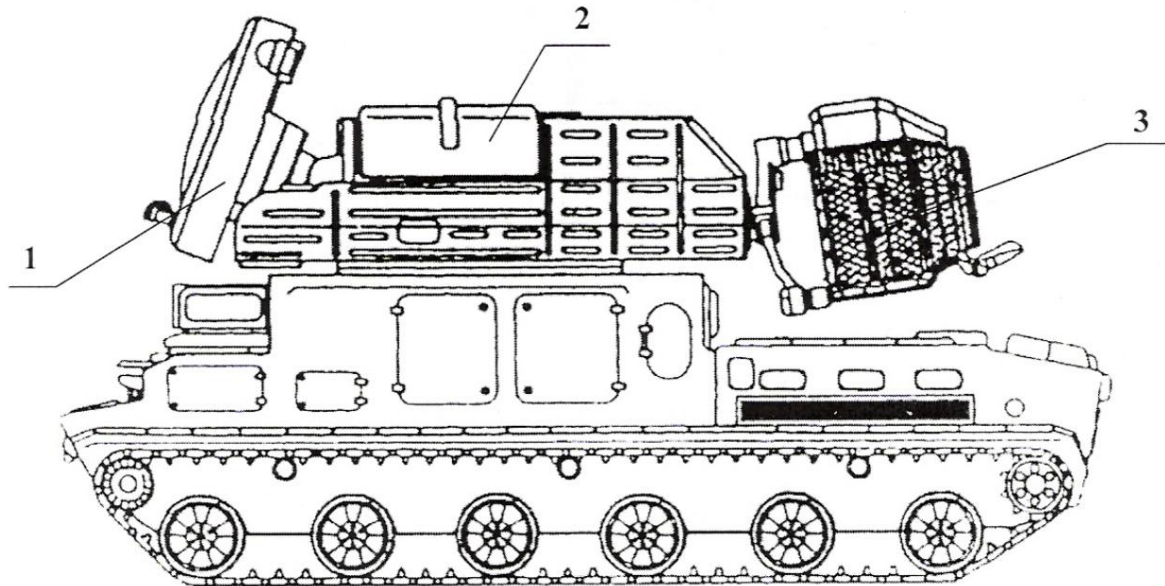


Fig. 2. Combat vehicle 9A331:

1 – Missile guidance radar (SVR, SPK); 2 – 9M334 module compartment; 3 – detection station

The 9A331 combat vehicle is based on:

- two 9M334 anti-aircraft missile modules (eight 9M331 missiles) (Fig. 2, 3);
- three-coordinate target detection radar station 3 (Fig. 2, 4) together with a ground-based radar interrogator;
- target tracking and guidance radar station 1 with a phased antenna with a scanning array (SVR and SPK) and an electronic beam control system (Fig. 2, 5);
- a duplicate television-optical sight (Fig. 5), providing automatic tracking targeting by angular coordinates;
- high-speed digital computing system;
- launch automation equipment (equipment for displaying information about the air situation and the combat operation cycle, as well as indication of the functioning of the combat vehicle systems and means, working consoles for the control commander and operators, auxiliary equipment);
- a system of telecode operational command radio communications;
- navigation, topographic referencing and orientation equipment;
- combat vehicle functional control system;
- autonomous power supply and life support system (primary power supply source with an electric generator drive from a gas turbine engine or the running engine of a self-propelled chassis).

To protect low-mobility military, as well as civilian and industrial facilities, design modifications of the combat assets of the SAM system have been developed: container, towed and wheeled versions. These modifications have the same tactical and technical characteristics, excluding mobility, but are somewhat cheaper than the self-propelled base.



Fig. 3. Module 9M334

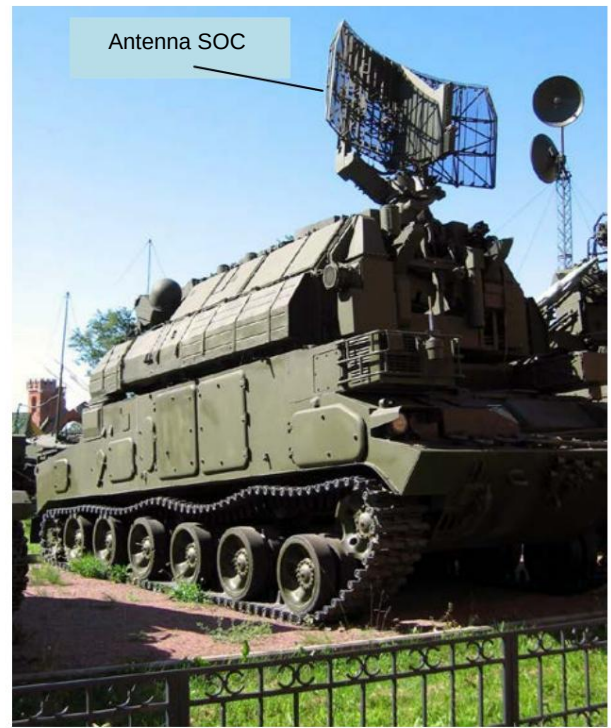


Fig. 4. Target detection station (TDS)



Fig. 5. Target tracking and missile guidance station (TTS and SPK) and backup television-optical sight

Detection radar 3 (see Fig. 2, 4) is a coherent-pulse all-round radar. It operates in the centimeter wave range with frequency beam control by elevation angle. The average transmitter power is 1.5 kW, the resolution is not worse than $1.5...2.0^\circ$ by azimuth, 4° by elevation and 200 m by range. Maximum errors in determining target coordinates are no more than half of the specified resolution values.

The station is capable of detecting F-15 aircraft flying at altitudes from 30 to 6000 m with a probability of no less than 0.8 at a range of 25...27 km. Unmanned aerial vehicles are detected with a probability of no less than 0.7 at a range of 9...15 km, and helicopters hovering in the air are detected with a probability of no less than 0.7 at a range of with a probability of 0.6...0.8 at a range of 13...20 km, helicopters with rotating propellers on the ground - with a probability of 0.4...0.7 at a range of 13...20 km. In this case, targets covered by active and passive interference can also be detected.

The detection radar provides multi-partial (8 partials – beams) [14] three-coordinate coverage of space at a high rate. The scanning period is 1 s, the beam width in the vertical plane is 4° . Scanning of the angular coverage space in the vertical plane is mechanically divided into two ranges from $0...32^\circ$ and $32...64^\circ$. This means that two batteries of the Tor-M1 SAM system can simultaneously view a zone in an angular raster of $0...64^\circ$. Provisions are made for increasing the signal energy by using a long pulse with intra-pulse modulation and a mode of concentrating all the radiation energy in one partial – three in one (Fig. 6).

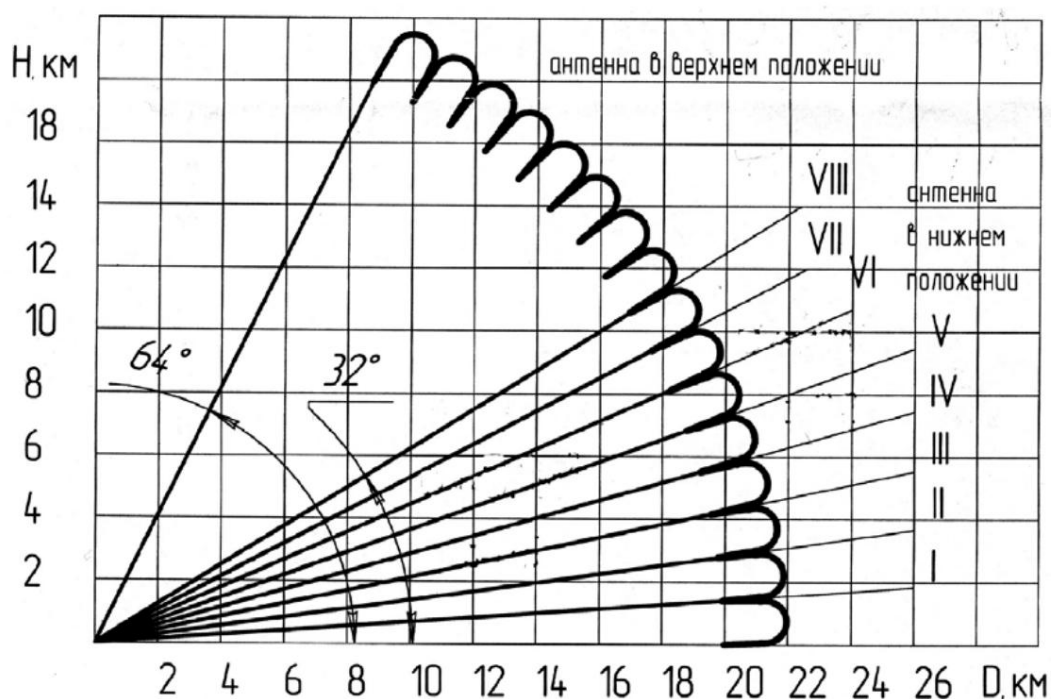


Fig. 6. Scanning range of the angular space of the detection radar

The use of digital signal processing allows for reliable detection of both high-speed and low-speed (up to 10 m/s) targets without "blind speeds" in difficult conditions of passive (natural and artificial) interference, taking into account the influence of the underlying surface.

Signal processing is carried out by special computers and a central computer, the computing and algorithmic capabilities of which make it possible to solve problems of air situation analysis, combat decision-making and other intellectual tasks of air force control operations.

The detection radar is linked to the target nationality identification system and automatically blocks (with a high probability) the possibility of hitting "friendly" aircraft. medical devices.

To ensure the station can operate while the BM is moving, the antenna position is stabilized.

Guidance radar (SVR, SPK) (see Fig. 2, 5) is a coherent-pulse (pulse-Doppler type) radar. It operates in the centimeter wave range, has a low-element phased antenna array (PAR) forming a beam, 1° wide in azimuth and in elevation, providing electronic scanning of the beam in the corresponding planes. Such an instantaneous (400...600 ms) transition to automatic tracking, as well as simultaneous tracking and firing of two targets in the PAR sector.

construction

systems allow

provide

practically

The station searches for a target using target designation data from the target detection station and captures one target for automatic tracking. With a probability of 0.5, the guidance station is capable of switching to automatic tracking of a fighter aircraft flying at a range of 23 km. With a decrease in range, this probability increases significantly, so at a range of 20 km it already amounts to 0.8 [14].

The tracking radar signal processing system – digital monopulse with pulse compression and a corresponding signal processing algorithm – ensures not only high accuracy and interference protection, but also target class recognition, which allows for optimization of the operating modes of the missile guidance system and its combat equipment.

The combat operation of the 9K331 SAM system follows a pattern typical for anti-aircraft missile systems. owls with a radio command guidance system.

The detection station, whether on the move or on the spot, performs a 360-degree view of the area, detects and identifies targets. The combat vehicle's computing facilities analyze the air target, select the most dangerous targets for firing, and generate target designation data for the guidance station (command transmission station, CTS).

Guidance station (missile sighting station plus command transmission station) based on The use of target designation data is carried out by:

- search and capture of one target for automatic tracking; – precise target tracking in three coordinates; – launch of one or sequentially (after 4 s) two missiles at the tracked target;
- capture of the rocket after launch by a separate coordinator and its introduction into the beam of a phased antenna array;
- precise missile tracking;
- control of missiles based on commands generated by the difference in coordinates between the missiles and the target in accordance with the selected guidance method, which corresponds to the most optimal minimum conditions for the missile to meet the target depending on its type, altitude and flight characteristics;
- issuing a command to the missile's radio fuse to delay its activation depending on the speed at which the missile approaches the target.

Main tactical and technical characteristics of the complex

Number of simultaneously detected targets.....	48;
Number of simultaneously tracked targets	2;
Detection zone boundaries:	
by range, km.....	27;
in azimuth, deg.....	360;
by elevation angle, degrees	0-32 or 32-64;
by altitude, km, not less than.....	23;
Boundaries of the affected area, km:	
by range, km.....	1-12;
by altitude, km by	0.01-6.0;
course parameter, deg.....	6;
Speed of hit targets, m/s.....	10-700;
Maximum lateral overload of the target being hit.....	10;
Minimum reflective surface of the target, m ²	0.1;
Complex response time (from target detection to missile launch), s:	
when shooting from a position.....	7.4;
when shooting from a short stop after movement	9.7;
Number of missiles on a combat vehicle.....	8;
Probability of being hit by one missile.....	
aircraft (type F-15).....	0.45-0.8;
helicopter	0.62–0.75;
cruise missile.....	0.93–0.97;
high-precision weapons	0.75-0.9;
Maximum speed of the basic version, km/h:	
by highway	65;
along a dirt road.....	45;
Combat vehicle weight, t	37;
Fuel range (with two hours of equipment operation), km.....	500;
Combat crew, including driver-mechanic.....	3

Operation of the complex is permitted at altitudes of no more than 3000 m above sea level, at any time of year and day, in various meteorological conditions in the ambient air temperature range from -50°C to $+50^{\circ}\text{C}$, in conditions of solar radiation and relative humidity of no more than 98% at a temperature of $(30 \pm 5)^{\circ}\text{C}$ and a wind speed of no more than 20 m/s [15].

The operating mode of the missile equipment during activation on the combat vehicle is cyclical: 10 minutes of operation - 10 minutes of break. After three activations there must be a break of at least one hour. At any time during the break, a single activation of the missile equipment for one minute is allowed to conduct the launch.

1.2. Anti-aircraft missile module 9M334

The 9M334 anti-aircraft missile module (see Fig. 3, 7) is a transport and launch container placed in a vertical position in a silo of a combat vehicle. Each module contains four missiles.

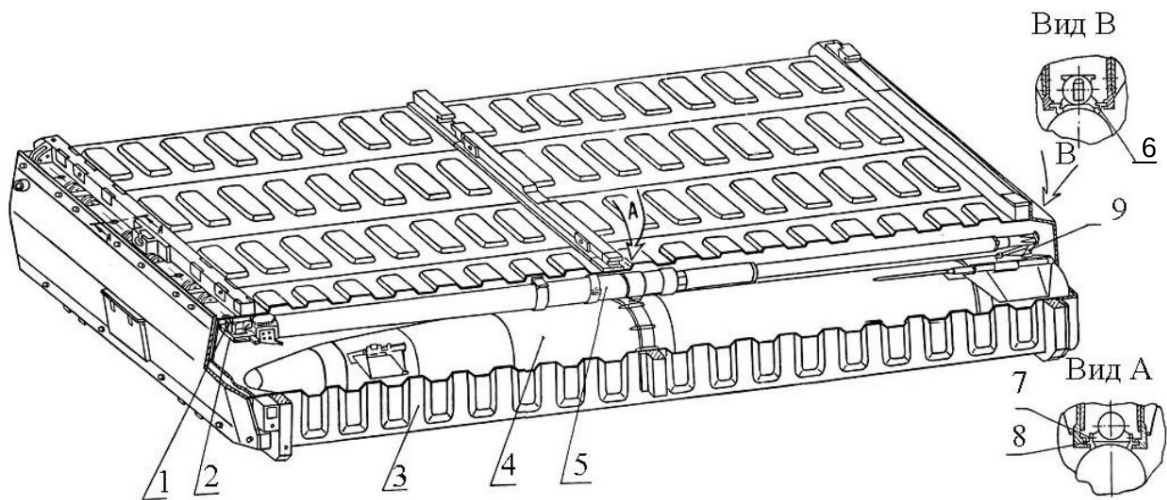


Fig. 7. Anti-aircraft missile module 9M334:
1 – jumper; 2 – earring; 3 – container; 4 – rocket; 5 – ejection device; 6,8 – yokes; 7 – shear bolt; 9 – lever

The 9M334 module is operated during its established service life without routine maintenance or inspection of the missile's onboard equipment.

Main parameters of the module

Mass of the module (TPK plus four missiles) with two beams, kg	Mass	1053
of the TPK with two beams, kg		333
Weight of one beam, kg	40	
Dimensions of the module with two beams, mm	539y1507y3005	

Each module is equipped with two special beams, with the help of which the modules can be assembled into multi-tiered packages (Fig. 8). In such packages, missiles are stored and transported at all stages of operation, except for operation on the transport and loading (TZM) and combat vehicle (CM). The transport vehicle carries two packages of four modules and has crane equipment for loading the module into the combat vehicle.

The BM is loaded using a transport and loading vehicle. First, the module is moved from a horizontal to a vertical position, then lowered into the BM shaft. The time for loading a combat vehicle with two modules is 25 minutes.

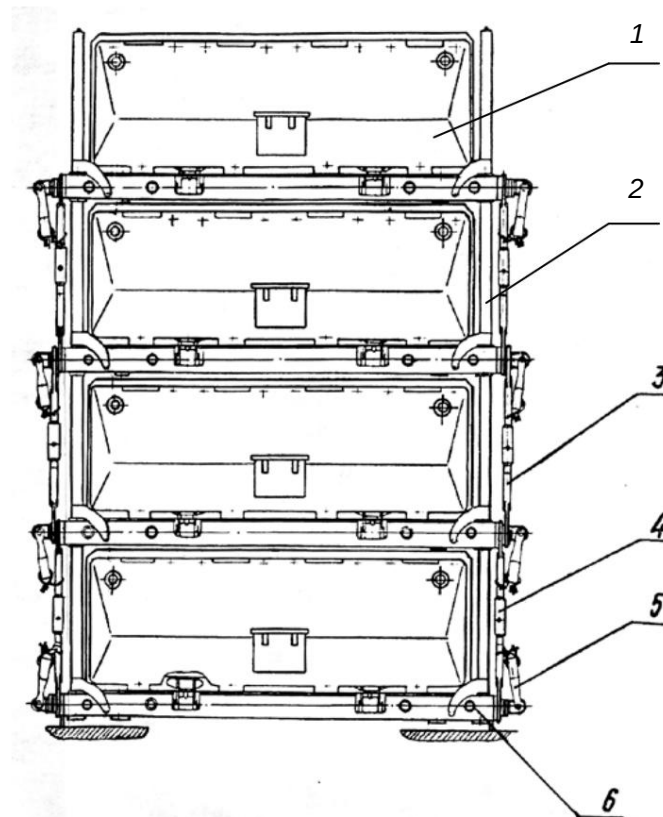


Fig. 8. Scheme of packaging of 9M334 modules:
1 – module 9M334; 2 – beam; 3 – coupler; 4 – coupling; 5 – bracket; 6 – socket

1.3. Missile 9M331

The general appearance of the rocket is shown in Fig. 9, 10. The rocket is made according to the aerodynamic configuration “duck”: the wings are located in the tail section of the rocket, the air rudders are in the nose section. The rudders provide control of the rocket flight along a given trajectory and its stabilization relative to the longitudinal axis. The wings together with the tail section of the body form a wing block, installed on the rocket body on a bearing. In flight, due to the asymmetric flow around the wings and body when the rudders are deflected and the rocket maneuvers, a “moment of oblique flow” occurs – a roll moment. Under the influence of aerodynamic forces, the block rotates freely relative to the longitudinal axis of the rocket, eliminating the occurrence of large roll moments.

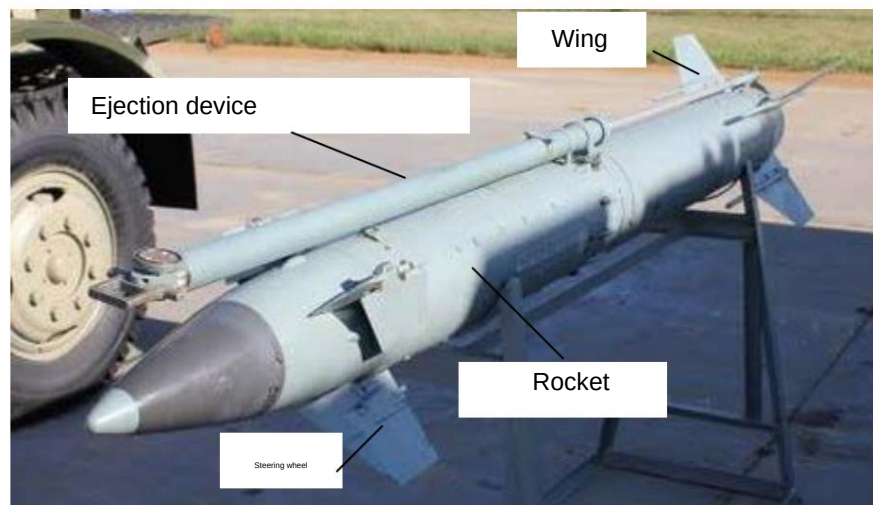


Fig. 9. General view of the 9M331 missile

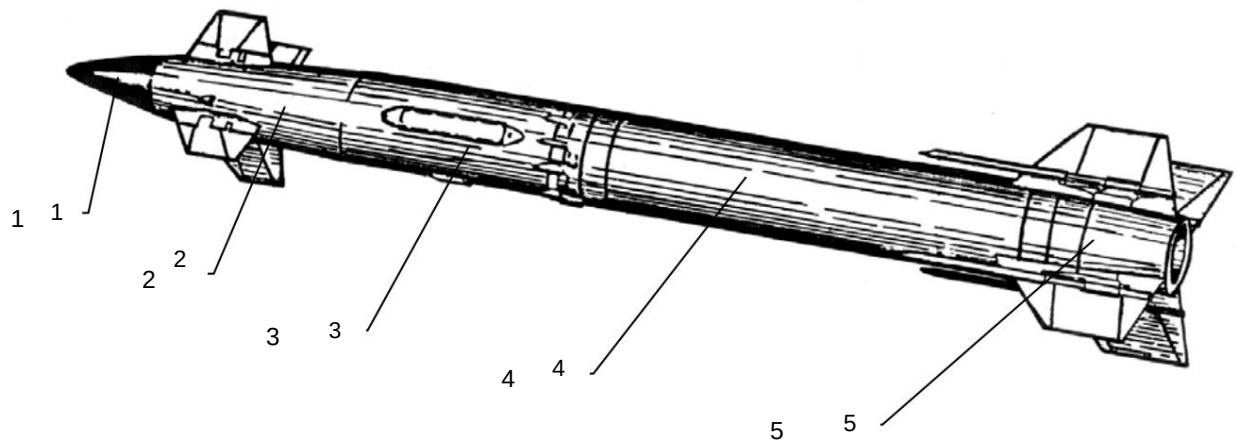


Fig. 10. Missile 9M331:

1 – radio-transparent fairing (first compartment); 2 – control compartment; 3 – instrument compartment; 4 – dual-mode engine; 5 – wing block

In order to minimize the transverse dimensions of the TPK, the rocket's wings and rudders are made foldable (Fig. 11, 12).

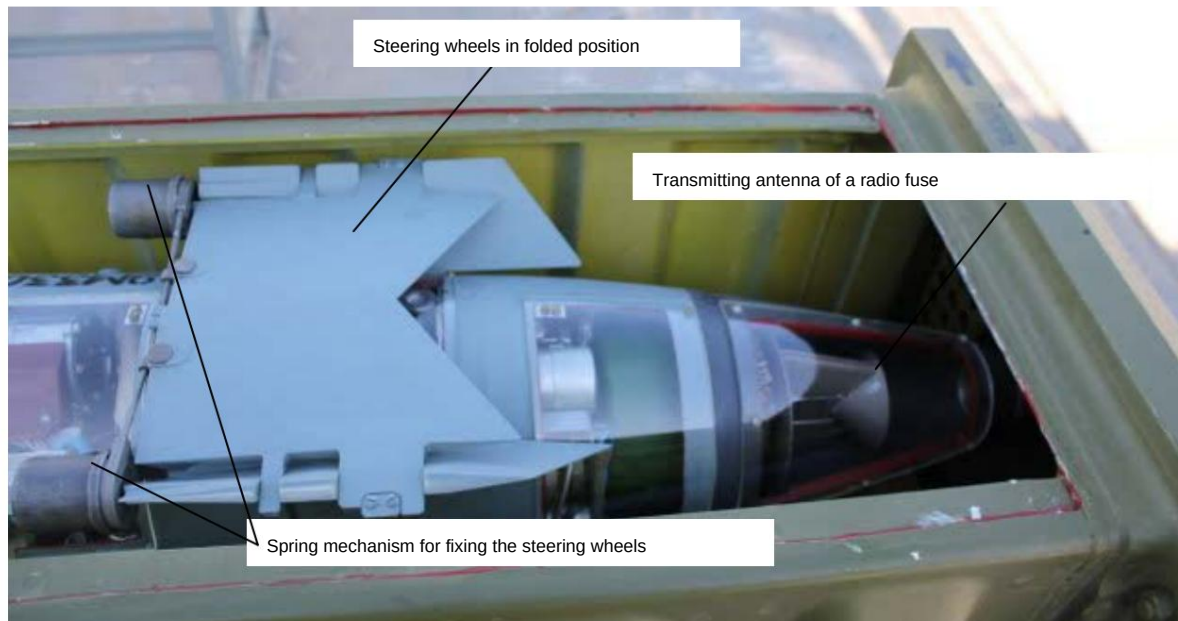


Fig. 11. The first and second sections of the rocket in the container

Each missile is equipped with a catapult device (CD) that ensures the missile launch from the TPK. Each missile is fixed and secured in the TPK in three places. The missile is secured from lateral movements by yokes 6 and 8 (see Fig. 7, 9) and the TPK guides along which it moves during launch. Longitudinal movement of the missile in the TPK is excluded by the catapult device 5, one end of which is secured to the TPK guide bracket by means of a clevis 2 with the help of a jumper 1, and the other rests against the end of the missile engine by means of a lever 9. Additional fixation is performed by two shear bolts 7. The missile and the catapult device have electrical connectors by which they are connected to the BM automation equipment via a harness and an electrical connector of the TPK.

The guided flight of the missile is provided by on-board radio control equipment (BRU), autopilot (AP) and command unit (CU) located on the rocket.

The target is destroyed by combat equipment consisting of an active radio fuse (RF), a safety-actuating mechanism (SAM) and a high-explosive fragmentation warhead (HE).

The onboard equipment is powered by a chemical current source and an electric current converter.

The gas supply to the rocket's executive control bodies is provided by two solid-fuel pouring gas generator.

The rocket's propulsion system is a solid propellant rocket engine, providing launch and cruise thrust modes.

Main performance characteristics of the 9M331 missile

Weight, kg	167;
Length, mm.....	2898;
Midship, mm	239;
Wingspan, mm.....	650;
Rudder span, mm.....	530;
Warhead weight, kg	14.8;
Warhead	High-explosive fragmentation;
Maximum rocket speed, m/s.....	700 – 850;
Minimum maneuvering speed up to, m/s	300;
Maximum available transverse overload	15-16;
Mass of ejection device, kg.....	9

Self-assessment questions for section 1

1. What is the 9M334 anti-aircraft missile module?
2. List the composition of the complex.
3. What is located on the base of the 9A331 combat vehicle?
4. Give a description of the detection radar.
5. Why is the detection radar a coherent pulse radar?
6. Purpose of the television optical viewfinder.
7. For what purpose and how is the vertical scanning of the radar detection space carried out?
planes?
8. How is the target's nationality identified?
9. List the functions performed by the guidance station.
10. Give a description of the guidance radar.
11. Phased array antenna (PA), what is it?
12. Why is the guidance radar a coherent pulse Doppler type radar?
13. List the functions performed by the guidance station.
14. What are the features of the operation of the complex and the operating mode of the onboard equipment?
15. What does the term "blind speed" mean in radar?
16. Main features of the 9M334 anti-aircraft missile module.
17. Advantages and disadvantages of the aerodynamic design of the 9M331 missile.
18. Explain the operation of the ejection device.
19. Provide a general description of the missile's performance characteristics.

2. ROCKET DEVICE

The layout of the rocket is shown in Fig. 12. The rocket body is divided into five compartments for convenience of its manufacture and subsequent installation of equipment.

The first section – nose cone 1 – is made of radio-transparent heat-resistant plastic AG-4V to ensure the operation of the transmitting antenna of the radio fuse, which is attached to the front frame of the second section.

The second section is the control section 32, consisting of two welded parts made of AMG-6 alloy. Four air rudders-ailerons 31 are installed on the section body. The section contains a block of hot gas sources 6, four gas steering machines 4 with gas wiring for their supply, and a transmitter of the radio fuse 3. Each rudder is driven by its own steering machine.

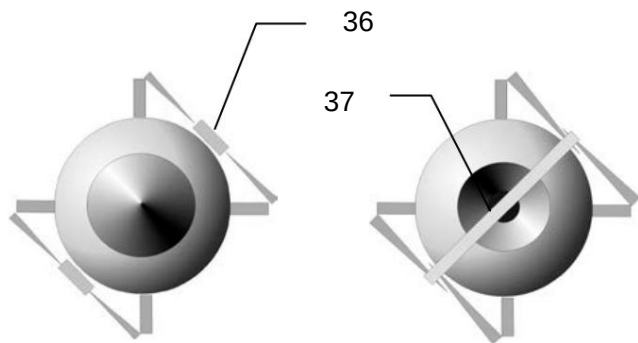
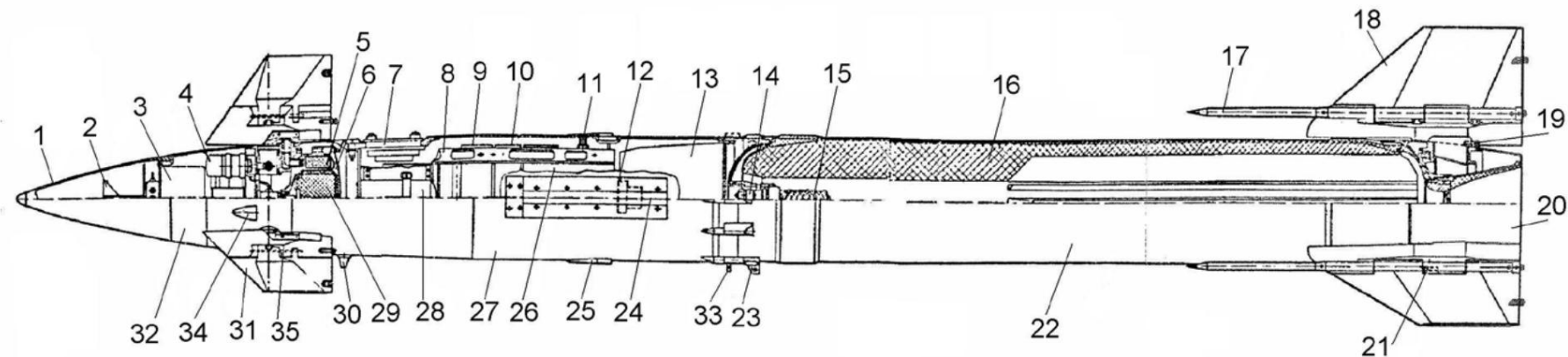


Fig. 12. Missile layout: 1 - radio-transparent fairing; 2 - radio fuse transmitting antenna; 3 - radio fuse transmitter; 4 - steering gear; 5 - declination system gas generator charge; 6 - hot gas source unit; 7 - on-board electrical connector; 8 - on-board equipment mounting stringer; 9 - autopilot; 10 - radio fuse receiver; 11 - letter switching device; 12 - safety actuator; 13 - warhead; 14 -

pressure indicator; 15 - igniter; 16 - solid propellant charge; 17 - torsion bar; 18 - wing; 19, 21, 23 - rocket yokes; 20 - tail section; 22 - solid propellant rocket engine; 24 - receiving antenna of radio fuse; 25 - antenna of on-board radio equipment; 26 - on-board radio equipment; 27 - equipment compartment; 28 - on-board power source; 29 - charge of gas generator for power supply of steering machines; 30, 33 - lever; 31 - rudder-aileron; 32 - control compartment; 34 - nozzle; 35 - locking mechanism, 36 - spring mechanisms of locking of rudders; 37 - jumper

The hot gas source block has two isolated chambers with solid fuel charges: a central one with a charge of 29 for powering the steering machines and an outer ring one with a charge of 5 to power the jet devices of the declination system.

From the central chamber of the block, gas enters the gas line and is distributed along the steering wheel. to the steering machines, and from the output of the steering machines it is discharged overboard the rocket through the nozzle 34.

From the annular chamber, the gas is discharged into the receivers of the jet devices formed in the body of the rudders 31. In the front

part of the compartment, there is a transmitter of the radio fuse 3, fixed to the front end frame. The transmitting antenna 2, installed on the body of the radio fuse, is located in the area of the first radio-transparent compartment.

The air rudder-aileron has a folding console. To hold the rudder in the folded position and to open it, a spring mechanism 36 is used, held by a shaped groove on a special bracket - the tip of the rudder at the trailing edge. The device of the spring mechanism is shown in Fig. 15. After the rocket is ejected from the TPK, the steering machines turn the rudders, the rudders open and the spring mechanism is

thrown off the tip. In the open position, the rudder is fixed by a spring lock (pin) 35 (see Fig. 12), located in the plane

steering wheel.

The third compartment – the instrument compartment (27) – is used to accommodate on-board equipment (except for the radio fuse transmitter), power supplies and electrical switching equipment, as well as the warhead with a safety actuator. The compartment hull shell is also made of AMG-6 alloy.

The equipment unit includes the autopilot 9, the radio fuse receiver 10 and the radio control equipment. The elements of the equipment unit, the chemical current source (two batteries) and the electric current converter 28 are combined into a single unit, being secured to the stringers 8. The stringers are attached to the compartment body with radial screws. The warhead 13 is console-mounted on the rear frame of the compartment. The safety

The actuator 12 is installed in the front part of the central channel of the BC.

In the front part of the third compartment there is an on-board electrical connector 7 of the missile. On the sides, flush with the hull, there are two receiving and transmitting antennas of the on-board radio control equipment 25. In the middle part of the compartment there is an approach to the device for switching the letter frequencies of the on-board radio equipment (11). In the front part of the compartment, at the bottom, there is a rotary lever 30. When the missile is launched, by turning the lever, the push-button switches in the on-board electrical system are triggered. At the bottom, in the rear part of the compartment, the second rotary lever 33 is used to duplicate the launch of the missile engine.

On the front and rear docking frames of the compartment there are bosses for connecting the compartment with the adjacent ones: the second (bolts) and the fourth (studs).

The fourth section is a dual-mode solid propellant rocket engine 22, consisting of a housing and a charge filled into it. The engine charge ensures engine operation in the launch and cruise sections of the rocket's flight. When the engine is running, the charge burns along the internal channel and slots. The presence of slots ensures a launch mode of increased thrust at the beginning of engine operation, followed by its reduction to the cruise mode. The duration of the launch section is 4 sec, the cruise section is 8 sec. The total operating time is 12 sec. The engine has a non-adjustable nozzle, ensuring normal operation in both modes throughout the entire range of rocket operating temperatures.

The igniter, pyrotechnic cartridges for igniting the igniter, and a pressure indicator in the combustion chamber, used to maintain the power supply circuit of the PIM in the initial state during the rocket ejection, are installed on the front bottom of the engine. To duplicate the engine start, an electric igniter with a one-second delay is installed in the front bottom. On the rear bottom there is a cylindrical landing belt on which the sub-

fifth compartment bearing.

The fifth section is the wing block 20, which forms the tail section of the rocket. Four folding wings 18 are fixed to the block body, and their opening occurs with the help of torsion bars 17. Two wings are fixed in the folded position by a crossbar 37, the other two are held from opening by fixed wings. After the engine is started, the crossbar is destroyed and the wings open. The general appearance of the wing block is shown in Fig. 13.

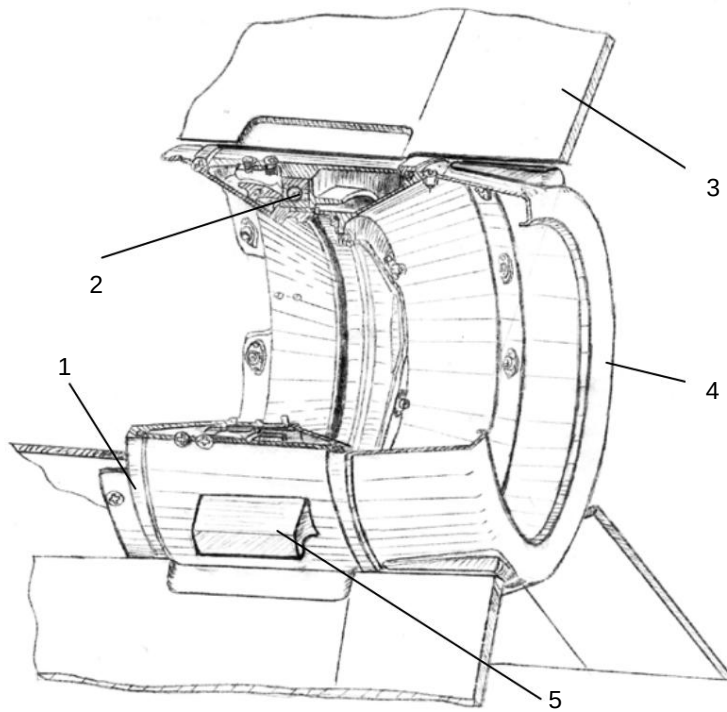


Fig. 13. Wing block of the rocket:
1 – annular belt; 2 – bearing; 3 – wing block;
4 – fairing; 5 – yoke

The engine body is equipped with yoke 21 for connection with the ejection device. In the front part, the compartment has a ball bearing, the outer race of which is fixed in the compartment, and the inner race is fixed on the bottom of the engine. Under ground operation conditions, the compartment is held from turning by the ejection device installed on the rocket body. During the rocket flight, the compartment rotates freely relative to the longitudinal axis under the influence of the flow of the

flowing wings.

The external views of the compartments are shown in Fig. 14.



The second compartment is the control compartment.



The third compartment is the instrument compartment.



The third compartment is the instrument compartment.



The fifth section is the wing block.

Fig. 14. External views of the rocket compartments

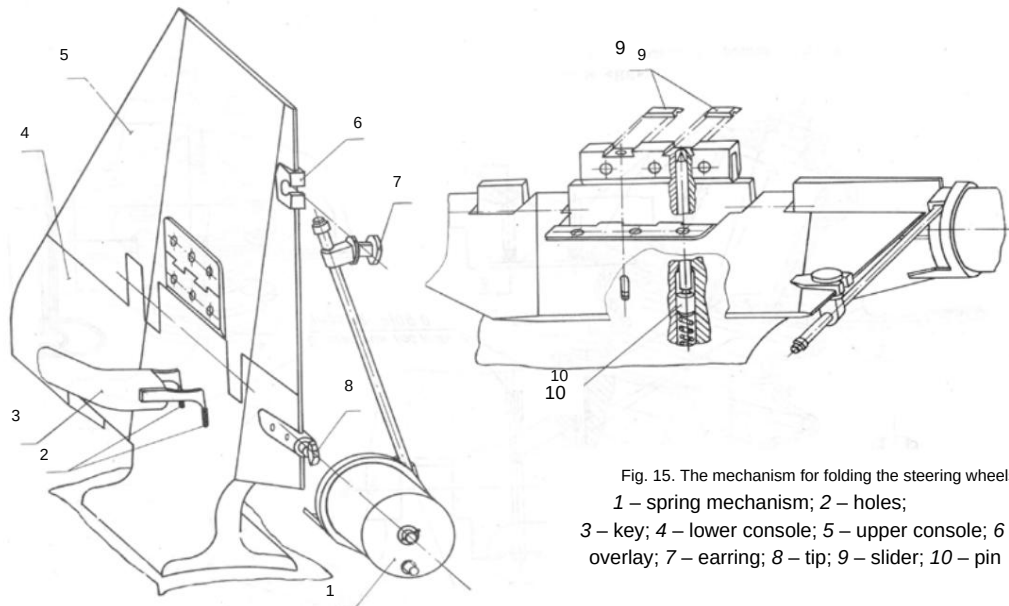


Fig. 15. The mechanism for folding the steering wheels:
 1 – spring mechanism; 2 – holes;
 3 – key; 4 – lower console; 5 – upper console; 6 – overlay; 7 – earring; 8 – tip; 9 – slider; 10 – pin

The rudder jet device (Fig. 16) has two receiving holes adjacent to the feed channel. It operates on the principle of a jet relay: when the rudder is deflected, the receiving holes

The nozzles are located asymmetrically relative to the feed channel and receive different amounts of gas. The outlet openings of the jet devices are made in the form of two oppositely directed supersonic nozzles. When gas flows out of the nozzle, a thrust is created proportional in magnitude to the amount of gas entering it and directed perpendicular to the plane of the rudder. The resulting transverse thrust of the nozzles ensures control of the rocket in the first second of flight, when the flight speed is low and the lifting forces of the rudders are insufficient to create the required control moment.

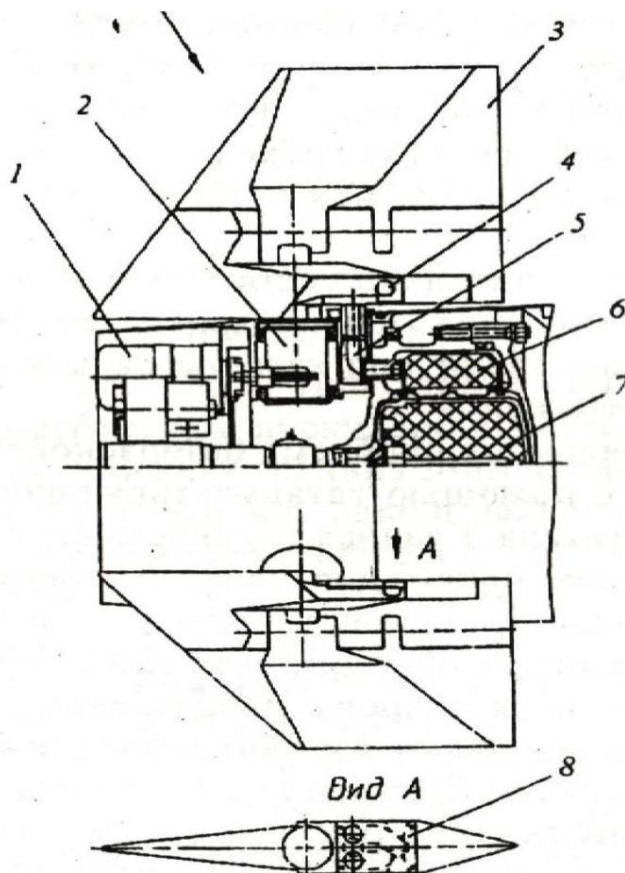


Fig. 16. Gas-dynamic device of rocket inclination: 1 – hot gas steering drive; 2 – steering axis; 3 – steering console; 4 – gas-jet nozzle; 5 – internal gas duct; 6 – inclination system gas generator; 7 – steering drive gas generator; 8 – outlet gas duct

The ejection device (Fig. 17) is designed to ensure the launch (catapulting) of the missile from the TPK.

The ejection device consists of two parts: fixed (cylinder 3) and movable (rod 6). On the front end of cylinder 3 there is a bracket 22 on which are mounted electrical connector for communication with the TPK electrical system, shackle 1 and stop 2. The cylinder is attached to the container using the shackle and clamp 4. Using the stop, when the rocket moves along the TPK guides, the front lever on the third compartment rotates, turning on the push-button switches of the rocket electrical system. At the other end of the cylinder there is a stop 23, designed to duplicate the activation of the engine using lever 24. Inside the cylinder there is an electrical harness 19, which is connected to the pyropatron 16 using an electrical connector 18.

Rod 6 is a turned pipe, with piston 13 attached to one end and brake pipe 5 and lever 10 to the other. Powder charge 14 is installed in the body of piston 13. Brake pipe 5 softens the load on the TPK body during braking of the ejection device.

The rocket is connected to the ejection device by means of shear bolts 20 and levers. chaga 10, pivotally connected to rod 6.

The ejection force is transmitted by the stop of the lever 10 to the end of the rocket compartment 4. To hold the rod in the working position until the rocket is launched, in the free volume zone of the cylinder 3 spring 12 is installed, maintaining a constant force on the rod.

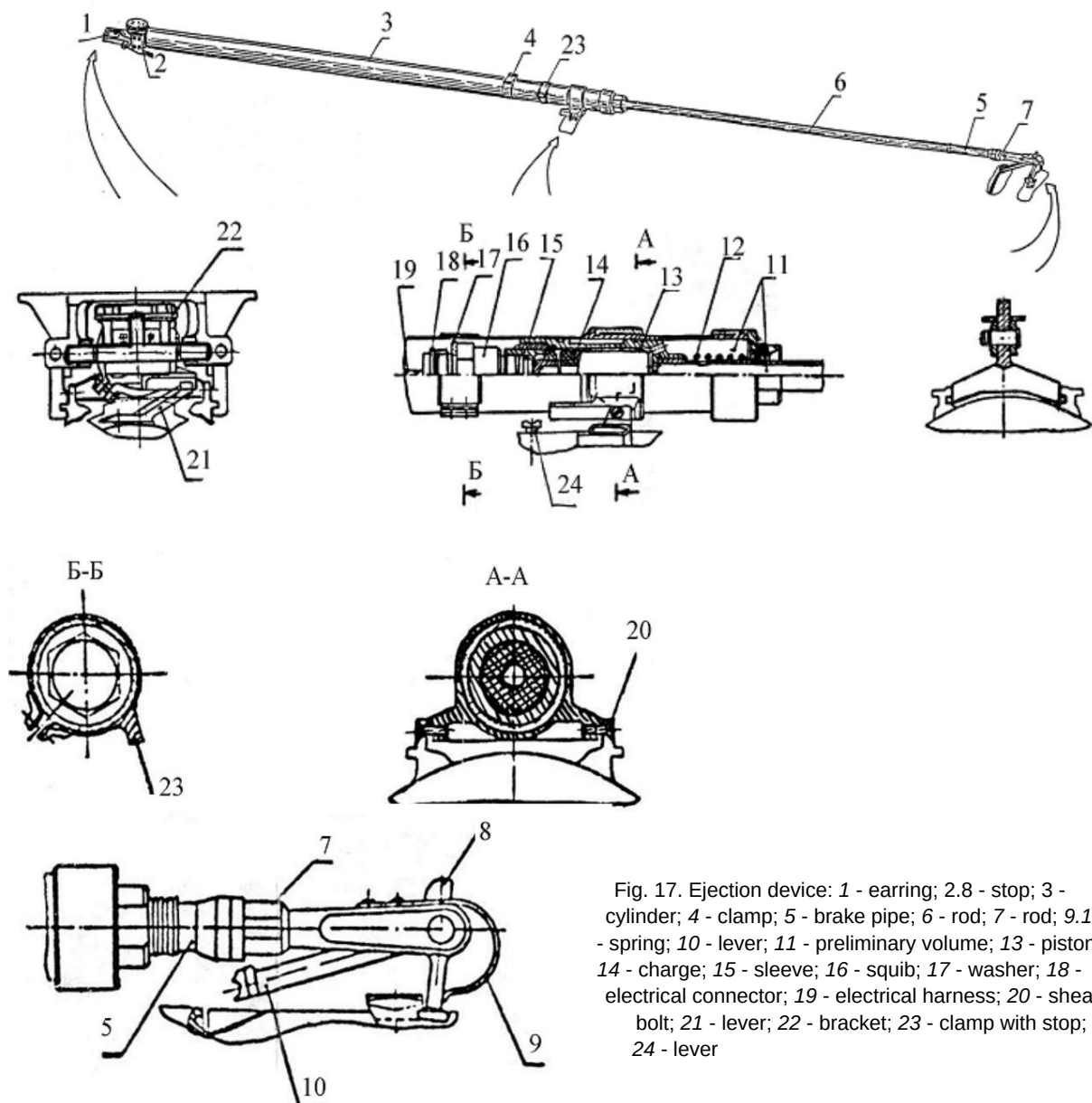


Fig. 17. Ejection device: 1 - earring; 2.8 - stop; 3 - cylinder; 4 - clamp; 5 - brake pipe; 6 - rod; 7 - rod; 9.12 - spring; 10 - lever; 11 - preliminary volume; 13 - piston; 14 - charge; 15 - sleeve; 16 - squib; 17 - washer; 18 - electrical connector; 19 - electrical harness; 20 - shear bolt; 21 - lever; 22 - bracket; 23 - clamp with stop; 24 - lever

Self-assessment questions for section 2

1. What is the first section of the rocket?
2. What is located in the second compartment of the rocket?
3. The purpose of the hot gas sources located in the second compartment of the rocket.
4. How does hot gas get into the steering nozzle?
5. Explain the operation of the spring mechanism for locking the steering wheels.
6. What is located in the third compartment of the rocket?
7. List the composition of the hardware unit
8. How and when are rocket letter frequencies installed?
9. List the composition of the hardware unit.
10. Composition and purpose of solid propellant rocket motor elements.
11. How are the wings held in the folded position and how and when do they open?
12. For what purpose is the wing block mounted on a bearing?
13. How is the rocket switched to on-board power supply?
14. Operating procedure of the ejection device.
15. Name the sequence of functioning of the rocket elements during ejection.
ni.
16. What happens to the rocket immediately after it leaves the launch pad?

3. COMBAT OPERATION OF THE ROCKET

Loading of the TPK into the BM is performed by a transport and loading vehicle. Before loading, the front removable cover is removed from the TPK, while the protection of the internal cavities of the TPK with missiles is provided by a single-use protective and sealing device.

The TPKs are installed in the BM shaft in a vertical position. The rockets are launched vertically. The TPK in the combat vehicle is secured in the longitudinal and transverse directions. The TPK is secured from longitudinal movements by securing the front frame of the TPK with the BM grips to the support plates. Transverse fixation is provided by the pins of the combat vehicle, which enter two openings in the front frame, and the BM guides and T-shaped grooves-yokes in the middle frame, into which the BM guides enter.

When loading, the missiles are assigned letter frequencies (missile addresses) and forging of electrical connectors for the TPK and combat vehicle.

The launch device in the BM ensures the missile's azimuth turn to align the missile's declination plane with the target's direction. The missile's flight during combat use has three characteristic

stages: ejection, turn in the target's direction (declination), and missile guidance to the target by radio commands from the ground-based facilities of the complex. The missile's flight is preceded by the operation of the onboard equipment in the "Preparation", "Waiting", and "Launch" modes.

3.1. Preparation for launch and launch

The rocket is prepared for launch using automatic launch equipment installed on the combat vehicle.

During the preparation process the following is produced:

- spinning up the onboard electric current converter and autopilot gyroscopes from a ground power source;
 - powering the onboard equipment with all types of consumed voltages; – entering into the autopilot declination commands, which are processed by the rocket in flight
- tone; –
- control of the safety chain of the safety-actuating mechanism;
 - control of cocking of catapult pyropatrons;
 - control of the initial state of on-board automation;
 - with the exit to the mode of the electric machine converter, switching the on-board output consumers via alternating current circuits;
 - receipt of a number of service commands and signals.

The design of the combat vehicle and the launcher with the TPK allow missiles to be guided azimuth to align the declination plane with the direction to the target.

If no "Launch" command is received immediately after completion of preparation, the rocket is transferred to the "Standby" mode, during which a launch or launch cancellation can be performed at any time with the shutdown of all supply voltages, commands and signals.

In the "Standby" mode, which, in principle, may be absent, within 0.05 s after the end of the "Preparation" mode, the power supply of the on-board radio equipment switches from the ground source of alternating voltages to the on-board one. The duration of the "Standby" mode can reach 60 s.

On the "Start" command the following is performed:

- activation of the on-board chemical current source (CCS);
- with the on-board HIT switching to the mode of powering the on-board equipment from it in parallel powered from a ground source;
- input, if required, to the radio fuse of special one-time commands that activate modes of operation against low-flying targets and in passive interference;
- memorization of entered declination commands in the autopilot;
- uncaging of gyroscopes;
- entering and recording the missile address in the radio control unit. Upon completion of these operations, a command is issued from the launch automation equipment to detonation of the catapult's pyropatron, which ignites the catapult's propellant charge.

As the force on the catapult rod increases and is transmitted to the rocket body, two locking bolts 7 (see Fig. 7) are cut off and the rocket begins to move along the TPK guides. With the start of this movement, the plug of the onboard electrical connector is disconnected, and then, due to mechanical contact with special stops 2 and 8 (see Fig. 17) on the ejection device, levers 30 and 33 (see Fig. 12) on the third compartment are turned, closing the contacts of the corresponding push-button switches. As a result, the following occurs

- the gas generator for powering the steering machines is started;
- in the command block, the device for generating time delays is switched on, which gives commands to start the declination gas generator and the rocket engine;
- a duplicate pyropatron is launched (duplicates the launch of the rocket engine), which has a one-second response delay.

At the end of the stroke, the catapult rod is braked by compressing the gas in cylinder 3 and crushing a special brake pipe 5 (see Fig. 17) to reduce the impact load on the container. The rocket, having gained the required speed, continues to move by inertia to a height of 15-20 m.

3.2. Rocket flight

When the rocket nose starts moving, the cover of the protective and sealing device of the TPK is destroyed. Immediately after the rocket leaves the TPK, the time relay issues a command to reset the steering machines. The autopilot, based on the information about the pitch and course angles received from the launch automation equipment, generates a control signal according to a specified algorithm and sends it to the steering machines. The rudders are deflected, and due to their rotation, the spring mechanisms 36 (Fig. 11, 12, 15) are released, and the rudder consoles open, and the spring mechanisms are dropped to the ground. By this time, the gas generator of the declination system is started, and the gas enters the gas-jet devices of the rudders, creating a reactive force on the deflected rudders. The missile body begins to turn (inclination) at an angle, the value of which depends on the trajectory of the subsequent radio-controlled flight (Fig. 18): from the minimum angle when firing at the upper near part of the target interception zone to the maximum when firing at a low-flying target at the near boundary of the interception zone.

The rocket engine is started by a signal from the command block. If there is no signal, the launch is duplicated by a pyropatron located in the igniter block on the front bottom of the engine. The flight time of the rocket with the engine running is 12 s. During this time, the rocket covers a distance of about 8 km, the maximum speed achieved at the starting section of the flight is 850 m/s (Fig. 19).

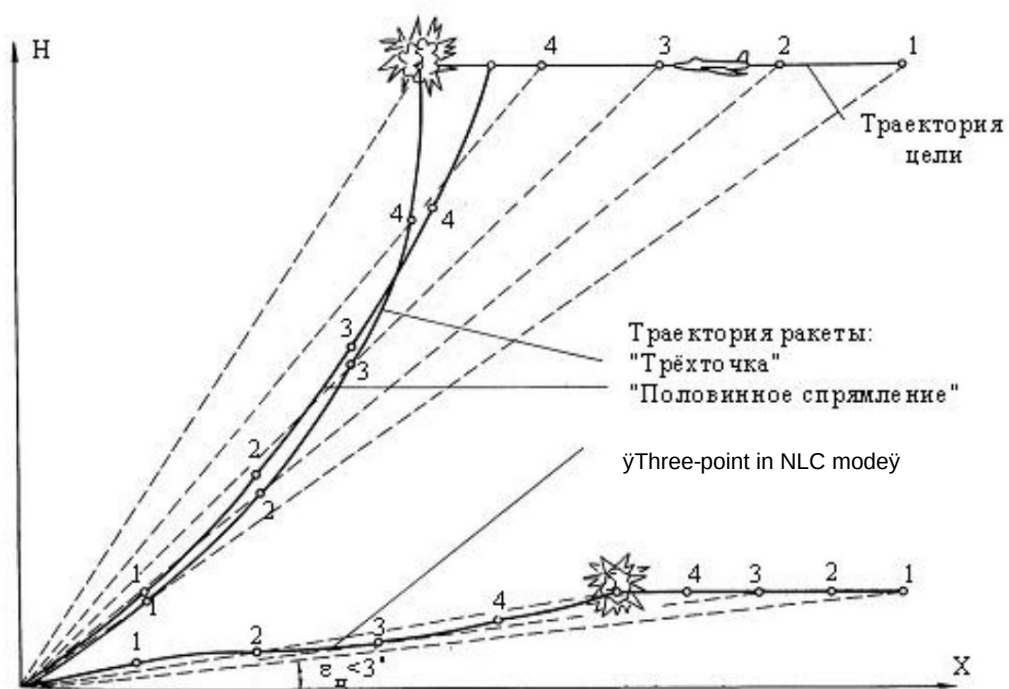


Fig. 18. Scheme of the missile flight trajectory with different guidance methods

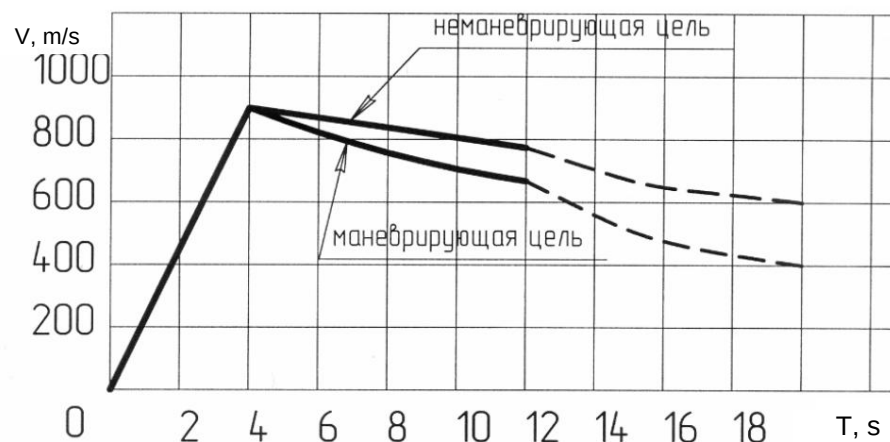


Fig. 19. Tor-M1 missile velocity profiles

The command block generates a signal to start the engine when the rocket reaches an inclination angle of 50° or one second after turning lever 30 (see Fig. 12), located in the rear part of the third compartment under the catapult.

The gas generator of the declination system finishes its work with some overlap after the engine starts. By this time the rocket gains speed and the air rudders-ailerons become effective. At the end of the work of the gas-jet devices, according to the pressure drop in the gas generator

The inclination system issues a command to the autopilot to switch the limitation of the rudder deflection angles to increase the maximum values of their deflection in the control area.

During the missile's flight in the initial section (100...150 m), it is captured for automatic tracking by the radar guidance station located on the combat vehicle, and then the transmission of control commands begins. At the same time, the SVR receives signals emitted by the BRU and determines the position of the missile by angular coordinates. The SPK emits signals with a code arrangement in accordance with the launch automation equipment installed on the missile.

Radio control commands are generated via two channels corresponding to the missile control planes and are transmitted by the command transmission station to the missile along with the "Request" signals (Fig. 20, pulse "Z").

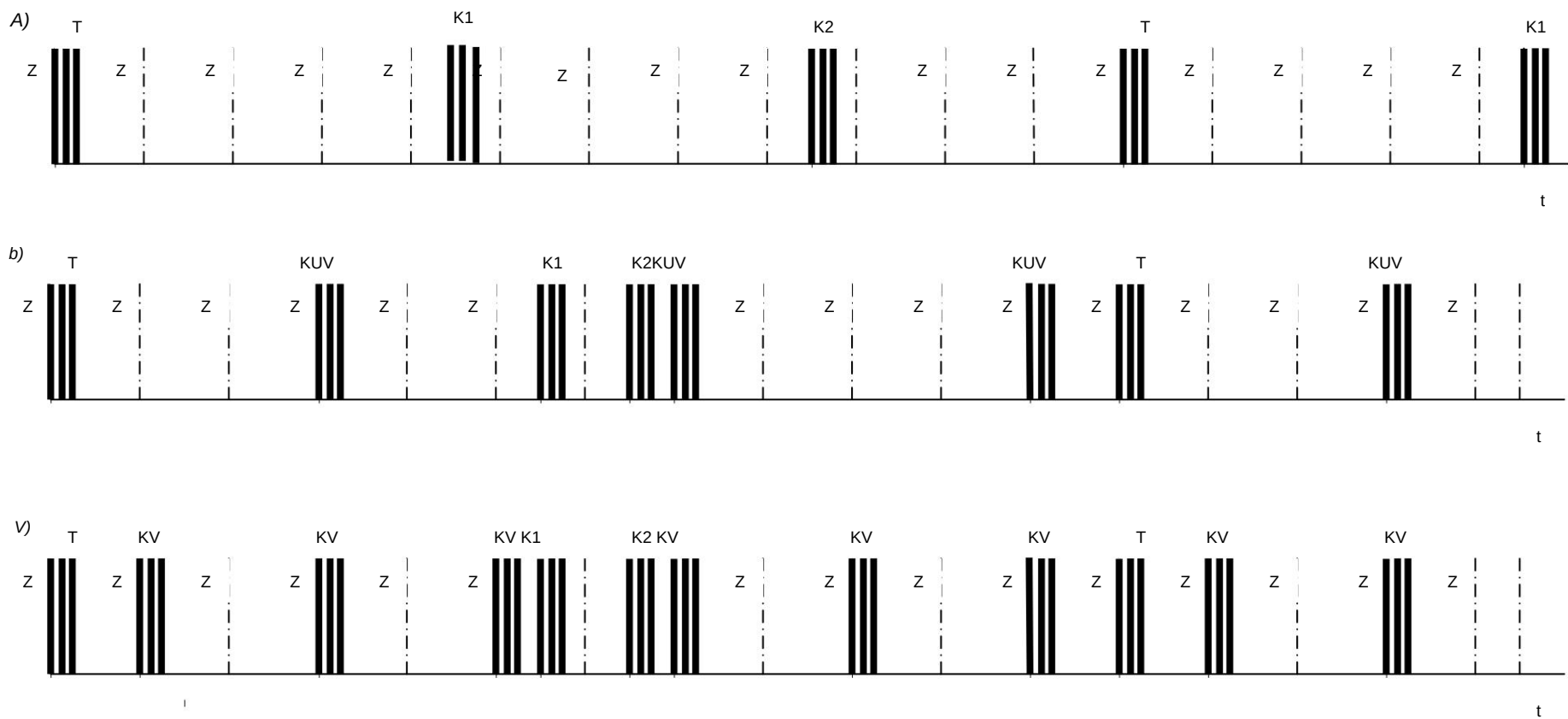


Fig. 20. Total signal at the input of the radio control and radio visualization equipment:

a – before the one-time commands KUV and KV are given, the code combinations of commands K1 and K2 to the left of the zero-middle of the clock interval are positive, while the voltage at the output of the main channel of the UPT block is positive, $K1 > 0$, $K2 > 0$ are shown conditionally; b – after the one-time command KUV is given; in – after the one-time command KV is given; "Z" – request pulse, T, K1, K2, KUV, KV – code combinations "Tact" and corresponding commands

Based on the request information from the guidance station, the on-board transponder transmits response signals, ensuring precise tracking of the missile along its entire trajectory.

The radio control and radio vision equipment installed on the missile receives radio commands K1 (pitch control), K2 (yaw control) from the command transmission station.

Commands K1, K2, KUV and KV are transmitted by the SPK in the form of code combinations (pulse groups) with different time arrangement. The transmission of control command levels (K1, K2) is carried out by the phase-pulse method. The magnitude of the command level is determined by the position of the command code combinations in a constant time interval - a clock. This interval is determined by the distance between two adjacent code combinations "Clock" ("T"). The frequency of transmission of commands K1, K2 is equal to the frequency of the code combinations "Clock".

The one-time command KUV is intended to turn on the transmitter of the radio fuse. The one-time command KV turns on the information processing unit and is intended to arm the radio fuse when the missile approaches the target. The command is transmitted at a frequency multiple of the "Tact" frequency during the transmission time of two ticks.

At the final section of the trajectory, when the missile's flight speed decreases and, accordingly, the efficiency of the rudders-aileron, a command is transmitted to the missile to switch the gain factors in the autopilot. When the missile approaches the target, the HF transmitter of the radio fuse

begins to irradiate the target upon the command to arm. After a certain number of pulses reflected from the target and received by the radio fuse have accumulated, a command to detonate is generated in the actuator circuit of the radio fuse, which is sent to the safety-executive mechanism (SAM).

The safety-executive mechanism, to which the warhead initiation circuits are connected, ensures reliable protection against its unintended detonation in all operating conditions and at missile launch – until the safety stages are removed in flight. At launch and normal operation of the missile engine, the PIM is armed by a signal about the presence of pressure in the combustion chamber (from the pressure indicator) and about the presence of longitudinal overload of the required duration (from the inertial stopper). After this, the PIM firing circuit is triggered by the detonation command from the radio fuse and the warhead is detonated.

In the event of a disruption to the normal flight of the missile, the guidance station may stop transmitting flight control commands. In this case, after a certain time interval, the onboard radio control equipment issues a command to eliminate the missile to the safety-executive mechanism, which detonates the warhead.

Questions for self-control in sections 3.1, 3.2

1. *How is the 9M334 module equipped?*
2. *Characteristic stages of a rocket's flight.*
3. *How does the missile turn in azimuth towards the target?*
4. *List the pre-flight operating modes of the onboard equipment.*
5. *What is done in the "Standby" mode?*
6. *What operations are performed in the "Preparation" mode?*
7. *What operations are performed by the "Start" command?*
8. *When is the command to launch a missile given?*
9. *What operations with the rocket are carried out during the operation of the ejection device? "stva"?*
10. *When is the command to reset the steering gears given?*
11. *How is the control signal to the steering machines generated after the rocket launch?*
12. *How do the rudders unlock?*
13. *How is the rocket controlled in the declination phase?*
14. *When and how does a rocket engine start?*
15. *Why are the rudder deflection angles limited in the initial phase of a rocket's flight?*
16. *When is the command to switch the rudder deflection angle limitation issued?*
17. *How is the rocket position determined by angular coordinates?*
18. *When and how are the SPK commands transmitted on board the rocket?*
19. *What commands are transmitted by the radio control and radio visualization equipment located on a rocket?*

20. What is the essence of transmitting control command levels ($K1$, $K2$) using the phase-pulse method?
21. What determines the time interval "Tact"?
22. When is the KUV command given?
23. When is the KV command given?
24. Purpose of PIM.
25. How and when does a missile self-destruct?

3.3. Rocket flight dynamics

- The rocket flight trajectory can be divided into the following main sections: – initial uncontrolled vertical section (10...15 m);
- the declination section, where the missile is controlled in azimuth and elevation angles in accordance with a given algorithm and target coordinates, embedded in the missile by the launch automation equipment (150...200 m);
 - a section controlled in accordance with the commands of the guidance station;
 - the final section, where the missile is eliminated in case of a miss. In the declination section, the missile is controlled by a gas-dynamic method (see Fig. 16).

The nature of the trajectory in sections with one or another method of control in flight is determined by the magnitude and direction of the control force. The vector of this force, in turn, is determined by the orientation of the aerodynamic planes of the rocket (rudders and wings) in space, which is two-plane (x-shaped). Since in this case the lift force can be created in any direction, a preliminary roll is not required to perform the maneuver. The direction and magnitude of the lift force required to perform the maneuver are provided by the corresponding combination of the angles of attack and slip. Therefore, the task of the computing device is to generate the corresponding commands for two symmetrical control planes: $K1$ and $K2$ (Fig. 21).

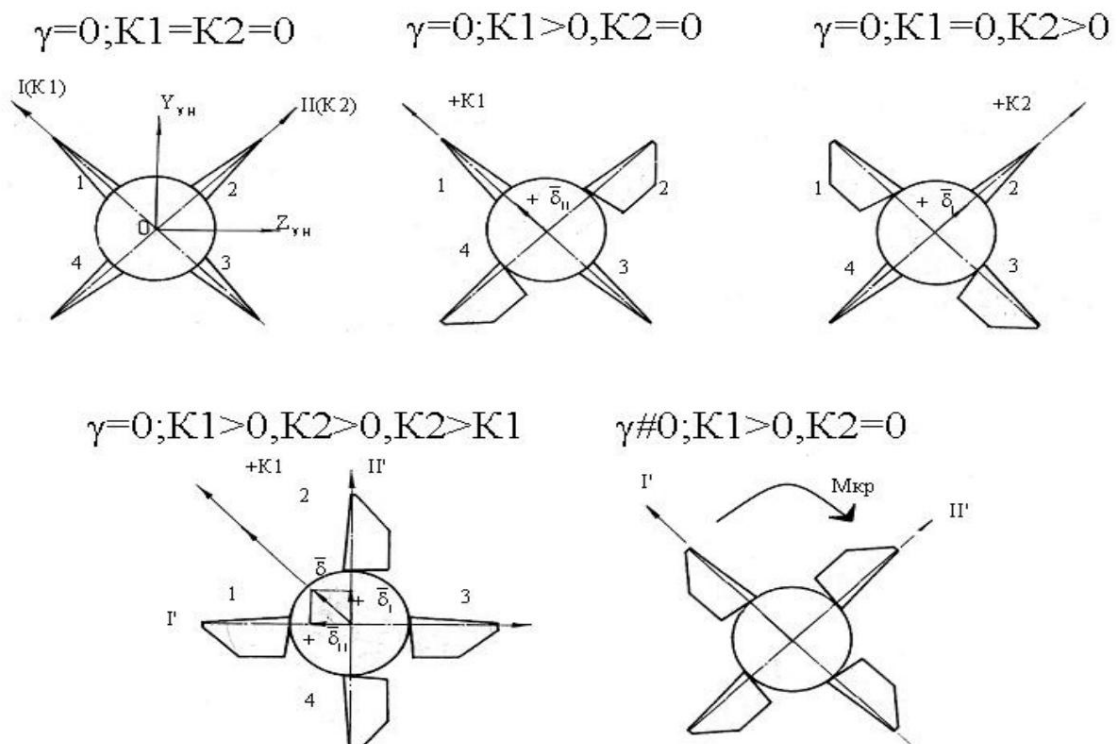


Fig. 21. Control principle of a roll-stabilized rocket

Guidance of a missile to a target consists of continuous measurement and elimination of the misalignment of the missile's position relative to the kinematic trajectory determined by guidance methods.

The guidance method (the "half-rectification" method or the "three-point" method) (see Fig. 18) is selected depending on the type of target and its parameters (speed, range, angular position).

When guiding using the "three points" method, the missile, moving along a curved trajectory, is kept on the line of sight of the target all the time. This method of guiding is the simplest in hardware implementation; instrumental errors of the hardware and fluctuations of the target and missile signals cause smaller random errors in guiding than with the "half-rectification" method. The disadvantages of the "three points" method include the large curvature of the kinematic trajectory and, as a consequence, large, increasing required lateral overloads of the missile as it approaches the target.

When guiding using the "half-rectification" method, the missile's movement at each moment in time is directed toward the anticipated point of the missile's meeting with the target. The essence of the method is that the current sighting angles of the missile and target (azimuth and elevation) are determined taking into account the lead correction:

$$\begin{aligned} \ddot{y} &= \ddot{y}_r + \ddot{y} & \kappa \frac{\ddot{y} \ddot{y}_c}{2}, \omega_{p,c} \frac{\ddot{y}}{\ddot{y}} \ddot{y}_{rt}(\cdot); \\ \ddot{y}_r &= \ddot{y} + K_c \frac{\ddot{y} \ddot{y}_c}{\ddot{y}}, \omega_{p,c} \frac{\ddot{y}}{\ddot{y}} \ddot{y}_{rt}(\cdot), \end{aligned}$$

where $\ddot{y}, \ddot{y}_c$ is $\ddot{y}, \ddot{y}_c$ - sighting angles of the missile and target (azimuth and elevation, respectively); $r(t)$ -

the current distance between the missile and the target; K is the lead coefficient, which determines the correction for the lead.

As a result, the trajectory when guiding using this method is flatter and the required lateral accelerations along the entire trajectory and at the point of meeting the target are significantly less than with the "three points" method. When developing the lead angle in the "half-straightening" method, the missile's approach speed to the target and the distance between them, as well as the angular velocity of rotation of the target's sighting line, are taken into account.

If the target is low-flying (the target elevation angle is less than 3°), then in the vertical plane the missile is guided to the target using the "three points in the NLC mode" ("hill") method, and in the horizontal plane - either by the "three points" method or by the "half-rectification" method.

When using the "three-point" guidance method when firing at a low-flying target, the missile is guided along a trajectory lying above the target's line of sight by a value of h_{db} , which depends on the distance between the missile and the target. Starting from a distance to the target of approximately 6 km, the value of h_{db} begins to gradually decrease, and when the distance between the missile and the target is reduced to 1.5 km, the missile will reach the target's line of sight.

The "three-point" guidance method when firing at a low-flying target ensures that the missile flies at an altitude that eliminates the possibility of the missile touching the earth's surface and the false triggering of the radio fuse from signals reflected from it.

During the entire flight of the rocket, the autopilot stabilizes it relative to the transverse and longitudinal axes using signals from the acceleration and angle sensors. Stabilization of the rocket and reduction of its overshoot in the angles of attack and slip and overload is accomplished by a stabilization circuit formed by introducing negative feedback in the angles of attack and slip and the bank angle. The structural diagram of the declination and control paths with stabilization circuits is shown in Fig. 22.

In the controlled section of the flight, the missile and the elements of the control system (SRP, SPK, SVR) form a closed control loop, the characteristics of which determine the accuracy of the missile's guidance to the target. The missile is controlled by two channels (pitch and yaw). Since both channels are identical, Fig. 22 shows a diagram of one channel - the pitch channel.

Throughout the entire controlled section of the flight, radio control commands are sent to the missile, proportional to the linear deviations of the missile and their derivatives from the kinematic trajectory, the determination of which requires measuring the angular coordinates of the missile and the target. Angular

the coordinates of the missile ($\dot{y}$) and target ($\dot{y}_c$) are determined by the coordinate units of the missile and target (SVR and SPK equipment).

Kinematic links 1 and 2 are not material links of the control loop, but express the relationship between the angle of attack (slip) and the input values of the measuring devices.

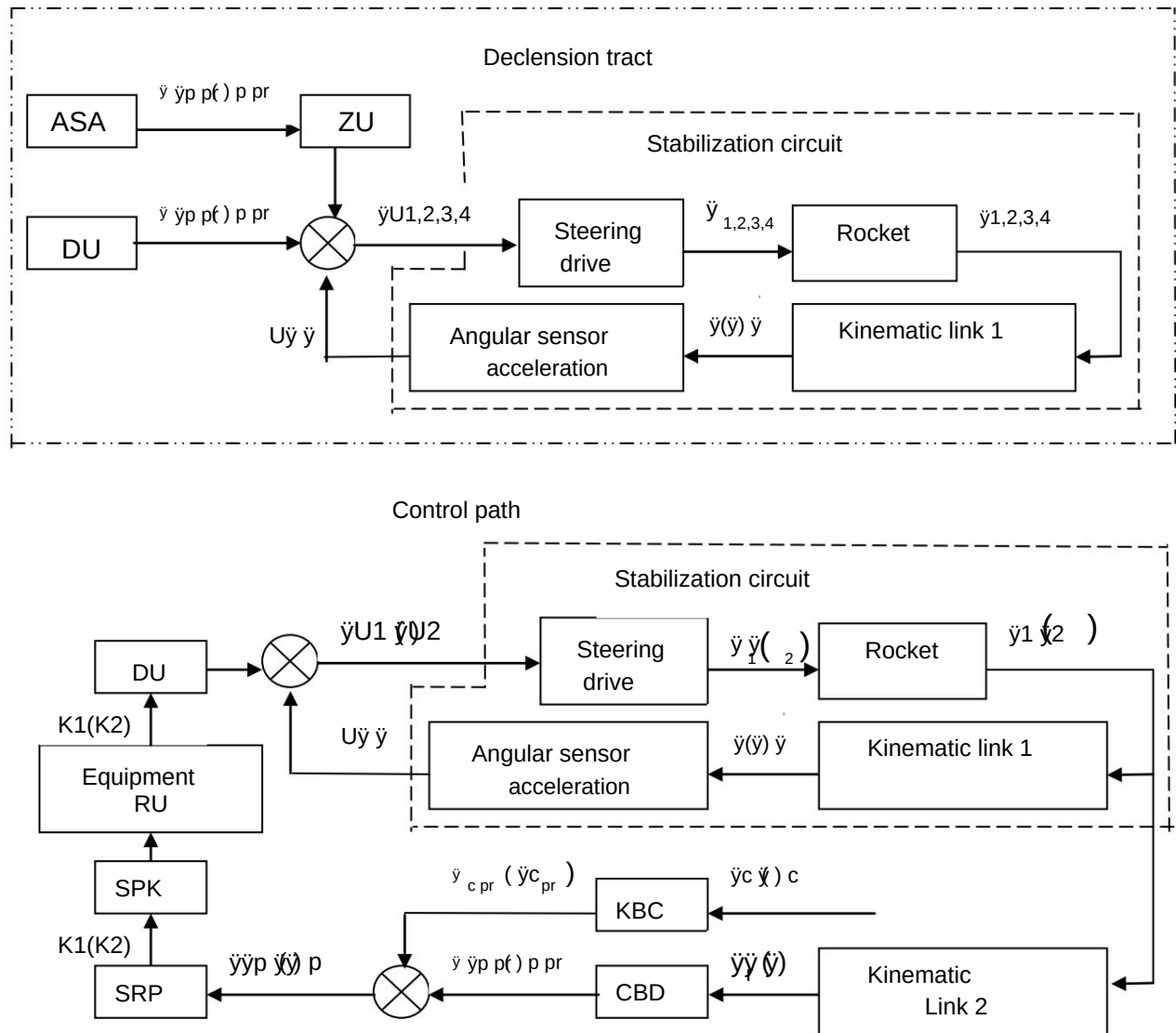


Fig. 22. Simplified diagrams of the missile declension and control paths:

$\dot{y}_1$ – deflection angle of the rudders; $(\dot{y}_{pp}, \dot{y}_{cp})$ – angular coordinates of the rocket; $\dot{y}_{\alpha}$ – angle of attack

goals; $(\dot{y}_{pr}, \dot{y}_{\alpha})$ – angular misalignment of the rocket relative to the kinematic trajectory; DU – sensor angle; ZU – storage device; ASA – automatic starter equipment; SPK – command transmission station; SRP – computing device; K1(K2) – control commands; U1n – feedback signal on

damping gyroscope targets; $\dot{y}(\dot{y})$ – angular velocity of the missile relative to the transverse axis; KBC – target coordinate unit; KBR – missile coordinate unit

Self-assessment questions for section 3.3

1. List the main sections of a rocket's flight trajectory.
2. How are the rocket's rudders oriented in flight?
3. How are commands for rudder turns generated at bank angles $\dot{y} = 0$ and $\dot{y} \neq 0$?
4. What is the essence of the three-point guidance method?

5. What is the essence of the "half-rectification" guidance method?
6. What are the advantages and disadvantages of the three-point and half-rectification methods?
7. Features of missile guidance when working on a low-flying target.
8. How is the possibility of a missile hitting the earth's surface and causing a false triggering of a radio fuse from signals reflected from it eliminated?
9. What is a kinematic link?
10. What feedback loops are involved in the rocket as an object of control?

4. ON-BOARD RADIO CONTROL EQUIPMENT

The onboard radio control equipment (BRU) is designed to receive information from the command transmission station, decipher it, analyze it, and issue commands to the rudders, and generate a response signal for the missile sighting station. In doing so, the radio equipment must receive, amplify, decipher, and generate the following commands in accordance with the signals received from the command transmission station and the launch automation equipment:

- two main control commands for the autopilot: K1 and K2;
 - commands for control and arming of the radio fuse: KUV and KV;
 - commands for prohibiting and permitting asynchronous start of the responder: KZAZ, KRAZ;
 - "Relative approach speed" commands in the form of four values KOS1 – KOS4;
 - the "Time" command to eliminate the missile in the event of a radio link failure time more than 1.5 – 2 s;
 - commands for switching the autopilot (AP) and "Control" operating modes. The onboard radio control equipment must also:
 - convert request pulses into trigger pulses for their subsequent generation radios and missile targeting signals;
 - issue zero values of control commands to the autopilot in the event of a radio line failure for a period exceeding 100 ms;
 - provide frequency selection of the received signal on any of the specified letters frequencies;
 - receive commands "Product address", "Record" from the automatic start-up equipment;
 - perform decoding of commands at eight addresses.
- The following main blocks, included in the app, serve to perform the specified functions:
- radio control and radio visualization equipment:
- receiving antenna-waveguide system;
 - receiver;
 - command generation unit (decoder);
 - responder (radio imaging unit);
 - transmitting antenna-waveguide system;
 - direct current amplifier (DCA).

4.1. Operation of on-board radio control equipment

When loading the TPK into the combat vehicle, the missiles are set to a predetermined value using a key. letter frequency.

Before launch, the onboard radio control equipment in the rocket operates in the "Preparation", "Waiting" and "Launch" modes, and after the "Preparation" mode, it is possible to immediately switch to the "Launch" mode.

In the "Preparation" mode, the radio equipment is supplied with power supply voltages of +20 V, -20 V, 3x36 V 1000 Hz from the ground power source, as well as +27 V to turn on the forced heating mode of the magnetron. At the same time, the address code is output and the DC voltage in the form of a logical "1" level is received through the "Record" circuit. The voltage of +20 V is received through the asynchronous start inhibition circuit.

The alternating voltage of 3x5.2 V 1000 Hz from the on-board source reaches the nominal value values after 1.0...1.5 s. As a result, the following mode is set on the radio equipment:

- forced heating of the magnetron filament occurs;

- issuing an asynchronous trigger pulse to the responder is prohibited;
- the block's counting and output circuits are reset using the "Write" command;
- in the initial state, the lower antennas are open and the upper ones are closed;
- the address code is set, issued by three circuits from the automatic starting equipment.

The "Preparation" mode lasts 5 seconds. By the end of the fifth second, the radio equipment is prepared for the "Standby" or "Start" modes.

In the "Standby" mode, within a time of at least 0.05 s after the end of the "Preparation" mode, the power supply of the onboard control equipment is switched from the ground source of alternating voltages 36 V 1000 Hz to the onboard one. All other supplied voltages correspond to the "Preparation" mode. The duration of the "Standby" mode can reach 60 s.

In the "Start" mode, in no more than 1 s, the on-board voltages +20 V and –20 V from the chemical current source reach nominal values and the power supply of the radio equipment through these circuits is transferred to the on-board sources. In the time of 0.1...0.2 s before the rocket launch, the voltage with the logical level "1" is removed from the "Record" circuit.

. This command prohibits rewriting for 1.5...2.0 s.

address code during the separation of the on-board connector during the missile descent. With the separation of the on-board connector, the voltage of +20 V is removed in the chain of the command for permission of the asynchronous launch of the transponder (KRAZ). According to this command, the on-board radio control equipment begins the asynchronous mode of the transponder launch and the transponder radio pulses are emitted into the air through the upper and lower antennas.

The missile sighting station receives signals emitted by the onboard radio control equipment, and determines the position of the rocket based on angular coordinates.

During communication sessions, the command transmission station transmits continuous and one-time command signals to the rocket, request pulses with a code arrangement corresponding to the address code on the rocket set by the launch automation (see Fig. 20).

At the beginning of each communication session, the KZAZ command terminates the asynchronous mode of the responder launch and pairwise switching of antennas begins, with either the two upper antennas (receiving and transmitting) or the two lower ones opening.

After double confirmation of the highest signal in one of the receiving antennas, this pair of antennas (upper or lower) opens and transmits the signal until the end of the communication session. This creates the best conditions for the operation of the radio link. Before the start of communication sessions and in between them, the transponder of the onboard radio equipment operates in an asynchronous mode. In this case, the missile sighting station determines the position of the missile by angular coordinates. During the communication session, the transponder is launched by interrogation pulses from the command transmission station and operates in a synchronous mode. In this case, the missile sighting station determines the position of the missile by angular coordinates and range.

During the first communication session and commands K1 and K2 on the BRU, the radio equipment generates a "Control" signal in the form of a direct current voltage of +27 V, which switches to the mode of processing control commands issued by the radio equipment (K1 and K2). The specified "Control" signal is blocked by the autopilot circuit, and if the radio equipment removes this signal, the autopilot does not switch to another operating mode.

During the flight, as the station issues one-time commands, the BRU issues signals:

- in the autopilot – the command "Switching the autopilot operating mode";
- to the radio fuse – four commands "Relative approach speed" in the form of a direct current voltage corresponding to the logical level "0", as well as the commands "Control of radio fuse arming" and "Arming radio fuse" in the form of a voltage of +27 V.

In case of radio link failure for up to 100 ms or absence of K1 and K2 commands in three consecutive communication sessions, the "Control" signal is not removed, and the K1 and K2 command levels are remembered. In case of radio link failure for more than 100 ms, the "Control" signal is removed, and the commands are set to zero. When the radio link is restored, the "Control" signal is restored, and the command levels correspond to the levels transmitted by the command transmission station.

In case of a radio line failure for a period of 1.5...2.0 s, the BRU issues the "Time" command, which Paradise closes the circuit to self-destruct the missile.

⁹ This means that the signals in the channel are transmitted in accordance with the high-speed signal protocol. According to this protocol, low-voltage transfer of nominal voltages of logical "0" (max) and logical "1" (min) is carried out with a specified difference relative to the nominal. For example, the nominal control voltage is +20 V, then logical "0" can be equal to +20.2 V, and logical "1" +19.8 V. Using low-voltage signals in the channel reduces energy consumption and electromagnetic interference.

4.2. Command Block

The command block is designed to generate commands for starting the rocket engine and the gas generator of the rocket's turn (inclination) system. It consists of two main parts: a time delay generation device and an algorithmic device.

0.31 s after turning the lever 30 (see Fig. 12), the time delay generation device generates a command to start the gas generator of the turn system and 1 s after turning the said lever, a command to start the rocket engine. The algorithmic device generates a command to start the engine when the rocket reaches, during the turn, angles in course or pitch equal to or greater than $\pm 50^\circ$.

Signals of current pitch and heading angles are sent to the command block from the autopilot.

To ensure the specified level of reliability, the device uses double redundancy of channels in the algorithmic device and quadruple redundancy of channels in the time delay generation device.

The command block is powered by a voltage of ± 13.5 V from a source included in the vehicle.
pilot.

4.3. Autopilot

The autopilot, as part of the rocket flight control equipment, performs:

- control of the missile's flight along the trajectory in accordance with the commands K1 and K2 received from the onboard radio control equipment;
- turning the rocket in pitch and course in a given direction after a vertical launch;
- stabilization of the rocket's motion relative to its three mutually perpendicular axes.

The autopilot consists of a control unit and four steering machines, which provide the necessary conversion and execution of control commands K1 and K2, as well as stability the rocket's relative position relative to its center of mass during its movement.

The control unit is a single unit, which contains sensitive elements (two linear acceleration sensors, three angular acceleration sensors, three free gyroscopes) and an electronic unit (voltage stabilizer, time relay and two boards with electrical and radio elements).

The starting coordinate system (conditionally fixed – Fig. 23) is used to determine the origin of the angles measured on board the rocket using gyroscopic instruments. Its center is located at the center of mass of the rocket at the moment of uncaging the autopilot gyroscopes. In this case, the OY axis at launch lies in the plane of declination of the rocket and is directed in the direction opposite to the direction to the target. The linked coordinate system is used to count the angles of course $\dot{\gamma}$, pitch $\dot{\theta}$ and bank $\dot{\psi}$ relative to the starting coordinate system. The origin of the linked coordinate system also lies at the center of mass of the rocket. At the moment of rocket launch, the linked coordinate system coincides with the starting gyroscopic system. The linked expanded coordinate system OXI (II) is used to measure the angular velocities of rotation of the rocket, the overloads acting on the rocket and the angles of deflection of the control surfaces.

The rocket is controlled and stabilized relative to the center of mass in the declination section at subsonic flight speeds using a gas-dynamic method, and then with the help of two pairs of air rudders. The arrangement of the rudders relative to the linked coordinate system and the conventionally fixed (launch) system is shown in Fig. 23. Each rudder is set in motion by its own steering machine.

When operating the autopilot, the following operating modes should be distinguished: before the start of the slope
nia, declension, control.

Before the start of the declination (height 15...20 m), the steering machines are reset to zero, which is achieved by disconnecting the inputs of their amplifiers from the control signals.

The autopilot's electrical circuits are supplied with voltages of +20 V, -20 V and +27 V from the ground power source simultaneously with the activation of the gyromotors on the "Prepare" command. From this time on, the storage devices in the "Tracking" mode track information about the pitch and course angles received from the launch automation equipment.

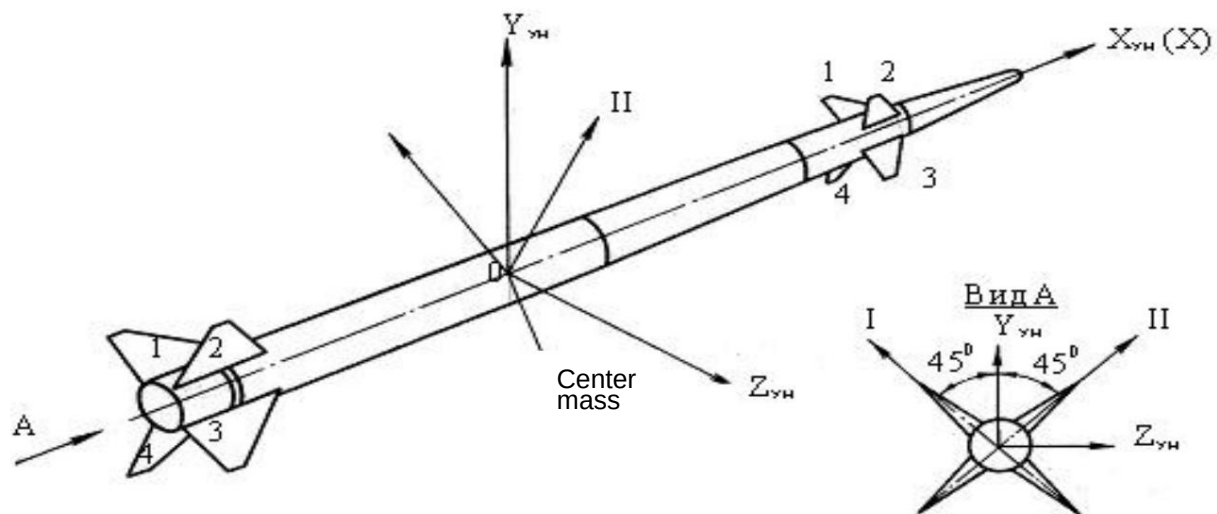


Fig. 23. Conditionally fixed and bound coordinate systems

After 5 s, the transition from the ground power source to the on-board AC power source occurs and, if the "Standby" mode follows (maximum time 60 s), then two gyroscopes operating in flight on coasting are powered from the ground source with a voltage of (36 ± 1) V (1000 ± 100) Hz. At any time during the "Standby" mode, the "Start" command can be issued, and after 0.9 s, commands are issued from the launch automation equipment to uncage the free gyroscopes and to switch the autopilot memory device to the information storage mode (the "Memory" command) on the value of the declination commands at the end of the rocket launch process.

0.9–1 s after the "Launch" command is given, the rocket launch begins, during which the on-board connector is disconnected and the transition from the ground power source to the on-board power source occurs via the +20 V, -20 V, and +27 V circuits. 0.25 s after the rocket's on-board connector is disconnected, the time relay is triggered and the steering machines are reset.

The declination path (see Fig. 22) of the autopilot, based on information about the pitch and heading angles received from the launch automation equipment and stored in the memory device in the "tracking" mode, generates a control signal according to a specified algorithm and sends it to the rudder.

The principle of forming control commands K_1 and K_2 from a conditionally motionless coordinate system into commands UNI and $UNII$ of the linked coordinate system (see Fig. 23) is as follows (Fig. 24). Control radio commands K_1 and K_2 in the form of two components of the main vector of command K are fed to the autopilot input in the "X" system, i.e. parallel to the planes of the rocket rudders. The autopilot is designed according to a two-channel scheme, each pair of rudders has its own control path. Channel I is intended to control the rocket movement in accordance with command UNI and its stabilization relative to axis OI , channel II is intended to control the rocket movement in accordance with command $UNII$ and its stabilization relative to axis II (see Fig. 13). At the moment the rocket leaves the container, the vectors of control commands K_1 and K_2 are parallel to the planes of the rudders. With a positive command K_1 , the trailing edges of the rudders of channel II deflect downwards (positive $\dot{\gamma}$) and the rocket flies left and upwards. With a positive command K_2 , the trailing edges of the channel rudders are at an angle of $2\dot{\gamma}$ and the rocket flies to the right and upward. Thus, the vector

I are deflected downwards (positive angle 1

rudders $K = K_1 + K_2$ $\dot{\gamma} = \dot{\gamma} \sqrt{\frac{2}{2} - 2}$ corresponds to the vector of the control command for deflection of the $\sqrt{\frac{2}{2} - 2}$.

To process the radio control command in the linked coordinate system and to align the processing plane with the radio command plane, command K_1 (K_2) is fed from the onboard radio equipment to the potentiometer of the free gyroscope 11 (see Fig. 26). The latter redistributes the received control commands between the planes of the rudder pairs. The command from the free gyroscope is summed with the stabilization circuit signal and fed to the steering drive, which deflects the rudders by an angle ($\dot{\gamma}$)

$$\dot{\gamma} \dot{\gamma} \cdot 2$$

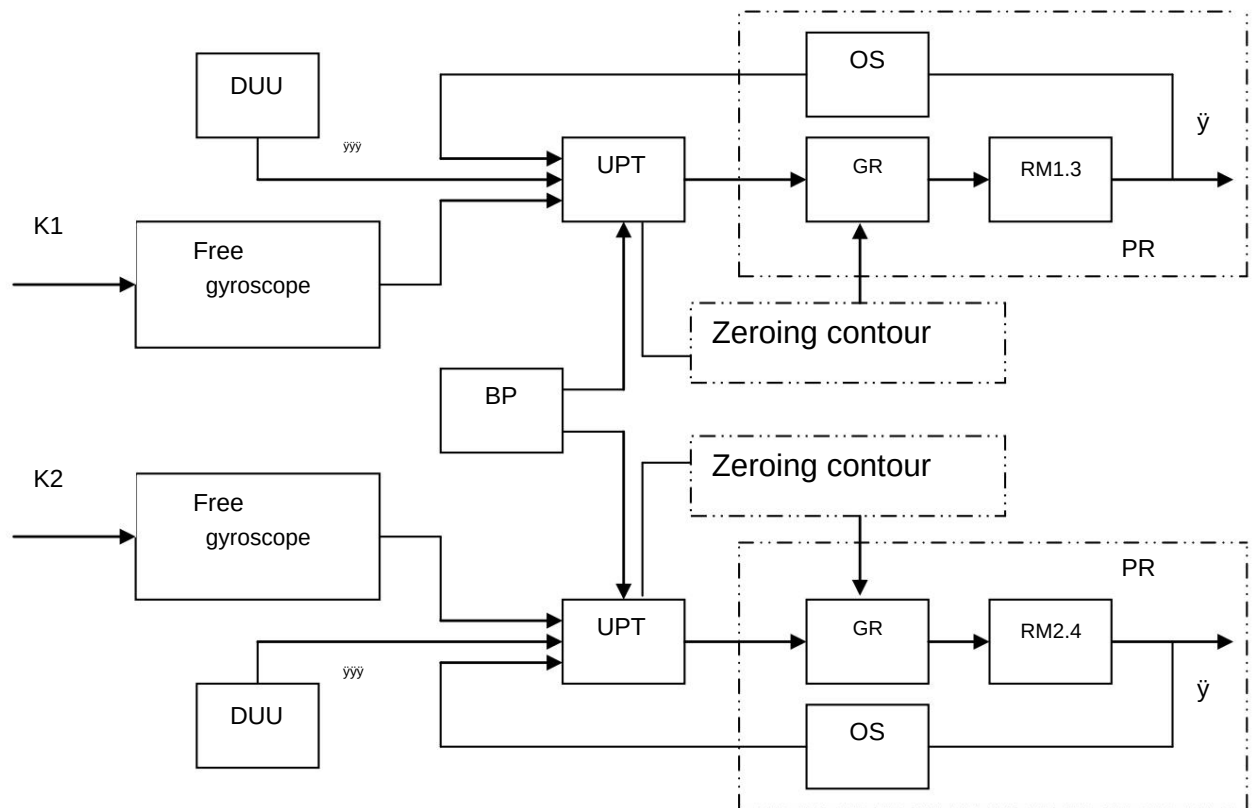


Fig. 24. Structural diagram of the autopilot (pitch and yaw control): RM – steering machine, OS – feedback (RM feedback potentiometer), GR – gas distributor (jet relay), DUU – angular acceleration sensor, DU – angle sensor, PR – steering drive

The deflection of the control surfaces changes the angle of attack (slip) of the missile and the transverse overload, under the action of which the missile moves in the transverse direction, changing its angular coordinates. The changed angular coordinates of the missile and target are sent to the coordinate units of the missile and target, thus the control loop is closed.

The command generated in each of the control paths is distributed to one pair of rudders, ensuring the rocket turns in the pitch and course planes. In flight, information about the current the missile's pitch and heading rotation angles are fed into the same declination paths from the free gyroscope angle sensors. After 1.5...2 s from the start of declination,

the "Control" command from the BRU switches the autopilot from the declination mode to the control mode. In this case, the "Control" command is blocked, disconnects the declination path and connects the control path to the steering machines.

In the control mode, the autopilot processes commands K1 and K2 received from the BRU according to a specified algorithm. At the same time, the signals to the steering machines are limited to a 15° rudder deflection level. Switching of the limitation level is performed by the “KPO” (“Switching of the rudder deflection angle limitation”) command from the pressure drop indicator in the gas generator of the declination system (or by the “Control” command from the BRU).

At the final part of the control section, when the missile speed drops to 400...600 m/s, the “KP” command transmitted from the missile guidance station via the BRU switches the autopilot gain factors.

During the entire period of inclination and control, the autopilot ensures stabilization of the rocket relative to three mutually perpendicular linked axes. Stabilization is carried out according to information coming from three angular acceleration sensors, the sensitivity axes of which are located along the corresponding linked axes of the rocket.

During the entire time, the rocket is stabilized by the bank angle. Stabilization is carried out by the autopilot bank channel (channel 3) using information from the bank angle sensor of the free gyroscope. The signal generated in the bank channel is distributed to all four steering machines, counteracting the rocket's movements relative to the longitudinal axis (Fig. 25).

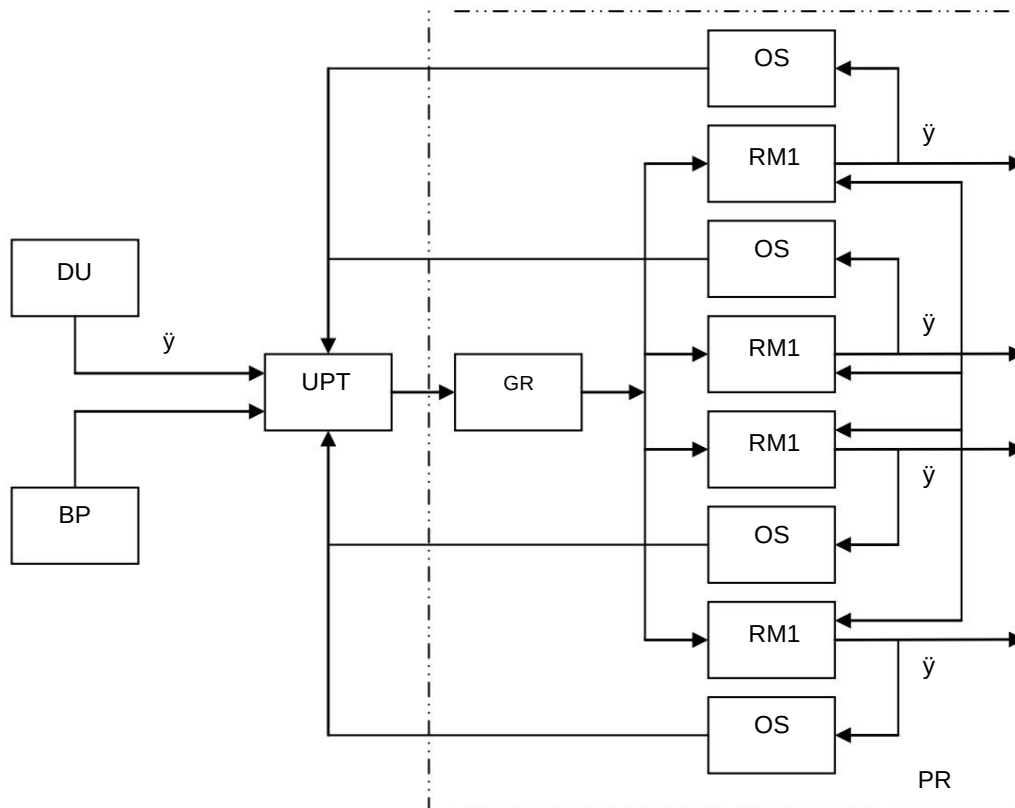


Fig. 25. Structural diagram of the autopilot roll stabilization channel (see Fig. 24)

Stabilization of the missile relative to the center of mass (relative to the transverse axes $O1$ and $O2$) is reduced to introducing additional artificial damping of the missile using the autopilot. This is necessary to meet the quality requirements for transient processes during control command processing, i.e. to limit overshoot in the angles of attack and slip. Although the missile is statically stable, its own damping is insufficient and transient processes during control command processing would have unacceptable oscillations and overshoot in the above-mentioned angles. To artificially increase the missile damping, the autopilot provides negative feedback on the angular velocity of the missile. Introduction of this feedback forms a closed stabilization loop. Due to the fact that during a controlled missile flight, there is always a dynamic error in the control loop (the missile lags behind the kinematic trajectory), a signal for compensating it is introduced

into the loop for all guidance methods. The dynamic error compensation circuit is not included in the closed control loop. It generates an additional command that changes the curvature of the rocket's flight path in the direction of reducing linear deviations relative to the kinematic trajectory.

The sensing elements of the autopilot are free gyroscopes, angular acceleration sensors (AAS) and linear acceleration sensors. *Free gyroscopes*

actually perform the functions of the autopilot's computing device. They are designed to convert control radio commands from a conditionally fixed coordinate system into control commands (voltages) of a linked coordinate system. The property of a free gyroscope to maintain the rotor axis position in space is used. The voltages taken from the brushes of the functional potentiometers of the free

free gyroscopes are distributed between the channels so that the control voltage $\ddot{\gamma}_I$ is supplied to the input of the direct current amplifier (DCA) of channel I, and the corresponding voltage $\ddot{\gamma}_{II}$ is supplied to the input of channel II. Thus, the DCA sums the signals of the control system and free gyroscopes, compares the resulting control signal with the signal of the feedback potentiometer of the steering machine and generates an error signal to activate the gas distributor via the power supply channels of the RM.

The steering drive forms two closed loops of tracking systems with the UPT: the drive loop itself and the zeroing system loop. The drive loop is designed to move the rocket rudders in accordance with the resulting control signal, the zeroing system loop is designed to compensate for the zero errors of the amplifier and the jet relay of the gas distribution device.

The design and kinematic diagram of a free gyroscope are shown in Fig. 26 and 27, The electrical circuit of the autopilot is shown in Fig. 28.

The main units of the free gyroscope are the gyro unit 15, the outer frame 14, the arrester mechanism 16-19 and the potentiometric sensors 11. The gyro unit uses a gyro motor, which is an asynchronous three-phase electric motor. The gyro motor is powered by alternating voltage (36 ± 1) V (1000 ± 100) Hz, which is supplied through the collector ring 10 to the axes of the outer frame (Fig. 26) and 8 to the axes of the inner frame. The number of revolutions of the gyro motor is not less than 51,000 rpm. The maximum angle of rotation of the inner frame 15, mounted on the outer frame, from stop to stop is $\pm 80 \div 85^\circ$. Readiness time (with afterburner) is no more than 10 s. To ensure the initial mutual position of the gyroscope frames before the rocket launch, a caging mechanism is provided, the design of which is clear from the figure. The gyroscope is arrested when the electromagnet 2 is de-energized.

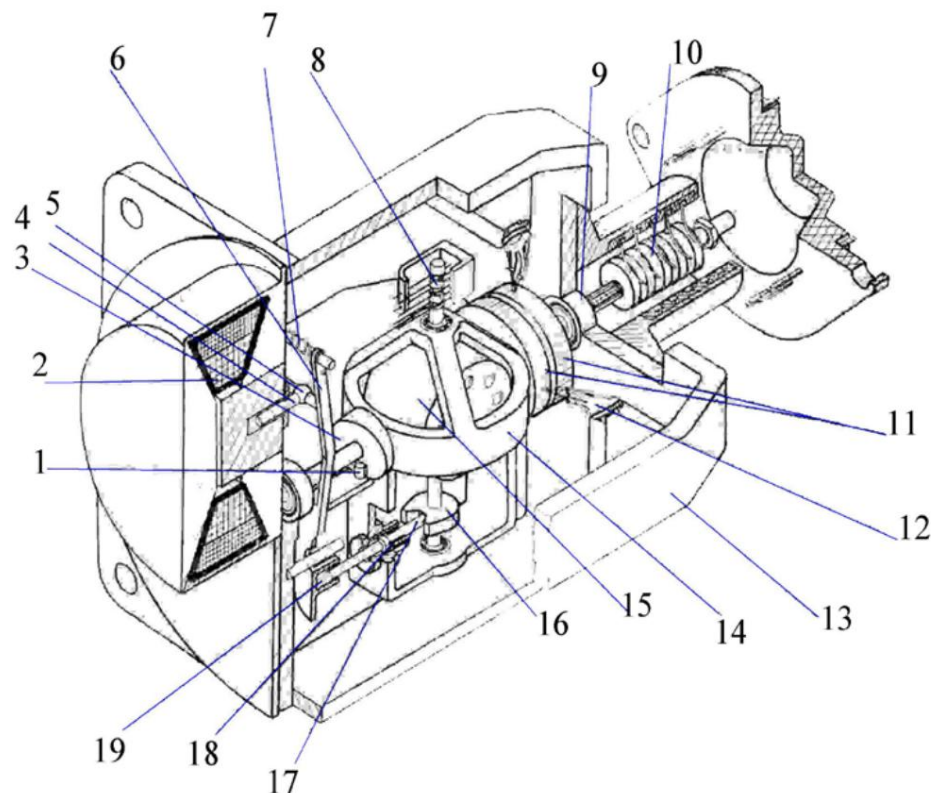


Fig. 26. Construction of a free gyroscope: 1 - roller; 2 - electromagnet coil; 3 - cam; 4 - movable anchor; 5 - clevis; 6 - lever; 7 - spring; 8 - collector ring on the axis of the inner frame; 9 - bearing; 10 - collector ring on the axis of the outer frame; 11 - potentiometer; 12 - contact brush; 13 - housing; 14 - outer frame; 15 - inner frame; 16 - profiling cam on the axis of the inner frame; 17 - pusher; 18 - spring; 19 - catcher

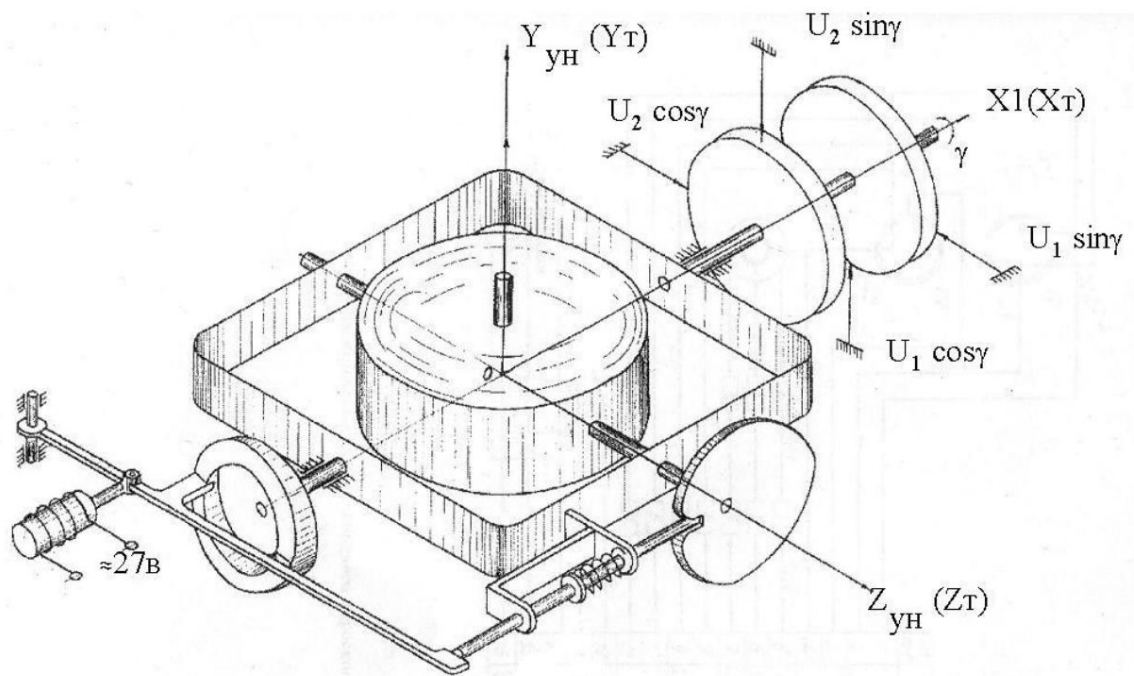


Fig. 27. Kinematic diagram of a free gyroscope

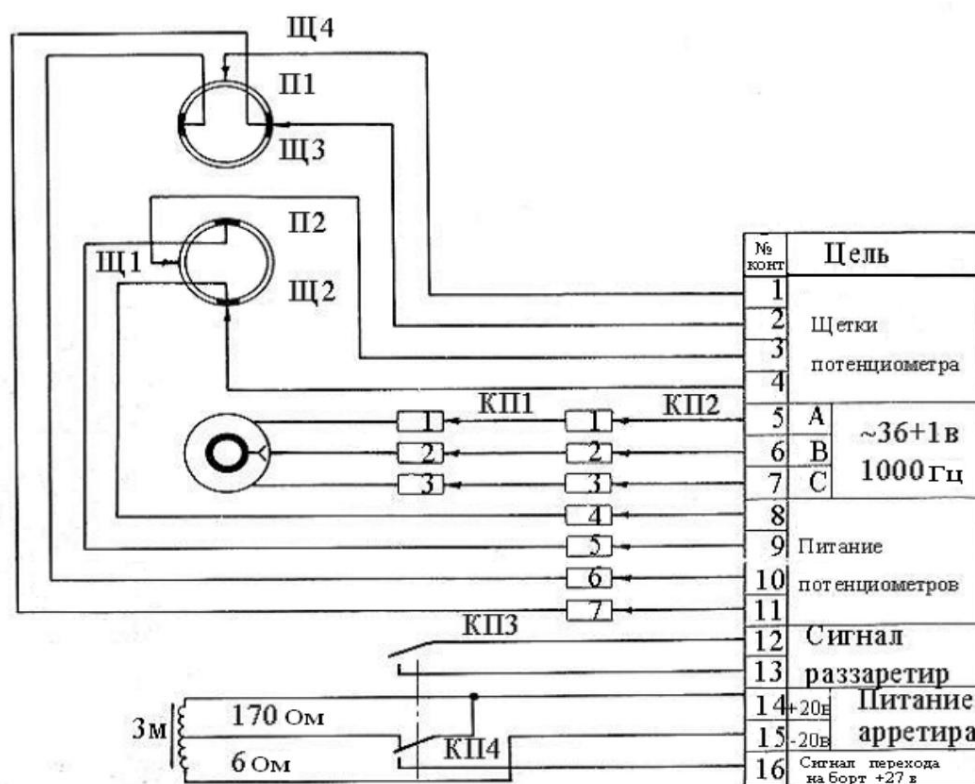


Fig. 28. Electrical circuit of a free gyroscope

Angular acceleration sensors (speed gyroscopes, Fig. 29) are designed to measure the angular acceleration of the rocket relative to the associated coordinate system and to generate signals in the form of DC voltage. These voltages correspond to the values of the angular acceleration component relative to each axis. The sensitive element of the sensor is a two-degree gyroscope. When the rocket rotates relative to the sensor's sensitivity axis, a precessional motion of the gyro unit occurs with a speed proportional to the angular acceleration.

rocket launch.

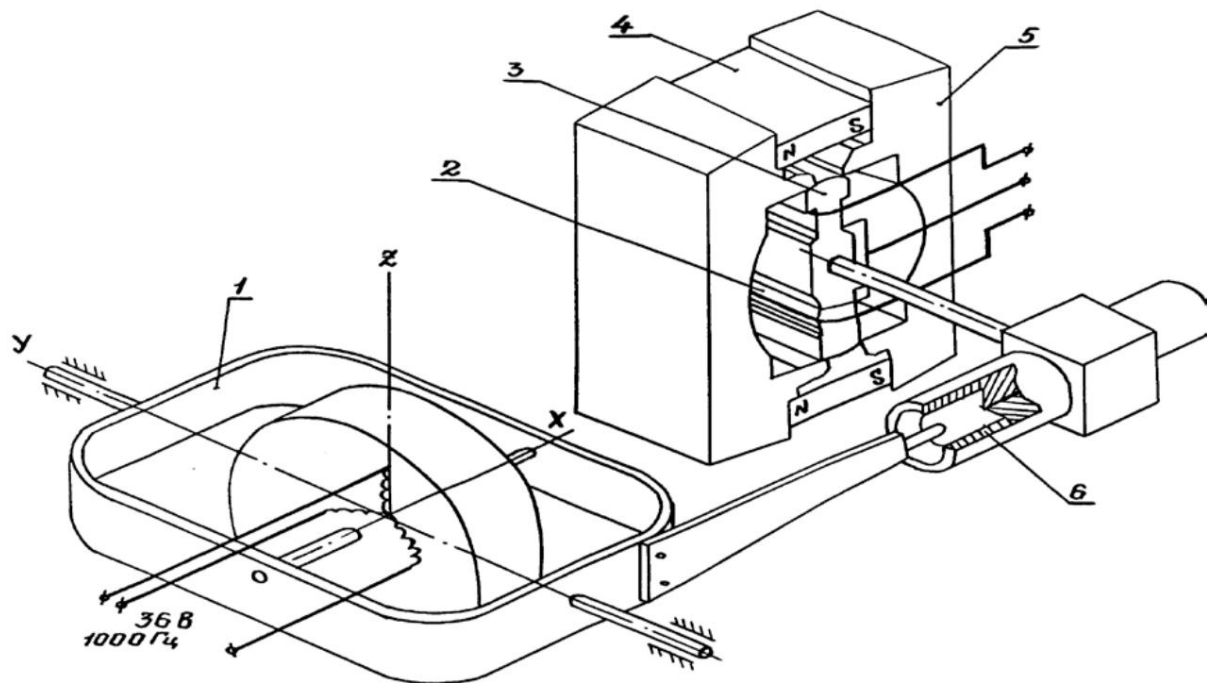


Fig. 29. Speed gyroscope (angular acceleration sensor): 1 – gyroscope, 2 – coil, 3 – rotor, 4 – magnet, 5 – magnetic circuit, 6 – drive gear

The *autopilot linear acceleration sensor* is designed to measure linear accelerations and generate a signal in the form of a DC voltage, the value of which is proportional to the linear acceleration acting in the direction of the measuring axis. The operating principle of the sensor is based on the property of a physical pendulum to be installed in the direction of the resulting forces acting on it in the direction of the measuring axis. The sensor is a structure that combines the electromechanical and electronic parts of the device in a single housing. The electromechanical part is a pendulum made of two cylindrical weights of different masses on a rocker. The pendulum deflection angle is limited by stops.

The DC amplifier is designed to form a control signal and amplify its power. Control signals in the form of DC voltages taken from the brushes of the functional potentiometers of the free gyroscopes are fed to the input of the amplifiers of channels I and II. At the input of the DC amplifier, the control signal is summed with the stabilization signal generated by the angular acceleration sensor, and the resulting control signal is compared with the feedback signal of the steering gear. The resulting signal is amplified in power and fed to the valve timing windings.

The general appearance of the control unit is shown in Fig. 30 and 14.

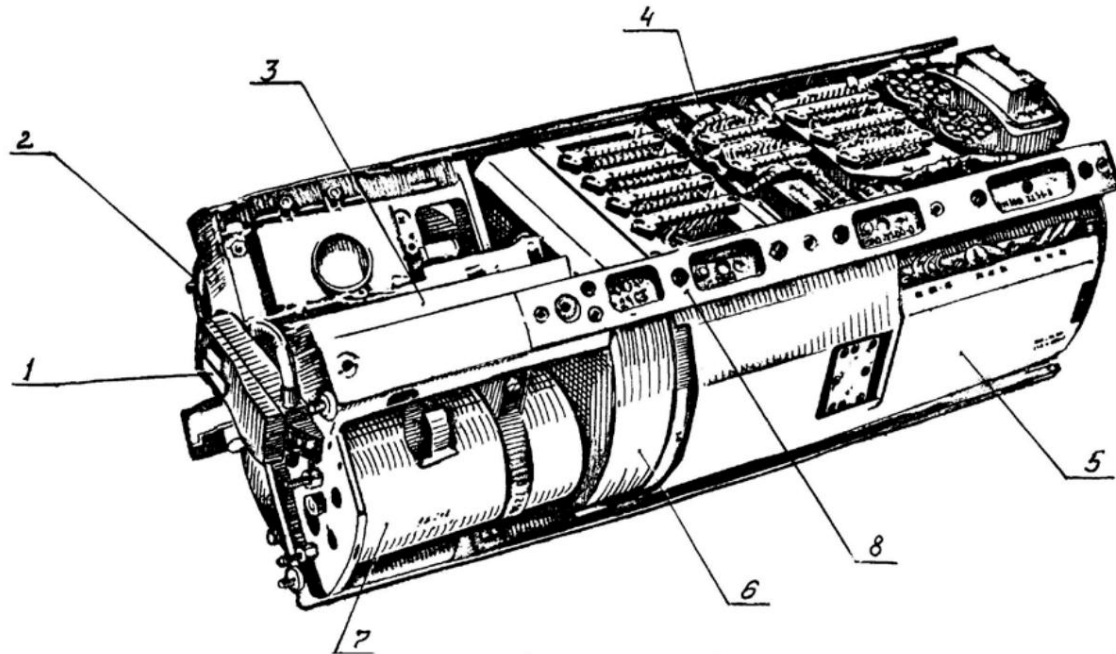


Fig. 30. Hardware monoblock: 1 – delay block, 2 – electric machine converter, 3 – command block, 4 – radio fuse receiver, 5 – on-board radio control equipment, 6 – autopilot, 7 – battery, 8 – stringer

Self-assessment questions for section 4

1. What should the onboard radio control equipment do?
2. What functions does the onboard radio control equipment (BRU) perform?
3. What commands does the command transmission station (CTS) transmit?
4. What tasks does the onboard radio control equipment perform by executing CTS commands?
5. List the main units of the onboard equipment.
6. Why are power supply voltages used to operate the BRU?
7. What is the meaning of using logical "1" or logical "0" levels?
8. How does the BRU function in the "Preparation" mode along the "Record" chain?
9. What functions does the BRU function in the "Standby" mode?
10. What functions does the BRU function in the "Launch" mode?
11. What functions does the missile sighting station (SVR) perform?
12. Purpose and operating modes of the onboard radio equipment transponder.
13. How is one of the two receiving antennas determined in the BRU during flight?
14. What does the synchronous and asynchronous operation mode of the BRU elements mean?
15. When is the "Control" command issued?
16. What signals does the BRU issue during flight?
17. How does the BRU operate in case of a radio link failure?
18. The meaning of using logical "0" and "1".
19. When is the command for missile self-destruction issued?
20. The purpose of the command block. How are commands generated?
21. What functions does the autopilot perform?
22. What is included in the autopilot?
23. Why does the autopilot include two linear acceleration sensors, three angular acceleration sensors, and three

rhenum and three free gyroscopes?

24. How is the loss of dynamic stability of the rocket prevented after exiting the TPK in the vertical flight segment?
25. When does the zeroing of the control machines occur?
26. What is the orientation of the aerodynamic planes (rudders and wings) of the rocket in flight?
27. How are control commands generated in the declination segment?

28. How are control commands implemented in the declination section?
29. How is the restriction on the rudder rotation angle removed at the initial part of the declination section?
30. Autopilot functions in control mode.
31. In what coordinate system are rocket control commands generated?
32. In what coordinate system are rocket control commands implemented?
33. How is the roll stabilization of a rocket accomplished?
34. What is a dynamic trajectory?
35. What is a kinematic trajectory?
36. What is the essence of missile guidance?
37. What is dynamic error?
38. Purpose of free gyroscopes.
39. The device of a free gyroscope.
40. Why is the maximum rotation angle of the inner frame of the gyroscope limited?
41. Purpose of the angular acceleration sensor.
42. Operating principle of the angular acceleration sensor.
43. Purpose and design of the linear acceleration sensor in the autopilot.

5. COMBAT EQUIPMENT

5.1. Radio detonator

The 9E337 radio fuse (RF) is an integral part of the missile's onboard control equipment and is a radar device designed to generate and issue commands to the PIM of the warhead detonation pulse at a point that ensures maximum coverage of the target with the striking elements (fragments) of the warhead.

The radio fuse is a small-sized non-contact radar active pulse fuse and operates in the following combat modes: LFT (low-flying target), NVTs (surface target), PP (passive interference) and standard.

The functional diagram of the RV is shown in Fig. 31 [16,17]. It includes an antenna system, transmitting and receiving devices.

The RV antenna system consists of a transmitting disk-cone antenna installed under the nose cone and two receiving slot antennas installed in niches in the body of the third compartment on the right and left sides of the missile. This arrangement of the antennas aligns the resulting radiation pattern in the plane normal to the missile axis and ensures the required

isolation between the receiving and transmitting paths.

The RV transmitter consists of a transmitting antenna, a magnetron and a magnetic modulator (magnetic pulse generator). The latter produces rectangular voltage video pulses with a duration of about 0.1 μ s with a repetition frequency of 10 kHz. These voltages

are the supply voltage for the magnetron, which generates high-frequency radio pulses. They are directed to a transmitting antenna, which radiates them into space.

In addition to the probing microwave pulse, the transmitter generates pulses (SI) that synchronize the operation of the microwave oven.

The receiving device contains a microwave unit with an amplifier and detector, a video amplifier and an information processing unit. The information processing unit, depending on the specified operating mode of the RV (NLC mode, PP mode, standard mode), controls the operation of the receiving device, automatically limiting the operating range of the RV and coarsening the sensitivity of the receiving path if necessary. Due to this, the selection of signals reflected from the target is carried out against the background of the earth's surface, as well as in the presence of interference. The processing unit also counts the pulses reflected from the target. High-frequency radio pulses reflected from the target are captured by antennas 1 and 2 and enter the detector sections, structurally combined with the antennas (see Fig. 31).

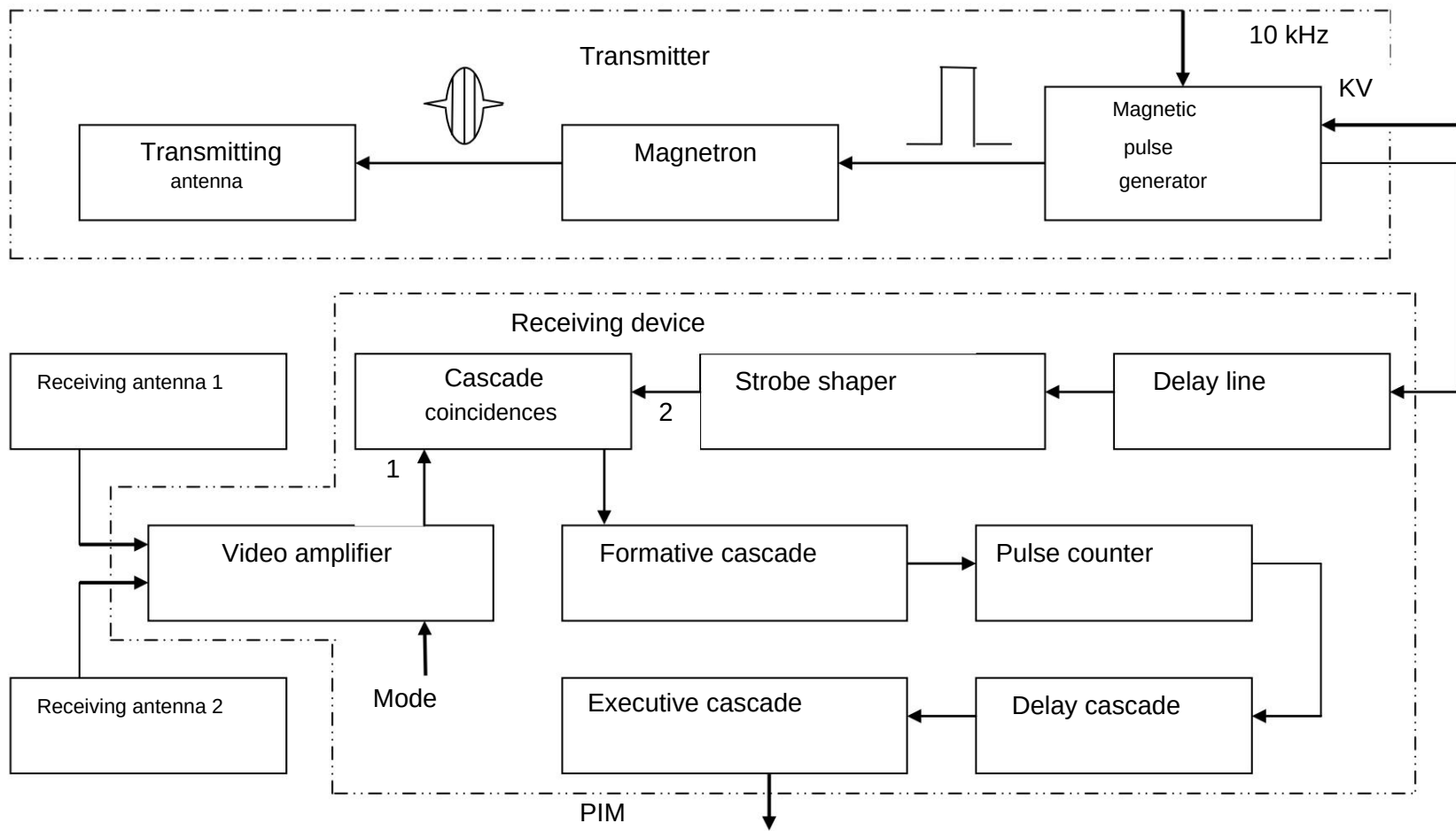


Fig. 31. Functional diagram of the RV

The video pulses obtained at the output of the detector sections are amplified by the video amplifier and fed to the input of the first coincidence cascade. The strobe pulses from the strobe former are fed to the input of the second cascade. The strobe pulses are formed from the synchronizing pulses coming from the magnetic pulse generator through the delay line. When the strobe and video pulses coincide in time at the coincidence cascade, a pulse is sent from it to start the shaping cascade, which produces pulses normalized in amplitude and duration that are fed to the counter.

Strobing in the RV is provided for the implementation of range selection and ensuring isolation from the earth's surface.

Time selection by accumulation of information (counting of a certain number of reflected pulses) increases noise immunity with respect to random noise emissions and passive interference. To provide additional isolation between the transmitting and receiving antennas, correction of the RV response area at small misses and increase of noise immunity to passive interference, the RV has a software sensitivity adjustment. To increase noise immunity to active interference, the video amplifier has a coarsening chain.

To effectively defeat a target, it is necessary to coordinate the RV actuation area with the area of dispersion of the main mass (80...90%) of warhead fragments. The RV actuation area is a part of the space in which the conditional centers of targets may be at the moment of its actuation. The RV actuation area is coordinated with the area of dispersion of fragments by delaying the warhead detonation pulse depending on the relative speed between the missile and the target. Information about the specified speed is entered into the RV by the KOS commands.

Before the launch of the rocket, commands are issued from the launch automation equipment on the launch vehicle, inducing one of its operating modes.

During a controlled flight, the ground control station determines and transmits to the missile commands for the relative velocity of the missile and target (KOS), which are received by the RV, and sets a certain delay between the moment the RV is triggered and the detonation of the warhead.

When the missile approaches the target at a specified distance, the missile transmitter is switched on by the command to arm the missile (KUV) received from the onboard radio control equipment. The probing pulses of the missile transmitter irradiate the target.

Upon the radio command for arming (KV), following the command for KUV, the information processing unit is activated, which, taking into account the operating mode of the RV, analyzes the pulses reflected from the target and, upon accumulation of a certain number of them, issues a triggering pulse.

5.2. Safety actuator

The safety actuator mechanism (SAM) 9E134 is intended:

- to ensure reliable protection against accidental explosion of the warhead of a loaded missile at all stages of storage, transportation and operation;
- to prevent the warhead from detonating during launch and in flight until commands are received KUV and KV;
- to generate a detonation impulse to detonate the warhead of the missile upon command from a radio detonator or from a control unit;
- for self-destruction of a missile in flight if the radio fuse fails to trigger on the target.

The PIM can only be triggered after the safety stages have been removed during the rocket's flight and the safety device has been armed.

PIM has three levels of protection, the removal of which requires the presence of the following factors:

- starting and entering the mode of the on-board chemical current source, from which the PIM is powered; – starting the engine – the presence of pressure in its combustion chamber, from which, with the help of a signal, the pressure analyzer closes the contacts in the PIM cocking circuit;
- axial overload during the rocket flight (immediately after launch) of at least 19 s with an action time of at least 1 s. Due to the action of the inertial force, the locking of the moving parts of the PIM on the firing chain line is removed and additional contacts in the chain are closed cocking.

During the launch cycle, the rocket's HIT is launched, and when it reaches the mode, it becomes possible to supply voltage to the PIM cocking circuit. This removes the first stage of the PIM safety (this circuit is additionally opened by the engine pressure indicator and the PIM inertial contactor). During the rocket ejection, the PIM is in the initial state, since the circuit for supplying voltage to the PIM is opened by the engine pressure indicator.

When the engine starts and the pressure in the combustion chamber increases, the pressure alarm is triggered, closing the contacts of the voltage output from the HIT to the PIM arming circuit. The second stage of its protection is removed.

Under the action of axial acceleration with the engine running for at least 1 s, the inertial contactor closes the PIM arming circuit. The third safety stage is removed, and the final arming of the PIM occurs. If the axial acceleration lasts less than 1 s, the removal of the third safety stage and the final arming of the fuse do not occur.

The activation of the PIM and, as a consequence, the detonation of the warhead occurs in two cases:

- near the target upon the "Activation" signal – to hit the target;
- in case of a miss (failure to receive a signal from the RV) by the electrical signal "Liquidation"

from BRU - to eliminate the missile.

5.3. Warhead

The high-explosive fragmentation warhead as part of the missile's combat equipment provides destruction of enemy air attack weapons by fragmentation striking elements, and near the target also by high explosive impact.

The warhead (Fig. 32) consists of a body, a fragmentation shell, an explosive charge (EC) and an additional detonator. The body is a welded structure consisting of a thin-walled outer shell 3 and two flanges 5 and 10. Along the axis of the warhead, inside the explosive charge, the body has a cavity formed by an inner shell 9 of a stepped-cylindrical shape. On the side of the large diameter, the PIM and an additional detonator (checker 4, nut 7, cover 6) are installed in the cavity, which is screwed into flange 5 using a thread.

The fragmentation shell is placed on the shell 3 of the warhead body between flanges 5 and 10. It consists of striking elements 1 made of high-density alloy. Electrical communications from the PIM connector are led out through cavity "D" beyond the warhead.

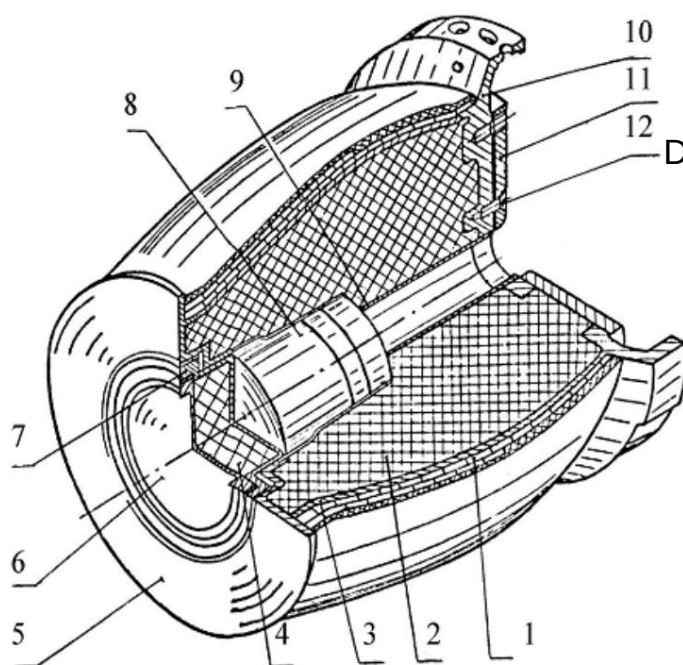


Fig. 32. Warhead: 1 – striking elements; 2 – explosive charge; 3, 9 – shell; 4 – checker; 5, 10 – flange; 6, 11 – cover; 7 – nut; 8 – PIM; 12 – screw

When the initiating pulse is applied from the PIM to the additional detonator, the charge 4 is triggered and causes the detonation of the explosive charge 2. The resulting shock wave and detonation products rupture the thin-walled shell 3 of the body and ensure the throwing of the striking elements. A circular fragmentation field with specified characteristics is formed in the transverse plane.

istics, hitting the target.

The angle of dispersion of striking elements in the longitudinal plane depends on the shape (curvature) of the shell 3 of the warhead body, which determines the shape of the explosive charge and the fragmentation shell. The value of this angle is selected based on the condition of covering the target with a stream of striking elements in various conditions of the missile's encounter with the target, also taking into account the angles of the missile's response.

The kinematic diagram of the coverage of the target by the warhead's area of destruction is shown in Fig. 33, distribution of initial velocities of fragments by the angle of dispersion (under static conditions) – in Fig. 34.

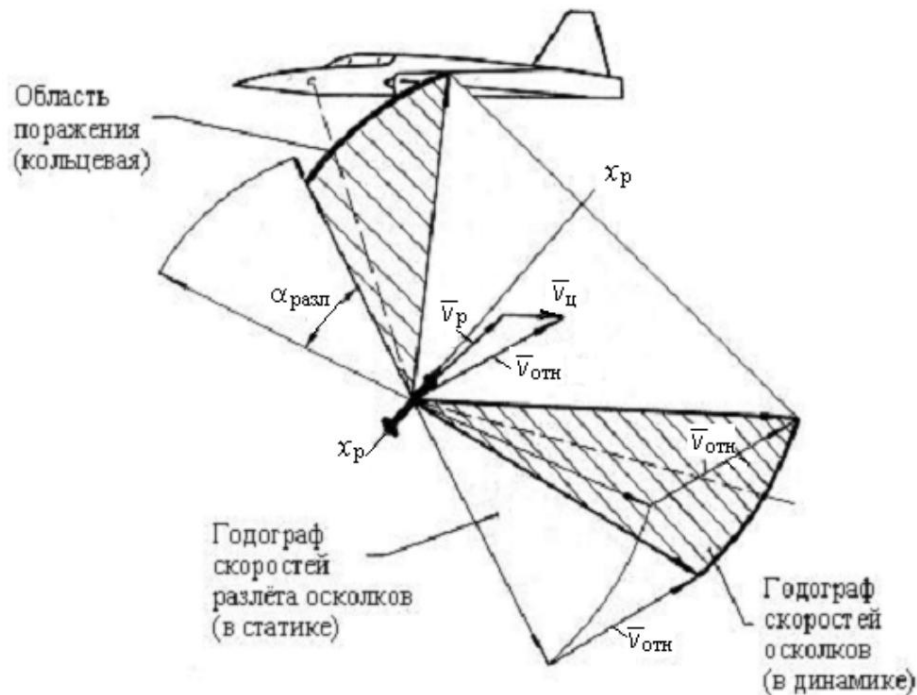


Fig. 33. Kinematic diagram of covering the target with the warhead destruction area: - - - - line of operation of the radio fuse; x_p – longitudinal axis of the missile; $\alpha_{\text{разл}}$ – angle of fragmentation; $\vec{V}_p, \vec{V}_{\text{ц}}, \vec{V}_{\text{отн}}$ – missile speed, target speed and relative speed

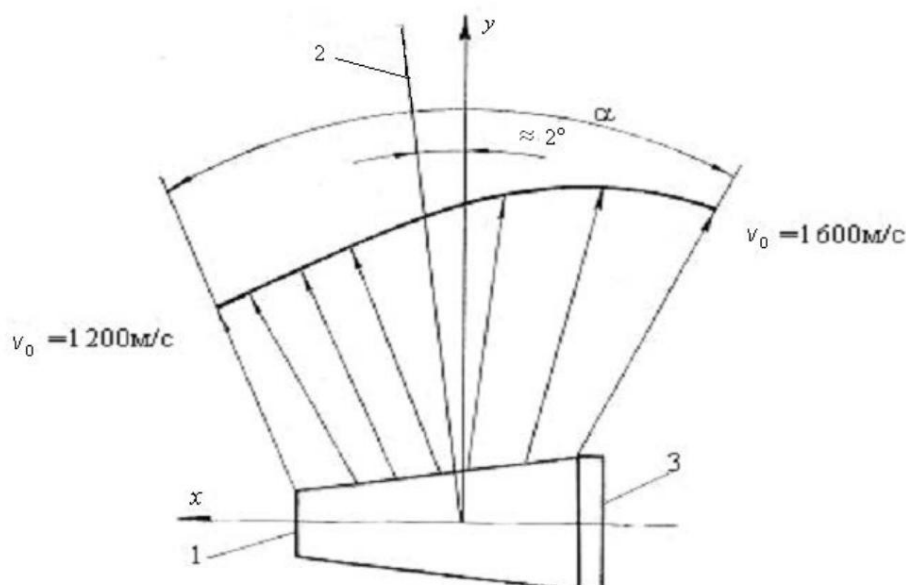


Fig. 34. Distribution of initial velocities of fragments by angle of dispersion (under static conditions): 1 – front end; 2 – direction of scattering of the main mass of fragments; 3 – rear end

The effectiveness of a missile, determined by the destructive effect of the warhead and the dispersion of its point of detonation relative to the target, must be high not only when affecting weakly protected targets such as airplanes and helicopters, but also when affecting high-precision weapons, which often have high strength. The typical appearance of the zones of destruction of various targets is shown in Fig. 35.

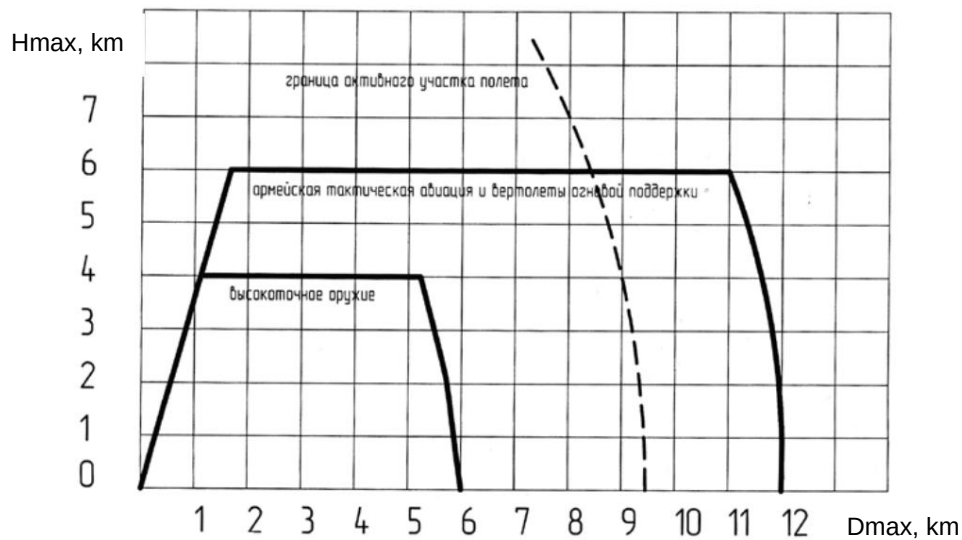


Fig. 35. The zone of destruction of various targets

Self-assessment questions for section 5

1. What functions are assigned to a radio fuse?
2. When and on what command does the radio fuse begin to irradiate the target?
3. How is the warhead detonation team formed?
4. List the design solutions that ensure coverage of the target with a stream of destructive elements in various conditions of the missile's encounter with the target.
5. In what modes does the radio fuse operate?
6. Operating procedure of the radio detonator.
7. Components of a radio fuse and their purpose.
8. What are the main components of a radio transmitter?
9. What is a magnetron for and how does it work?
10. Components of the radio detonator receiver and their purpose.
11. What functions does the information processing unit of the radio explosion receiving device perform?
12. Where do the pulses come from to the counter of the information processing unit of the receiving device?
13. How do you understand the phrase "coincidence in time of strobe and video pulses on the cascade" "coincidences"?
14. Why does a radio fuse have a software adjustment for sensitivity and a roughening chain?
15. How is maximum effectiveness of the warhead achieved in the area?
16. What determines the area of operation of a radio fuse?
17. How is the coordination of the radio fuse triggering area with the scattering zone achieved?
18. How is the noise immunity of the RV increased?
19. What is the purpose of the safety actuator mechanism (SAM)?
20. What functions does the SAM perform?
21. When and by what command is the SAM activated?

22. List the conditions under which the safety steps are removed in the PIM.
23. How is the first stage of protection in PIM removed?
24. How is the second stage of protection in PIM removed?
25. How is the third level of protection in PIM removed?
26. In what cases does PIM trigger?
27. Name the conditions necessary and sufficient for hitting a target. 28. List the design solutions that ensure coverage of the target with a stream of damaging elements in various conditions of the missile's encounter with the target.
29. How to build a zone of destruction of any target?

6. Rocket engine

The rocket engine (Fig. 36) is a dual-mode, single-chamber rocket solid propellant engine (SPRE), which consists of:

- a body including a cylindrical part 7, a front bottom 1, a rear bottom 9 and a nozzle block 11 with an insert 13 and a plug 12;

- a charge of solid fuel 6 - a single-channel pellet of mixed fuel, having slotted cuts on the side of the engine nozzle.

The presence of eight slots allows to significantly increase the surface of fuel combustion and ensure the starting mode of engine operation. In the cruising mode, after the charge burns out in the slot zone, combustion occurs only on the surfaces of the single-channel pellet;

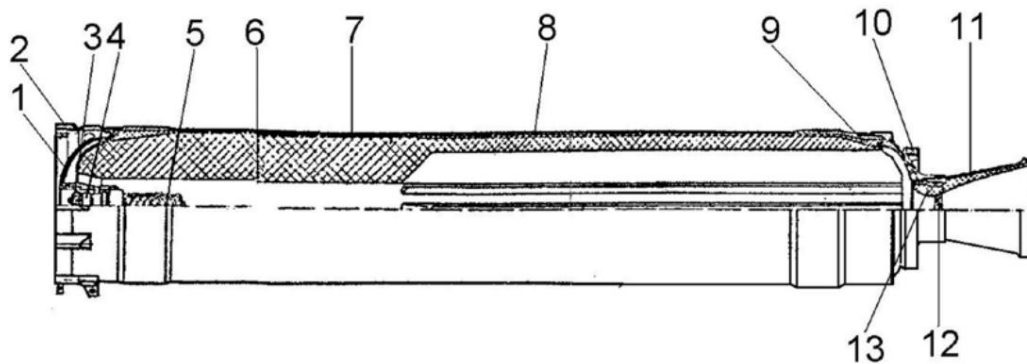


Fig. 36. Rocket engine: 1 – front bottom with heat-protective bowl; 2 – power frame; 3 – pressure indicator; 4 – pyropatrons; 5 – igniter; 6 – solid fuel charge; 7 – body; 8 – heat-protective coating; 9 – rear bottom with heat-protective cup; 10 – seating belt; 11 – nozzle block; 12 – plug; 13 – insert

– igniter 5 with two pyropatrons for its ignition and pressure indicator 3 in the combustion chamber. All elements are located on the front bottom of the engine housing. The indicator is used in the combat equipment safety system to issue a command to arm the radio fuse. The second pyropatron serves to duplicate the engine start. The duplicate pyropatron is triggered one second after the engine is started, ensuring ignition of the igniter in the event of failure of the first pyropatron.

The front and rear bottoms (elliptical in shape) of the hull are made of high-strength. On the outer cylindrical part of the steel type KVN with $2 \cdot 190 = \text{kg mm} / \dots$ bottoms, special thrust threads are cut and flanges are machined, with the help of which, when screwing, the bottoms are centered relative to the front frame and the thickening of the rear end of the cylindrical part of the hull.

The cylindrical part of the hull is a thin-walled cylinder 7 made of high-strength sheet steel, to the front end of which a power frame 2 is welded. Ten bosses are placed on the power frame and a special thrust thread is cut for screwing the front bottom. In the bosses, made as a single unit with the frame,

threaded holes are cut and studs are installed in them for connection with the instrument compartment of the rocket. A special hole is also cut in the thickening of the cylindrical part of the rear end

thrust thread for screwing in the nozzle block, holes are drilled for joining with the wing block compartment.

A cylindrical seating belt 10, on which the bearing of the fifth compartment is installed, and a cylindrical nozzle in which a thread is cut for are welded to the rear bottom.

The tightness of the internal cavity of the engine body during storage and operation is ensured by pressing the rubber gaskets between the cylindrical part and the bottoms. The gaskets are pressed by the front bottom and the nozzle block during assembly. The igniter-pyroc cartridge block joint is sealed with a copper gasket.

A rubber bag is glued to the inside of the cylindrical body, which plays the role of a protective and fastening layer 8 between the engine body and the charge. In the front and rear parts, the bag has thickenings in the form of cups, designed to ensure the required stress-strain state of the charge during temperature fluctuations.

The rear bottom has a seat on the outside for the wing compartment bearing and a thread. used for its fastening. The inner part has a thread for installing the nozzle.

The nozzle consists of a body with a cylindrical threaded part and a conical outlet. The critical section of the nozzle is formed by a nozzle insert screwed into the body.

The graphite insert is glued into the internal cavity of the body from the rear bottom side; the tightness of the joints during assembly of the nozzle block is ensured by rubber gaskets.

A glass is welded into the center of the front bottom, which has seats for installing pyropatrons, as well as internal threads for installing the igniter body. A nozzle for installing a pressure indicator is welded into the spherical part of the front bottom. The igniter is three cylindrical rods of solid fuel and a charge of coarse-grained black powder, enclosed in aluminum cases and placed in a mesh body. The igniter body, fixed to the front bottom, partially enters the charge channel.

The pyropatron (Fig. 37) is designed to ignite the igniter. It consists of a body 2, a bottom 5, a suspension of the igniting composition 3, a suspension of the pyrotechnic composition 4, a connector for connecting a pyroprotective device and an electric fuse 1, made in the form of a wire bridge. Reliable operation of the igniter is ensured by two pyropatrons.

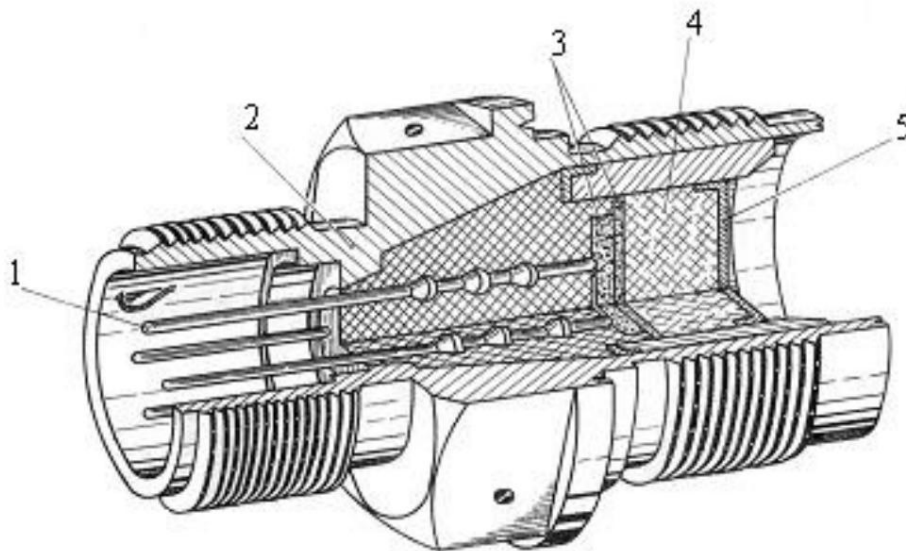


Fig. 37. Pyrocartridge: 1 – electric fuse; 2 – body; 3 – igniter composition; 4 – pyrotechnic composition; 5 – bottom

The engine starts to operate from the moment the launch command is sent through the rocket's electrical system circuit in the form of a current with a voltage of (+27 V) to the wire bridge of the pyropatrons. Wire the bridge instantly heats up, causing the ignition of the pyrotechnic composition of the pyropatron, as a result of which its pyrotechnic composition also ignites. The resulting gases, breaking through the

pyropatron, ignite the main igniter attachment. The mass of the gases formed and their thermal energy is sufficient to create the necessary conditions for ignition of the fuel charge and its stable combustion. The charge burns along the internal channel, cracks and ends. The pressure in the chamber in the starting mode is no more than 1.5 kPa. Engine thrust in the starting mode is ≈ 3200 kgf. Engine operating time

in the starting mode is ≈ 4 s [14], in the cruising mode ≈ 8 s. Total engine operating time is ≈ 12 s.

Self-assessment questions for section 6

1. Give a description of the rocket engine.
2. Calculate the engine thrust in starting and cruising modes.
3. List the main components that make up a rocket engine.
4. How is the engine's dual-mode capability ensured?
5. How and by what command is a dual-mode solid propellant rocket motor started?
6. How is the independence of the engine operating parameters from the ambient temperature ensured?
7. Draw a curve of pressure change in the engine chamber over time.
8. How is the engine sealed?
9. How is the charge attached to the body?
10. How is the required stress-strain state of the charge ensured?
11. What is a charge igniter?
12. How is a pyropatron constructed and how does it work?
13. For what purpose is a decay alarm used in a rocket in addition to a pressure alarm?

7. PRESSURE ALARM

The pressure alarm (Fig. 38) is a single-action device and provides a constant voltage of +27 V to the PIM to remove the second stage of protection and to the command block to prepare the autopilot connection circuit when the pressure in the engine chamber is reached

of at least 0.3 kPa [16]. The pressure alarm consists of a steel housing 1, in which an insulator 5 with two reinforced pin contacts 6 is located, and steel piston 3.

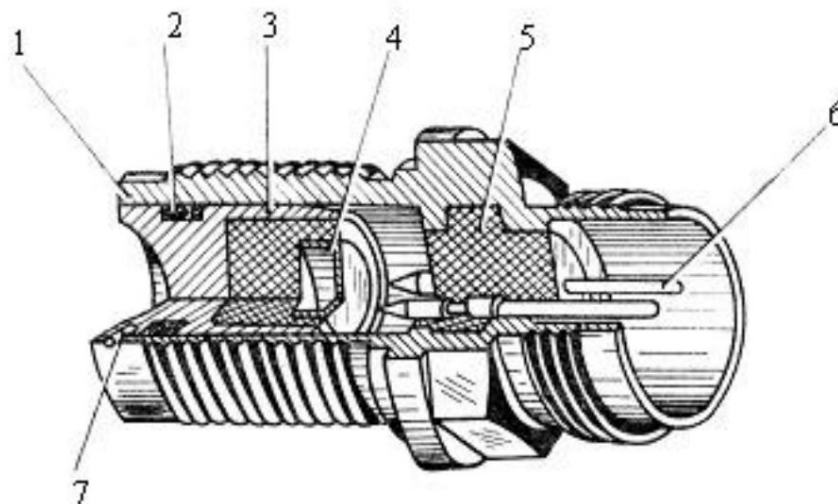


Fig. 38. Pressure alarm: 1 – body; 2 – ring; 3 – piston; 4 – bottom; 5 – insulator; 6 – contact; 7 – pin

The piston has a brass bottom 4, isolated from the piston, and a rubber-fluoroplastic sealing ring 2. In the rocket's electrical system, the indicator is connected by pin contacts using a socket of the electrical network plug connector. In the initial (far left) position, the piston is held by a steel pin 7, while contacts 6 are open. Under the effect of hot gases coming from the engine chamber on the piston, the pin is cut off and the piston moves to the extreme right position, closing the contacts with the brass bottom 4.

Questions for self-assessment in section 7.

1. For what purpose is a decay alarm used in a rocket along with a pressure alarm?
pressure?
2. How does a pressure alarm work?

8. STEERING DRIVE

The drive is designed to move the rudders in accordance with the resulting signals. control and stabilization signals generated in the autopilot circuit.

The operating principle of the drive can be seen in the electro-kinematic diagram shown in Fig. 39. The drive with a zeroing system consists of two circuits: the drive circuit itself and the zeroing circuit. Both circuits are formed on the basis of a common UPT and a gas distributor (jet relay).

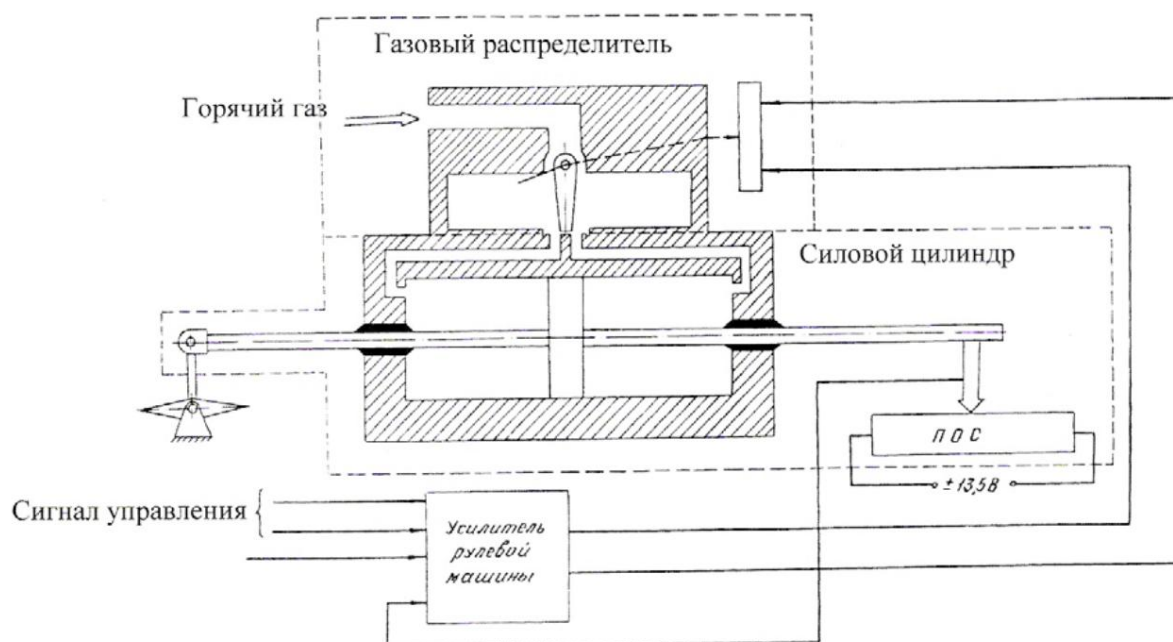


Fig. 39. Electrical diagram of the steering drive

The drive circuit of each of the autopilot control channels includes the UPT, gas distribution distributor and power cylinder (steering machine) with feedback potentiometer.

The steering machine (Fig. 40,41) uses the energy of compressed gas directed into its cavities to move the actuator (rudders).

The feedback potentiometer (FPP) generates a feedback signal in the form voltage proportional to the displacement of the steering gear rod (RM).

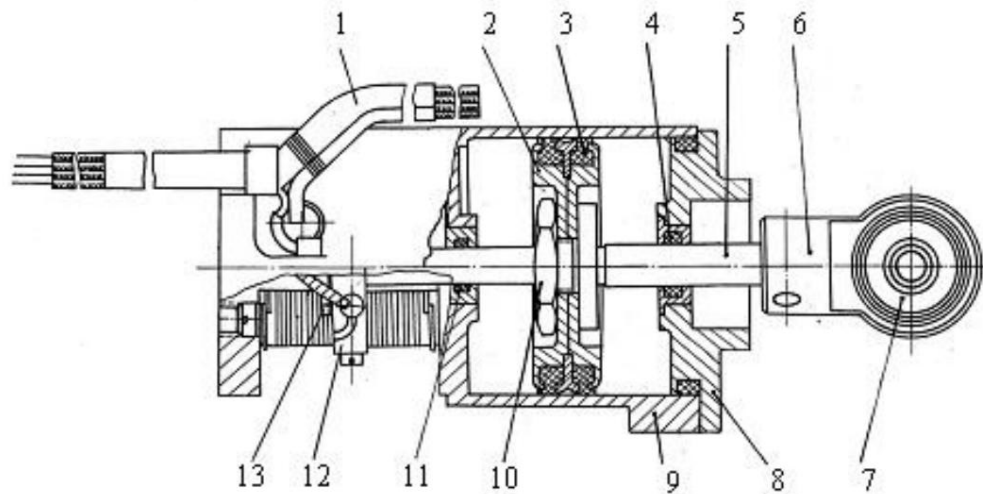


Fig. 40. Steering gear: 1 – electrical harness; 2 – disk; 3 – rubber cuff; 4,11 – seals rings; 5 – rod; 6 – earring; 7 – bearing; 8 – cover; 9 – housing; 10 – nut; 12 – feedback potentiometer; 13 – contact plate; 14 – potentiometer slider

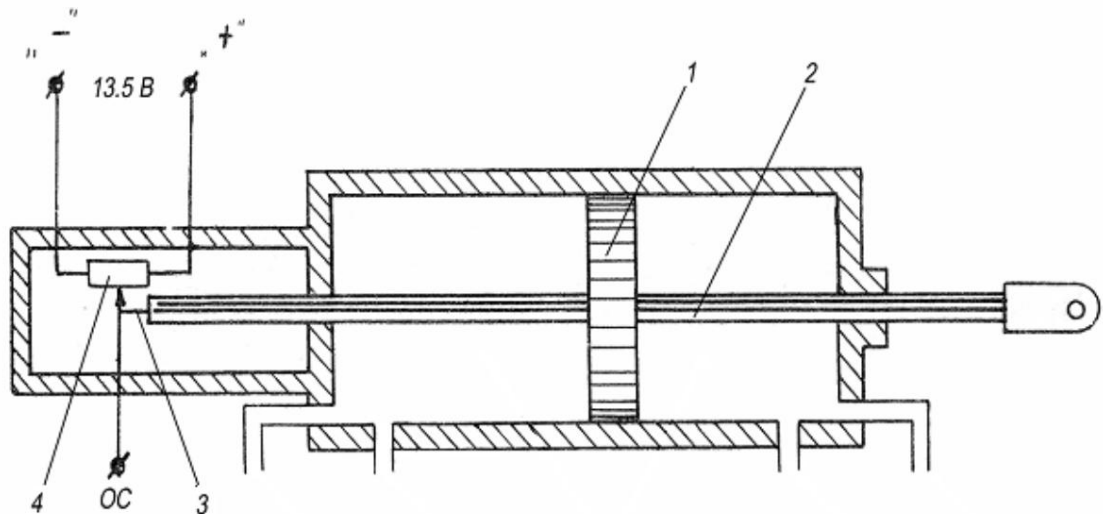


Fig. 41. Electrical diagram of the power cylinder: 1 – piston; 2 – rod; 3 – current collector; 4 – feedback potentiometer

8.1. Gas distributor

The gas distributor (Fig. 42) directs, with the help of a rotary device, into the cavities of the RM that amount of high-pressure gas that is required in accordance with the magnitude of the amplified UPT signal. The polarity of the signal determines the direction of the gas entering one or another cavity.

The gas distributor is designed to convert the electrical signal coming from the UPT into a gas pressure difference in the cavities of the power cylinder RM proportional to this signal.

High-pressure gas supplied from the gas generator to the gas distributor enters the nozzle, and from it – into two receiving openings, each of which communicates with the corresponding cavity of the power cylinder RM (Fig. 42).

The nozzle 11 is rigidly connected to the anchor of the electromagnetic rotating device of the gas distributor and has the ability to rotate at an angle of $\pm 3^\circ$ from the neutral position.

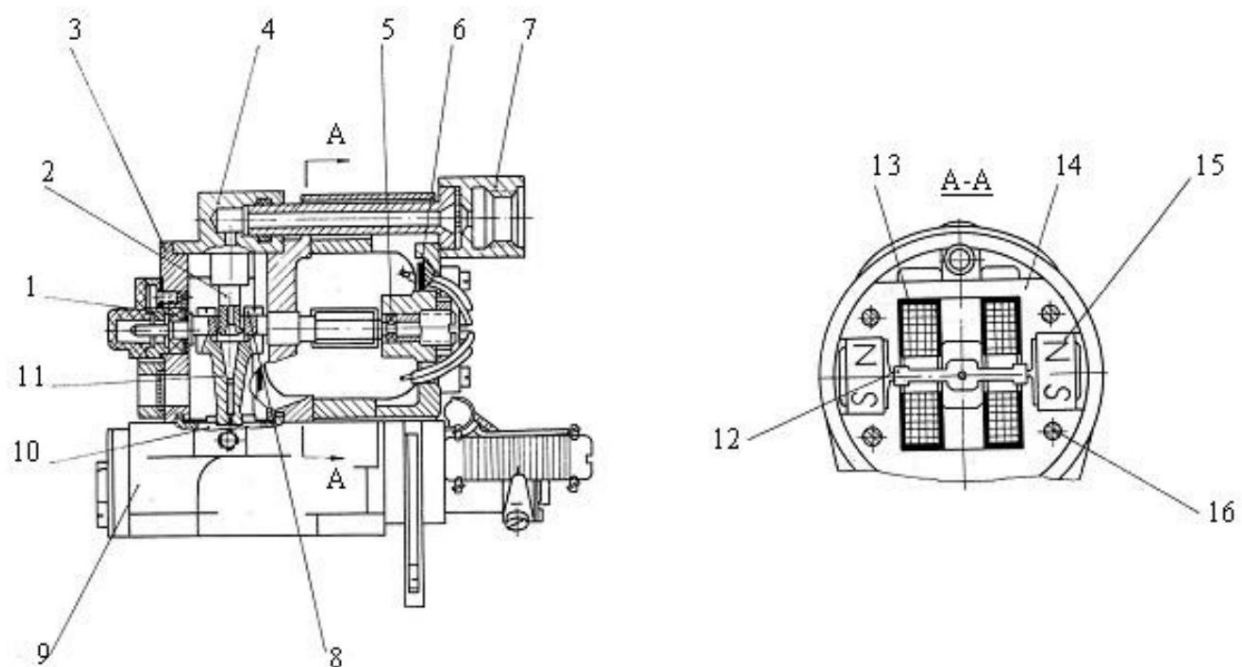


Fig. 42. Construction of gas distributor: 1.5 – bearings; 2 – nipple; 3 – body; 4.6 – flanges; 7 – bracket; 8 – screw; 9 – bracket; 10 – receiving part; 11 – nozzle; 12 – anchor; 13 – coil; 14 – magnetic circuit; 15 – permanent magnet; 16 – stud

The nozzle position is considered neutral when it is located symmetrically relative to the receiving holes. In this case, when gas flows out of the nozzle, equality of static pressures p_1 and p_2 is established in the cavities of the power cylinder. The equality is violated in the case of deviation of the nozzle from the neutral position. Redistribution of pressures in the cavities leads to displacement of the piston in the power cylinder.

The nozzle apparatus of the gas distributor is designed so that at the middle position of the nozzle its passage opening is completely blocked by the receiver jumper, and the receiver openings are completely blocked by the nozzle walls. This design of the nozzle apparatus brings it closer to spool-type distributors, ensuring low gas consumption in the neutral position of the nozzle and a greater steepness of the distributor's pressure characteristic.

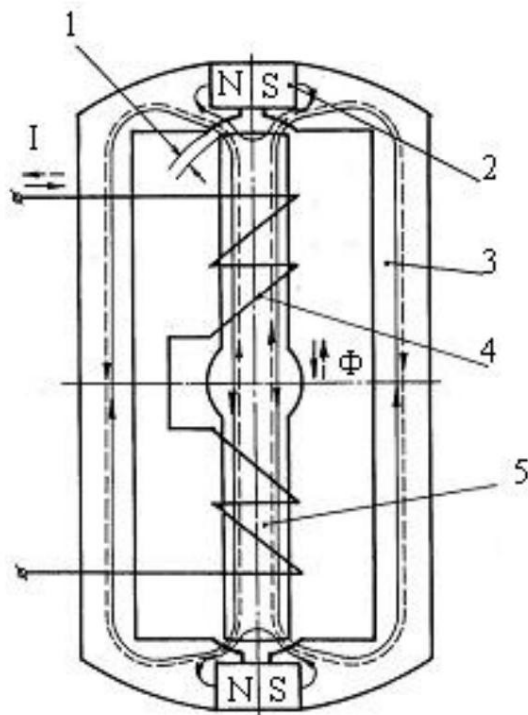


Fig. 43. Electromagnetic polarized device: 1 – working gap; 2 – permanent magnet; 3 – magnetic circuit; 4 – control winding; 5 – anchor

In the gas distributor, an electromagnetic polarized rotary device is used to control the rotation of the nozzle (Fig. 43).

It consists of two U-shaped magnetic cores, rectangular permanent magnets, an anchor with which the nozzle is rigidly connected, and a control winding.

Permanent magnets create the main magnetization flux Φ_0 , which closes along the magnetic circuits, passing through the working gaps and the end parts of the anchor.

If the anchor is in the middle position and the current in the control winding is zero (i.e. the flux Φ created by the control winding is zero), then the magnetic field strength in the gaps at the north and south poles is the same. Due to the equality of the field strengths in both gaps, the electromagnetic forces acting on the anchor are mutually compensated.

If we assume that under the action of an external moment the anchor deviates from the average position, then the cross-sectional areas of the gaps will change with an unchanged value of the magnetic flux of the permanent magnet. The equality of the field strength under the north and south poles will be violated. As a result, the equality of the electromagnetic forces acting on the anchor will also be violated and a moment will arise that tends to turn the anchor in the direction with greater strength (to restore the lost balance).

Thus, the anchor of the gas distributor with equal currents in the control windings stably maintains the average position, i.e. the action of the magnetic flux of permanent magnets on the anchor is similar to the action of the centering spring. The magnitude of the restoring moment of the "magnetic spring" is proportional within certain limits to the angle of deviation of the rotor from the neutral position.

When current passes through the control winding, a magnetic flux arises in the gas distributor anchor, which passes through the anchor and then branches into two $F/2$ flows, which close through the working gaps and magnetic circuits. The direction of the control magnetic flux depends on the direction of the current I , i.e., the polarity of the control signal at the amplifier input.

cal.

The magnetic flux $\vec{y}$, created by the control winding, interacts with the field of the permanent magnets $\vec{y}_0$ - the fluxes are added or subtracted in the working gap depending on the sign of the current I . This leads to an inequality of the field strengths in the working gaps and the appearance of a torque that tends to equalize the field strengths in both gaps. The anchor will deflect until the strengths become equal.

In the deflected position, the intensities are equal and there is no torque acting on the anchor. Thus, the position of the anchor of the gas distributor depends on the magnitude of the control current in the control winding of the relay.

In the gas distributor, the configuration of the magnet poles and the shape of the anchor are selected in such a way that the angle of rotation of the anchor relative to the neutral position is proportional to the magnitude of the current in the control winding. The direction of rotation of the anchor corresponds to the sign of the current in the control winding.

The nozzle of the gas distributor rotates together with the anchor. When the anchor is deflected, the overlaps of the receiving openings are also made unequal, as a result of which the gas stream flowing out of the nozzle creates a pressure difference in the cavities of the power cylinder RM.

The design of the gas distributor is shown in Fig. 42. Magnetic cores 14 with permanent magnets 15 installed on them are glued to flange 6 and fixed to it with studs 16. This unit together with anchor 12 and coils 13 is inserted into housing 3. The anchor rotates in bearings 1.5. Gas is fed into nozzle 11 through bracket 7, flange 4 and nipple 2.

The nozzle is fixed to the anchor axis using screws 8. The nozzle outlet is a slit. The receiving part 10, called the plug, is pressed into the bracket 9. By drilling in the bracket, the holes in the plug are brought out to plane "A", which is in contact with the plane of the power cylinder. The plug has two slit-like openings with a bridge between them. Sections

The holes in the nozzle and receiver are selected based on the specified gas flow rate and the drive operating speed. The gap between the nozzle and the plug is selected using gaskets so that the limit the operating pressure in the drive gas generator did not exceed the maximum.

8.2. Pressure drop alarm

The pressure drop indicator (Fig. 44) in the declination system gas generator is designed to issue the "KPO" ("Switching the rudder deflection angle limitation") command. In response to this command, the indicator supplies a constant voltage of +27 V to the autopilot control unit to switch its operating mode at the final section of the trajectory, when the pressure in the combustion chamber drops to (0.1 ± 0.01) kPa. In this case, the signals to the steering machines are limited to a level of 15° of rudder deflection. The pressure drop indicator consists of a steel housing 5 with a contact block 2 with spring contacts fixed in it, a steel piston 7 and a spring 6. A contactor with a contact ring 3 is

fixed on the piston. The spring contacts of the block 2 are connected to wires ending in a plug connector 1 for connecting the indicator to the rocket electrical system.

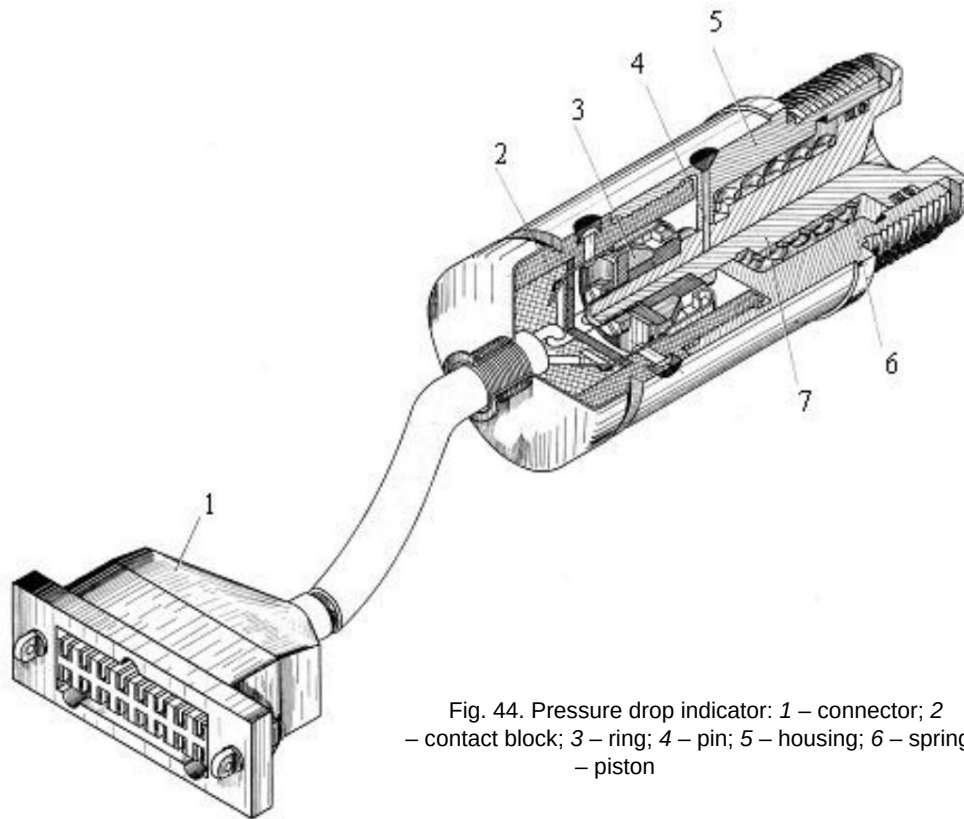


Fig. 44. Pressure drop indicator: 1 – connector; 2 – contact block; 3 – ring; 4 – pin; 5 – housing; 6 – spring; 7 – piston

In the initial position, the contacts of the indicator are open, since the piston is held in the middle position by pin 4, passing through the hole in the body and piston. At the moment of starting the gas generator of the declination system, the pressure of the gases on the piston cuts off pin 4 and the piston moves to the extreme left position. As the pressure decreases, the piston, under the action of the force of spring 6, moves to the right and closes the spring contacts of block 2 with ring 3. To ensure reliable operation, the contact block has two pairs

contacts.

Self-assessment questions for section 8

1. Explain the operation of the steering gear.
2. Explain the structure of a gas distributor.
3. What elements does the gas-dynamic drive of the steering wheel consist of?
4. Purpose and design of the gas distributor of the gas-dynamic drive of the steering wheels.
5. Why is a feedback potentiometer used in a steering machine?
6. What determines the polarity and magnitude of the signal that controls the operation of the gas distributor?
litella?
7. What function does the electromagnetic polarized regulator perform in the gas distributor?
le?
8. Explain the operating principle of an electromagnetic polarized relay.
9. For what purpose is a decay alarm used in a rocket along with a pressure alarm?
pressure?
10. How does a pressure drop alarm work?

9. CHEMICAL CURRENT SOURCE

As a chemical current source – a power source for the rocket's onboard equipment – DC batteries are used, which can be in one of the following states:

- in non-working mode, in which the current pulse on the filament of the electric igniter does not supplied, the electrolyte is in a solid state;
- in the working one, during which the pyroelectric heaters have been activated, the electrolyte is melted;
- unfolded.

The batteries are brought into working condition by applying a voltage of 20...35 V direct current from a ground power source to the electric igniters for at least 0.01 s. The batteries, brought into working condition, provide the following direct voltages during the flight: +20 V; -20 V; ± 27 V; +25 V.

10. ELECTROMECHANICAL CONVERTER

The electric current converter is designed to supply alternating voltage to the rocket's onboard equipment.

The converter is launched from a ground source of direct current with a power of at least 0.5 kW in steady-state mode and up to 2.0 kW in starting mode, providing a voltage of 27...31 V at the contacts of the rocket's on-board connector. The converter is powered in flight from an on-board chemical source of direct current.

The converter's ramp-up time to the mode is 1.5 s at idle when starting from ground source.

During operation, the converter supplies the following three-phase current to the rocket's on-board network: alternating voltages: 36 V 1000 Hz; 5 V 1000 Hz.

11. TRANSPORT AND LAUNCH CONTAINER

The transport and launch container (TLC) is designed to store, transport and launch missiles (Fig. 45). It consists of a hull 2 divided by diaphragms 18 into four cavities designed to install missiles, a single-action protective and sealing device 28 and two covers: front 1 and rear 12. The hull 2 is a welded ribbed structure made of aluminum alloy and reinforced with three frames. On the front frame there is a bracket 8 ensuring the fixation of the TLC on the TZM during loading of the combat vehicle, and a plate 7 for fixing the TLC on the BM. Fixation of the TLC from lateral movements during loading on the BM guides is ensured by T-shaped grooves-yokes 9 on the middle and yokes 11 on the rear frames. The tightness of the hull is ensured by rubber gaskets along all butt joints.

There are two holes on the front cover of the housing, closed with plugs. The holes are intended to check the tightness of the cavity under the front cover during the manufacturing of the TPK and to equalize the pressure in this cavity with the environment before removing the front cover.

Inside the cavities of the TPK, on the upper wall, guide beams 17 are attached to the frames using bolts and steel brackets, on which the missiles are installed. On the lower surface of the TPK body there are four hatches, which are closed by covers 22 with mechanisms for disconnecting electrical connectors 15, providing communication between the electrical circuits of the missiles and the launch automation and disconnection of the TPK electrical connectors from the onboard electrical connectors of the missiles during launch, and plugs 21, providing access for switching the letter frequency.

29 – gasket; 30 – desiccant (silica gel); 31 – cover; 32 – screw

To connect the TPK to the launch automation, there are on-board electrical connectors 16, located on the sides of the front frame of the TPK. To purge the internal cavities of the TPK with dry air during manufacture, there are purge holes in the rear cover and on the lower surface of the TPK, near the front frame, closed by plugs 20 and 23. The rocket is loaded into the TPK from the front end of the TPK. The rocket is fed into the TPK up to the stop of the ejection device earring in the grooves of the TPK guide. The missile is fixed in the TPK in the axial direction by a removable crossbar 1 (see Fig. 7). The missile is transversely fixed by brackets 9,11 (see Fig. 45), which are located in the grooves of the brackets on the middle and rear frames.

Questions for self-control on sections 9-11

1. What is a chemical current source? 2. What state can DC batteries be in at different stages of their functioning? 3. The purpose of the electric machine current converter. 4. What is included in the control channel drive circuit? 5. How is the rocket fixed in the TPK?

12. SEQUENCE OF OPERATION OF THE ROCKET AND ITS ELEMENTS

“Preparation” mode (the mode duration can be 5...5.5 s): – steering machines are reset; – voltage is supplied from the ground power source; – autopilot gyromotors are turned on; – operation of the AP memory devices in the “tracking” mode; – storing information on pitch and heading angles received in the ASA; – output of the address code to the radio equipment; – logical “1” along the “Record” circuit; – voltage +20 V along the KRAZ circuit; after 1...1.5 s: – forced heating of the magnetron filament is in progress; – output of the asynchronous start pulse of the transponder is prohibited; – the counter and output circuits of the unit are reset by the “Record” command; – the lower antennas are open and the upper ones are closed in the initial state; – the address code issued along three circuits from the start automatic equipment is set. **“Standby” mode** (the mode duration can be 5.5...60 s): – switching the BRU 36 power supply to 1000 Hz from ground to onboard; – two gyroscopes operating on coasting are powered from the ground source (if the standby continues for more than 5 s). **“Start” mode** (the command can be issued at any time starting from 6.5 to 6.1 s): – the heat exchanger output is at nominal value; – switching the BRU power supply from ground to onboard; – during 0.1...0.2 s, the voltage with the logical “1” level is removed along the “Record” circuit; – after 0.9 s, the ASA issues a command to uncage free gyroscopes; – switching the AP memory device to the information storage mode (the “Memory” command) on the declination command value at the end of the launch process;

– prohibition on rewriting the address code for 1.5...2.0 s; – command from the ASA to detonate the catapult pyropatron.

Rocket launch:

– separation of the on-board connector; – rotation of levers 30 and 33 (see Fig. 12);

- closing of push-button switches;
- launch of the steering gear;
- launch of the GG declination system 0.31 s after turning lever 30 on command from the command block time delay device; – exit of the missile from the TPK;

- opening of the rudders;
- 0.25 s after the on-board connector is disconnected – the time relay is triggered and the steering is reset outgoing machines;

- generation of control signals according to a given algorithm for the operation of rudders (declension path);

- vertical flight of the rocket to a height of 10...15 m.

Rocket flight:

- rocket inclination (flight controlled by pitch and heading channels under the action of aerodynamic and gas-dynamic forces on the rudders-ailerons);

- launch of the rocket engine 1 s after turning the lever 30 on command from the time delay device of the command block or 1 s after the lever is actuated;

- operation of free gyroscopes (determination of current pitch and heading angles);

- the UPR command from the BRU switches the autopilot from the declination mode to the control mode (1.5...2 s after the start of declination);

- blocking the UPR command, disabling the declination;

- connection of steering machines to the control path.

Control mode:

- removing the +20 V voltage in the KRAZ circuit;

- asynchronous mode of starting and operating the responder;

- radiation through the upper and lower antennas;

- determination of the missile position by angular coordinates using the BRU signals;

first communication session:

- request pulse from the SPC;

- teams K1 and K2;

- generation of the control signal by radio equipment in the form of a direct current voltage of +27 V;

- switching the autopilot to the control command processing mode (K1, K2);

Current communication sessions:

- request pulse from the SPC;

- on command KZAZ, termination of the asynchronous mode of starting the responder, transition to synchronic support;

- pairwise switching of antennas;

- double confirmation of the highest signal in one of the receiving antennas;

- switching reception to the selected pair;

- transmission of continuous and one-time command signals to the rocket;

- in the autopilot control unit – switching the autopilot operating modes (switching the autopilot gain factors, +27 V);

- into the radio fuse – KOS1-KOS4 in the form of direct current voltage corresponding to the logical level "0";

when approaching the target:

- The KUV from the onboard radio equipment turns on the RV transmitter;

- the HF radio command turns on the information processing unit (+27 V);

- the unit, taking into account the operating mode of the RV, issues a trigger pulse;

- formation of delays by commands KOS1-KOS4;

– detonation of the warhead.

In case of a miss (failure to receive a command from the RV or a radio link failure for 1.5...2.0 s), the onboard radio control equipment issues the "Liquidation" command accordingly and "Time", which closes the circuit for the self-destruction of the rocket. The rocket self-destructs.

During the flight, as the SPK issues one-time commands, the onboard radio equipment controls leniya gives out:

- in autopilot – "Switching the autopilot operating mode";

- to the radio fuse – four commands "Relative approach speed" in the form of a DC voltage corresponding to the logical level "0", as well as the command

"Radio fuse arming control" and "Radio fuse arming" in the form of voltage +27 V.

In case of radio link failure for up to 100 ms or absence of K1 and K2 commands in three consecutive communication sessions, the "Control" signal is not removed, and the K1 and K2 command levels are remembered. In case of radio link failure for more than 100 ms, the "Control" signal is removed, and the commands are set to zero. When the radio link is restored, the "Control" signal is restored, and the command levels correspond to the levels transmitted by the command transmission station.

mand.

In case of a radio line failure for a period of 1.5...2.0 s, the on-board radio control equipment The system issues the "Time" command, which closes the circuit for the rocket's self-destruction.

Questions for self-control

1. What is the 9M334 anti-aircraft missile module?
2. List the composition of the complex.
3. What is located on the base of the 9A331 combat vehicle?
4. Give a description of the detection radar.
5. Why is the detection radar a coherent pulse radar?
6. Purpose of the television optical viewfinder.
7. For what purpose is the vertical scanning of the radar detection space carried out?
planes?
8. How is the nationality of the target identified? 9. Give a description of the guidance radar.
10. Phased array antenna (PA), what is it?
11. Why is the guidance radar a coherent pulse Doppler type radar?
12. List the functions performed by the guidance station.
13. What are the features of the operation of the complex and the operating mode of the onboard equipment?
14. Why is the probability of detecting hovering helicopters higher than the probability of detecting
weapons of helicopters with rotating propellers on the ground?
15. What does the term "blind speed" mean in radar?
16. Main features of the 9M334 anti-aircraft missile module.
17. Advantages and disadvantages of the aerodynamic design of the 9M331 missile.
18. Explain the operation of the ejection device.
19. Provide general information about the missile's performance characteristics.
20. What is the first section of the rocket?
21. What is located in the second compartment of the rocket?
22. Purpose of hot gas sources located in the second compartment of the rocket.
23. How does hot gas get into the steering nozzle?
24. Explain the operation of the spring mechanism for locking the steering wheels.
25. What is located in the third compartment of the rocket?
26. List the composition of the hardware unit.
27. How and when are rocket letter frequencies installed?
28. Composition and purpose of solid propellant rocket motor elements.
29. How are the wings held in the folded position and how and when do they open?
30. For what purpose is the wing block mounted on a bearing?
31. How is the rocket switched to on-board power supply?
32. Operating procedure of the ejection device.
33. Name the sequence of functioning of the rocket elements during ejection.
34. What happens to the missile immediately after it leaves the launch vehicle? 35. How
is the 9M334 module equipped?
36. Characteristic stages of a rocket flight.
37. How does a missile turn in azimuth towards the target?
38. List the pre-flight operating modes of onboard equipment.
39. What is done in the "Standby" mode?
40. What operations are performed in the "Preparation" mode?
41. What operations are performed by the "Start" command?
42. When is the command to launch a missile given?

43. What operations with the rocket are carried out during the operation of the ejection device?
"stva"?

44. When is the command to reset the steering gears issued? 45. How is the control signal to the steering gears generated after the rocket launch? 46. How are the rudders released? 47. How is the rocket controlled during the declination phase? 48. When and how is the rocket engine started? 49. Why are the rudder deflection angles limited in the initial phase of the rocket flight? 50. When is the command to switch the rudder deflection angle limitation issued? 51. How is the rocket position determined by angular coordinates? 52. When and how are the SPK commands transmitted to the rocket? 53. What commands are transmitted by the radio control and radio visualization equipment located

on a rocket?

54. What is the essence of transmitting control command levels (K1, K2) using the phase-pulse method? 55. What determines the "Tact" time interval? 56. When is the KUV command given? 57. When is the KV command given?

58. Purpose of PIM. 59. How and when is the missile self-destructed?

60. List the main sections of the missile flight trajectory? 61. How are the missile rudders oriented in flight? 62. How are the rudder rotation commands generated at bank angles? 63. What is the essence of the three-point

guidance method? 64. What is the essence of the "half-rectification" guidance method? 65. $\ddot{y} = 0$ and $\ddot{y} \neq 0$?

Advantages and disadvantages of the "three points" and "half-

rectification" methods? 66. Features of missile guidance when working at a low-

flying target? 67. How is it possible for a missile to hit the earth's surface and cause a false triggering of the radio fuse from signals reflected from it excluded? 68. What is a kinematic link? 69. What feedback

loops does a missile have as a control object? 70. What should the onboard radio control equipment do? 71. What functions does the onboard radio control equipment (BRU) perform? 72. What commands does the command transmission station (CTS)

transmit? 73. What tasks does the onboard radio control equipment solve by executing CTS commands?

74. List the main units of the onboard equipment. 75. Why are different power supply

voltages used to operate the BRU? 76. What is the point of using logical "1" or logical "0" levels? 77. How

does the BRU function in the "Preparation" mode along the "Record" chain? 78. What

functions does the BRU function in the "Standby" mode? 79. What functions does the missile sighting station (MSS) perform in the "Launch" mode? 80. What functions does the missile sighting station

(MSS) perform? 81. Purpose and operating modes of the onboard radio equipment transponder. 82. How is

one of the two receiving antennas determined in the MSS during flight? 83. What do synchronous and

asynchronous operating modes of the MSS elements mean? 84. When is the "Control" command

issued? 85. What signals does the MSS generate during flight? 86. How

does the MSS operate in the event of a radio link failure? 87. When

is the missile self-destruction command issued? 88. Purpose of the command block. How

are commands generated? 89. What functions does the autopilot perform? 90. What is included in the

autopilot? 91. Why does the autopilot include two linear acceleration sensors, three angular acceleration sensors, and three

niy and three free gyroscopes?

92. How is the loss of dynamic stability of the rocket prevented after exiting the TPK during the vertical flight phase? 93.

What is the orientation of the aerodynamic planes

(rudders and wings) of the rocket during flight? 94. How are control commands generated during the declination phase?

95. How are control commands implemented in the declination section? 96. How is the restriction on the rudder rotation angle removed in the initial part of the declination section? 97. Autopilot functions in control mode. 98. In what coordinate system are the missile control commands generated? 99. How is the missile stabilized in roll? 100. What is a dynamic trajectory? 101. What is a kinematic trajectory? 102. What is the essence of missile guidance to a target? 103. What is a dynamic error? 104. Purpose of free gyroscopes. 105. The design of a free gyroscope. 106. Why is the maximum rotation angle of the inner gyroscope frame limited? 107. Purpose of the angular acceleration sensor. 108. Operating principle of the angular acceleration sensor. 109. Purpose and design of the linear acceleration sensor in the autopilot. 110. What functions are assigned to the radio fuse? 111. When and on what command does the radio fuse begin to irradiate the target? 112. How is the command to detonate the warhead formed? 113. List the design solutions that ensure that the target is covered by a stream of destructive

elements in various conditions of the missile's encounter with the target. 114. In what modes does a radio fuse operate? 115. The operating procedure of a radio fuse. 116. The components of a radio fuse and their purpose. 117. What are the main components of a radio transmitter? 118. Why is a magnetron needed and how does it work? 119. The components of a radio fuse receiver and their purpose. 120. What functions does the information processing unit of a radio fuse receiver perform?

vatela?

121. Where do the pulses come from to the counter of the information processing unit of the receiving device? the properties of a radio detonator? 122. How do you understand the phrase "coincidence in time of strobe and video pulses on the cascade?" where are the coincidences"? 123. Why does a radio fuse have software sensitivity adjustment and a coarsening chain? 124. How is maximum warhead efficiency achieved in the target area? 125. What determines the triggering area of a radio fuse? 126. How is the triggering area of a radio fuse matched to the dispersion zone?

fragments?

127. How is the noise immunity of the RV increased? 128. What is the purpose of the safety actuator mechanism (SAM)? 129. What functions does the SAM perform? 130. When and by what command is the SAM activated? 131. List the conditions under which the safety stages are removed in the SAM. 132. How is the first safety stage removed in the SAM? 133. How is the second safety stage removed in the SAM? 134. How is the third safety stage removed in the SAM? 135. In what cases does the SAM operate? 136. Name the conditions necessary and sufficient to destroy the target. 137. List the design solutions that ensure coverage of the target with a stream of damaging

elements in various conditions of the missile's encounter with the target. 138. How to construct a zone of destruction of any target? 139. Provide characteristics of a rocket engine. 140. Calculate engine thrust in starting and cruise modes. 141. List the main elements that make up a rocket engine. 142. How is dual-mode engine operation ensured? 143. How and by what command is a dual-mode solid propellant rocket motor launched?

144. *How is the independence of engine operating parameters from temperature ensured?
environmental conditions?*
145. *Draw a curve of pressure change in the engine chamber over time.*
146. *How is the engine sealed?*
147. *How is the charge attached to the body?*
148. *How is the required stress-strain state of the charge ensured?*
149. *What is a charge igniter?*
150. *How is a pyropatron constructed and how does it work?*
151. *For what purpose is the spam alarm used in addition to the pressure alarm in the rocket?
yes pressure?*
152. *How does a pressure alarm work?*
153. *Explain the operation of the steering gear.*
154. *Explain the structure of a gas distributor.*
155. *What elements does the gas-dynamic drive of the steering wheel consist of?*
156. *Purpose and design of the gas distributor of the gas-dynamic drive of the rudders.*
157. *Why is a feedback potentiometer used in a steering machine?*
158. *What determines the polarity and magnitude of the signal that controls the operation of the gas distributor?
divider?*
159. *What function does the electromagnetic polarized valve perform in the gas distributor?
relay?*
160. *Explain the operating principle of an electromagnetic polarized relay.*
161. *How does a pressure drop alarm work?*
162. *What is a chemical current source?*
163. *What state can DC batteries be in at different stages of operation?*
164. *Purpose of the electric machine current converter.*
165. *What is included in the control channel drive circuit?*
166. *How is the missile secured in the launch vehicle?*

14. COMMENTS ON THE TEXT AND SOME QUESTIONS

1. To questions about the operation of the radar (questions 4 - 15)

The physical basis of radar is the scattering of radio waves by objects whose electrical characteristics differ from the corresponding characteristics of the environment when they are irradiated. Moreover, the intensity of scattering or reflection of radio waves (RW) (the intensity of the secondary field) depends on the degree of difference between the electrical characteristics of the object and the environment, on the shape of the object, on the ratio of its dimensions and wavelength, on the polarization of the RW.

The resulting secondary electromagnetic field consists of a reflection field, propagating in the direction of the irradiating primary field, and a shadow field, propagating behind the object (in the same direction as the primary field).

Using a receiving antenna and a receiving device, it is possible to receive part of the scattered signal, transform it and amplify it for subsequent detection. Thus, the simplest radar can consist of a transmitter that forms and generates radio signals, a transmitting antenna that emits these radio signals, a receiving antenna that receives reflected signals, a radio receiver that amplifies and transforms the signals, and, finally, an output device that detects the reflected signals. As a rule, the amplitude (or power) of the received signal is small, and the signal itself is random.

The low signal power is explained by the large distance to the object (target) and the absorption of signal energy during its propagation on the one hand, and the dimensions

targets – on the other. The random nature of the signal is a consequence of the fluctuation of the reflected signal due to the random movement of the elements of the target of complex shape during the reflection of the radio waves, chaotic changes in the amplitude of the signal during propagation and a number of other factors. As a result, the received signal in its appearance, intensity and nature of change is similar to noise and interference in the receiving path.

Therefore, the first and main task of the radar is to detect a useful radio signal, i.e. to make a decision about the presence of a useful signal in the mixture of a useful signal with interference coming to the input of the receiving path. This statistical problem is solved by a special device included in the radar device - a detector, in which they try to use an algorithm of optimal (best) detection. The quality of the detection process is characterized by the probability of correct detection, when the signal present in the input implementation is detected, and the probability of a false alarm, when a useful signal is taken for

there is interference, but the signal itself is absent.

Most of the parameters of the received signal are unknown a priori, therefore, when detecting when installing, it is necessary to search for the required parameter of the radio signal that distinguishes it from associated noise and interference.

3. *Classification of radar stations [18]*

Radars are classified according to the following characteristics:

- the origin of the radio signal received by the radar receiver (active radars (with active and passive response), semi-active and passive radars);
- the used range of radars (decameter, meter, decimeter, centimeter and millimeter range radars);
- the type of probing signal (radar with continuous (unmodulated or frequency-modulated) and pulsed (non-coherent, coherent-pulse with large and small duty cycle, with intra-pulse frequency or phase modulation) radiation);
- the number of channels used for emission and reception of signals (single-channel and multi-channel (frequency or spatial division multiplexing);
- the number and type of measured coordinates (one-, two- and three-coordinate);
- a method of measuring, displaying and taking coordinates of an object;
- the location of radar installation (ground, ship, aircraft, satellite).

4. *The operating principle of the simplest radar station*

The number of simultaneously detected and tracked targets is determined by the speed of the systems for receiving and processing information about them. The design of a radar station is based on three components: a transmitter, an antenna, and a receiver.

The transmitter (transmitting device) is a source of high-power electromagnetic signal. It can be a powerful **pulse generator**. For centimeter-range pulse radars - usually a **magnetron** or pulse generator. The transmitter usually operates in pulse mode, generating repetitive short powerful

electromagnetic pulses. Short probing

pulses are emitted into space through **an antenna**.

The antenna (radar) focuses the transmitter signal and forms **the directional pattern**, and also reception of the signal reflected from the target and transmission of this signal to the receiver. If transmission and reception are combined in one antenna, then these actions are performed **alternately**, and so that the powerful signal leaking from the transmitting transmitter to the receiver does not blind the weak echo receiver, a special device is placed in front of the receiver, closing the receiver input at the moment of emission of the probing signal.

The receiver (receiving device) ensures the reception, processing and allocation of information from the received signal. Thus, the

radar serves to detect targets by illuminating them with an electromagnetic wave and then receiving the reflections (echoes) of this wave from the target. Since the speed of electromagnetic waves is constant (**the speed of light**), It becomes possible to determine the distance to the target based on the measurement of various parameters of signal propagation.

If there is an object (target) in the path of radio wave propagation, part of the electromagnetic energy is reflected back towards the radar. The reflected signal enters the receiver via the antenna, is amplified and enters the output device for indication and/or processing (detector, measuring devices).

5. **Magnetron** [19] – a multi-resonator device for generating microwave electromagnetic oscillations based on the interaction of electrons moving in a magnetic field along curvilinear trajectories with an excited electromagnetic field. The magnetron anode is a massive

a hollow cylinder, in the inner part of which volumetric resonators are cut out with slits that go out onto the inner surface of the cylinder (Fig. 46). The cathode is located along the axis of the cylinder. Under the influence of a magnetic field directed along the axis of the cylinder, the trajectory of electrons flying out from the cathode is curved. When oscillations are excited in the resonators, an alternating electric field arises near the slits. Under the influence of a microwave field and crossed static electric and magnetic fields, electrons flying out from the cathode form bunches ("spokes", Fig. 46).

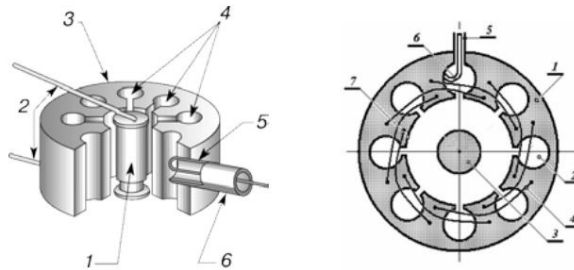


Fig. 46. 1 – cathode; 2,6 – terminals; 3 – anode block; 4 – resonators; 5 – coupling loop

Electrons in bunches, when interacting with the braking microwave field, give the field potential energy and approach the anode. They get to the anode, having given almost all the energy to the electromagnetic field, which determines the high (up to 90%) efficiency. Magnetrons are capable of generating oscillations up to the millimeter range of electromagnetic waves and giving power up to thousands of kilowatts in pulse mode.

The magnetron was developed as a powerful generator of electromagnetic microwave oscillations. range for use in radar systems.

6. Phased antenna array (PA) – an antenna formed by a group of antenna radiators in which the relative phases of the signals change in a complex manner so that the effective radiation of the antenna is amplified in one desired direction and suppressed in all other directions.

Phase control (phasing) allows:

- to form the required antenna radiation pattern (RP) (for example, a highly directional beam-type RP);
- change the direction of the beam of a fixed antenna, thus implementing fast (in some cases, practically inertia-free) scanning – beam swinging;
- control the shape of the DN within certain limits – change the beam width, intensity (levels) of the side lobes, etc. As a result, the radar consists of a phased array antenna, a synthesizer-synchronizer, an analog processor (receiver), a digital processor and an information display device.

7. Pulse radars are modern tracking radars. A pulse radar [20] transmits a signal for only a very short time, in a short pulse (usually about a microsecond), after which it switches to receive mode and listens for the echo reflected from the target while the transmitted pulse propagates through space.

Since the pulse travels away from the radar at a constant speed, between time, There is a direct relationship between the time from the moment the pulse is sent to the moment the echo response is received and the distance to the target. The next pulse can only be sent after some time, namely after it comes back (this depends on the radar detection range, transmitter power, antenna gain, receiver sensitivity). One of the main problems with pulse radars is getting rid of the signal reflected from stationary objects: the earth's surface, high hills, etc. If, for example, an airplane is located against the background of a high hill, the reflected signal from this hill will completely block the signal from the airplane. For ground-based radars, this problem manifests itself when working **with low-flying objects**. For airborne pulse radars, it is expressed in the fact that the reflection from the earth's surface shadows all objects lying below the aircraft with the radar.

Methods of eliminating interference use, in one way or another, **the Doppler effect** (the frequency of the wave reflected from an approaching object increases, from a departing object it decreases).

A radar with moving target selection (MTS) can detect a target in interference: it compares the reflections from two or more pulse repetition intervals. Any target that moves relative to the radar produces a change in the signal parameter (stage in the sequential MTS), while the interference remains unchanged. Interference is eliminated by subtracting the reflections from two successive intervals. In practice, interference elimination can be performed by algorithms in software.

An unavoidable drawback of MTCs operating with a constant PRF is the impossibility of detecting targets with specific circular speeds (targets that produce phase changes of exactly 360°). The speed at which a target becomes invisible to the radar depends on the operating frequency of the station and on the PRF. To eliminate this drawback, modern MTCs emit several pulses with different PRFs. The PRFs are selected in such a way that the number of "blind" speeds is minimal.

Unlike MTI radars, **pulse-Doppler radars** use a different, more complex method of eliminating interference. The received signal, containing information about targets and interference, is transmitted to the input of the Doppler filter block. Each filter passes a signal of a certain frequency. At the output of the filters, derivatives of the signals are calculated. The method helps to find targets with specified speeds, can be implemented in hardware or software, and does not allow (without modifications) to determine the distance to targets. To determine the distance to targets, the pulse repetition interval can be divided into segments (called range segments) and the signal can be fed to the input of the Doppler filter block during a given range segment. The distance can only be calculated with multiple repetitions of pulses at different frequencies (the target appears at different range segments at different PRFs).

An important property of pulse-Doppler radars is **signal coherence** – phase dependence. the stability of sent and received (reflected) signals.

Pulse-Doppler radars, unlike radars with MDT, more often detect low-flying ing goals.

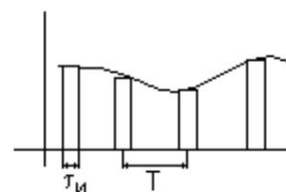
8. **Coherence** is the coordinated flow in time and space of several oscillatory or wave processes, which manifests itself when they are added together. Oscillations are called coherent if the difference in their phases remains constant (or changes regularly) over time and when the oscillations are added together determines the amplitude of the total oscillation.

9. **Intra-pulse modulation (frequency or phase). Modulation** is the physical process of controlling carrier oscillations. Types of modulation (amplitude, frequency, phase) are

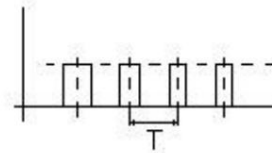
- characterized by: – the depth of amplitude modulation – the maximum relative deviation of the amplitude there from the average;
- frequency deviation – the maximum frequency deviation both at frequency and at phase modulation;
- angular modulation index – maximum phase change.

For undistorted transmission of a message, it is necessary that the spectrum width of the message be small compared to the carrier, and for this it is necessary that the relative change in the modulating function be small over one period of the carrier oscillation. In pulse modulation, a sequence of rectangular pulses is used as a carrier of control signals. In radio signals, this sequence is applied to a high-frequency oscillation (double modulation).

Pulse amplitude modulation (PAM), when the control law the increment of the signal changes decrease in pulse amplitude.

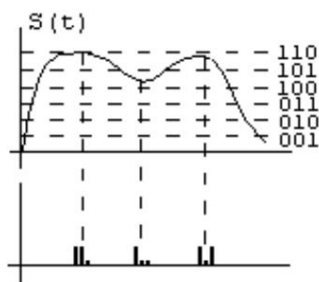
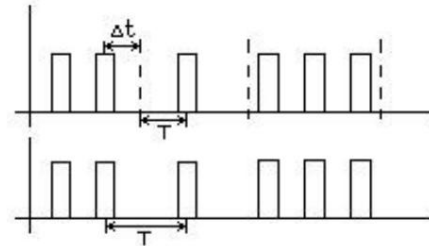


Pulse width modulation (PWM), when the pulse width changes according to the control signal law. Sometimes this type of modulation is called *PWM*



pulse-width modulation (PWM).

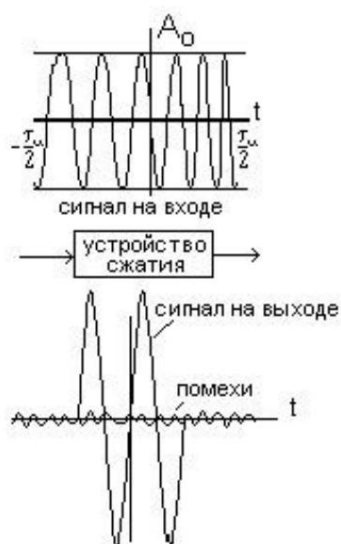
Temporal pulse modulation 3) (TPM), when, according to the law of the control signal, a shift of pulses occurs along the time axis (can be phase (PPM) or frequency (FPM)).



Pulse code modulation (digital).

Each signal level (quantized) is assigned a specific number (code), usually in the binary system. Instead of transmitting the signal value $s(t)$ at the moments of counting the function $s(t)$, a number is transmitted (in the form of a combination of narrow pulses), corresponding to the signal level number at the given moment.

Signals with intrapulse frequency modulation have become widespread in radar. Most often, a linear increase in frequency from the beginning to the end is used.



The practical significance of linear frequency-modulated (LFM) signals lies in the possibility of significant signal compression upon reception with an increase in its amplitude above the noise level.

This is achieved by a delay device with a delay time that decreases with increasing signal frequency. In this case, the output will be a summation of low-frequency oscillations related to the beginning of the pulse and higher-frequency ones observed at its end.

10. To question 15. **Blind speed** is the radial speed of movement of the object of radar observation, at which the Doppler frequency shift of the signal reflected from the object is equal to or a multiple of the repetition frequency of the pulses emitted by the radar, which excludes the possibility of measuring the speed of the object by the radar station.

Radars, as a rule, operate on the principle of receiving a reflected wave. Active radars in passive mode record all objects that reflect waves: thunderclouds and flocks of birds. But this also allows for better knowledge of the air situation. The accuracy of radars in range is not worse than 150 m. The measure of time in radar is 1 μ s. In 1 μ s the radar beam travels 300 m. $S = c \cdot t = (3 \cdot 10^8) \cdot (10^{-6}) = 3 \cdot 10^2 = 300$ m. The range is determined by the delay time of the reflected signal (Fig. 47):

$$D = c \cdot \tau_{\text{zap}} / 2 = (3 \cdot 10^8) \cdot 10^{-6} / 2 = 150 \text{ m}$$

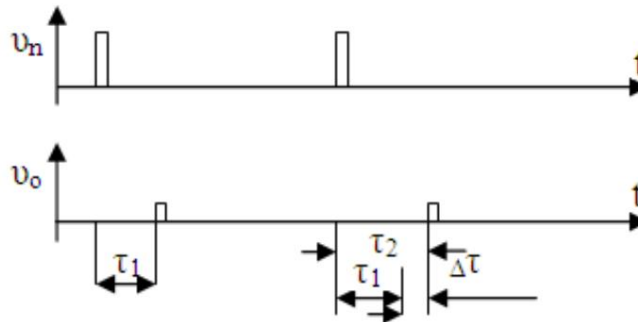


Fig. 47. Reflected signal delay time

By accumulating time changes in the reflected signal on the radar screen, the operator determines the change in the distance of the target from the radar antenna. A similar effect underlies the Doppler principle, which is used in the radar operating mode - MTC (SPC), that is, the selection of moving targets (selection of moving targets). This mode is used to isolate targets against the background of passive interference. However, it also has disadvantages. The operating range of the radar in MTC is 60 ... 75% of the passive. A significant disadvantage of MTC is the manifestation of "**blind speeds**". These are flight speeds at which the aircraft (AC) moves in space during the sending time. The phase shift of the Doppler frequency will be constant when the AC moves during the time between pulse sendings equal to the passage of a half-wave and a multiple of the value of K (Fig. 48).

$$S = K \cdot \dot{y} / 2,$$

where $K = 1.2 \dots$ will determine the first, second ... fifth "blind speeds".

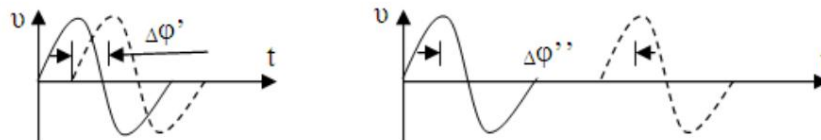


Fig. 48. Doppler frequency phase shift

In general, the value of "blind speed" can be determined from the following relationship: $W_{\dot{y}} = 0.018 K \cdot \dot{y} \cdot Fn$, where 0.018 is a constant coefficient, Fn is the number of pulses per second.

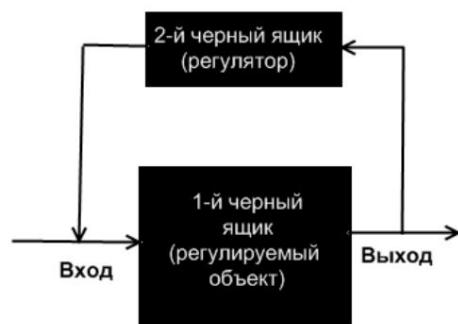
Synchronicity of the movement of the electron beam sweep on the radar screen with the rotation of the antenna is achieved by the coincidence of the mechanical and electrical parts through the selsyn sensors of the antenna and the selsyn receivers of the signal amplifier of the converting equipment. The accuracy of the mechanical part of the best radars - both domestic and foreign - reaches 0.1° , and the resolution by the width of the antenna radiation pattern is 0.5° .

11. **To question 27. Letter frequency** is the frequency of the receiver and transmitter, set placed on the rocket with a key in order to increase the noise immunity of communication lines.

12. **To question 29. A torsion bar** is a spring in the form of a shaft that works by torsion.

13. **To question 69. Feedback. Positive and negative feedback.** Any system can be represented as a black box. The **feedback** concept (Fig. 49) assumes that a change in the output signal of one black box is transmitted to the input of the first through some transfer function of the second black box. As one of the implementation options, the entire output signal of the first black box is fed (added, subtracted, multiplied, divided, etc.) to its input. The entire signal is also just a change in the signal relative to the zero output level.

Negative **feedback** is understood as feedback in which a change in the output signal is transmitted to the input of the black box in such a way as to suppress (compensate) this change. That is, negative feedback "keeps" the output parameter unchanged.



Very important feedback parameters, even with the correct choice of transfer function, are

sya:

- the speed of response to a change in the output signal (time delay). If this parameter is selected incorrectly, then either the system enters the self-oscillation mode (the response time is too short), or the regulation does not keep up with the process (the response time is too long);

Fig. 49. Feedback concept

- sensitivity of the system to changes in the output signal. If this parameter is selected incorrectly, either the system enters the self-oscillation mode (too high sensitivity), or the regulation does not keep up with the process (too low sensitivity);

- the possibility of changing the parameters of the transfer function for tasks is provided, in which external regulation of the output signal level (output control) is also required.

Positive feedback (PFF) is understood as feedback in which a change in the output signal is transmitted to the input of the black box in such a way as to amplify (increase) this change. That is, positive feedback "accelerates" the change in the output parameter. Among the practical general engineering applications of PFF, the use of an output signal with a time delay for system excitation should be singled out. A huge number of electrical engineering solutions (amplifiers, self-oscillating systems, signal generators) are based precisely on the PFF phenomenon.

14. **To question 83. Asynchronous and synchronous modes of operation of BRU elements.**

When exchanging data between network nodes, three methods of data transmission are used:

- simplex (unidirectional) transmission (television, radio);
- half-duplex (reception and transmission of information are carried out alternately);
- duplex (bidirectional), each station simultaneously transmits and receives data

nye.

Serial transmission is most often used for data transmission in networks. The following serial transmission methods are widely used: asynchronous and synchronous.

naya.

In asynchronous transmission, each symbol is transmitted as a separate packet (Fig. 50). Start bits warn the receiver about the beginning of transmission. Then the symbol is transmitted. To determine the reliability of the transmission, a parity bit is used (parity bit = 1 if the number of ones in the symbol is odd, and 0 otherwise). The last bit – the "stop bit" signals the end of transmission.

Advantages: simple, proven system, inexpensive (compared to synchronous) interface equipment.

Disadvantages of asynchronous transmission: a third of the bandwidth is lost on the transmission of service bits (start/stop and parity bits); low transmission speed compared to synchronous; in the case of multiple errors, it is impossible to determine the reliability of the information received using the parity bit.

Asynchronous transmission is used in systems where data exchange occurs sporadically and high data rates are not required. Some systems use a parity bit as a symbol bit, and information control is performed at the level of data exchange protocols.



Fig. 50. Asynchronous data transfer

When using the synchronous method, data is transmitted in blocks (Fig. 51). To synchronize the operation of the receiver and transmitter, synchronization bits are transmitted at the beginning of the block. Then the data, error detection code, and end-of-transmission symbol are transmitted. During synchronous transmission, data can be transmitted both as symbols and as a bit stream. A cyclic redundancy code (CRC) is usually used as an error detection code. It is calculated based on the contents of the data field and allows the reliability of the received information to be determined unambiguously.

Advantages of the synchronous method of information transmission: high efficiency of data transmission; high data transmission rates; reliable built-in error detection mechanism.

Disadvantages – the interface equipment is more complex and therefore more expensive.



Fig. 51. Synchronous data transmission

14. To question 90. About the composition of the autopilot

Two linear acceleration sensors are used to measure linear accelerations and generate a signal in the form of a DC voltage, the magnitude of which is proportional to the linear acceleration acting in the direction of the measuring axis.

Three angular acceleration sensors are used to measure the angular acceleration of the rocket relative to a relatively related coordinate system.

Three free gyroscopes are used to determine three angles (heading – γ , pitch – θ and roll – ϕ) relative to the starting coordinate system.

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